

Trigonometry

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Trigonometry was developed in ancient civilisations to solve practical problems such as building construction and navigating by the stars. We will show that trigonometry can also be used to solve some other practical problems. We can use the trigonometric ratios to solve problems in two dimensions that involve right-angled triangles.

As revision the three trigonometric ratios can be defined for right-angled triangles as:

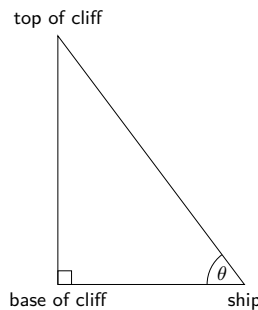
$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} \quad \cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}} \quad \tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

We will use these three ratios and the theorem of Pythagoras to help us solve two-dimensional problems.

11.1 Two-dimensional problems

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In two-dimensional problems we will often refer to the angle of elevation and the angle of depression. To understand these two angles let us consider a person sailing alongside some cliffs. The person looks up and sees the top of the cliffs as shown below:



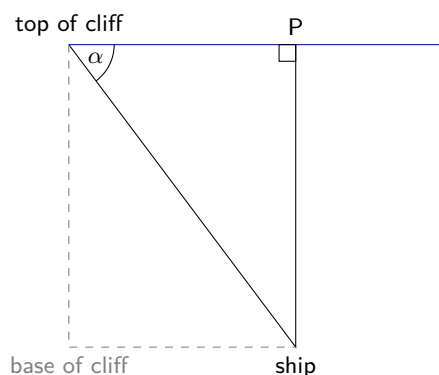
In this diagram θ is the angle of elevation.

DEFINITION: *Angle of elevation*

The angle of elevation is the angle formed by the line of sight and the horizontal plane for an object above the horizontal plane.

In our diagram the line of sight is from the ship to the top of the cliffs. The horizontal plane is from the ship to the base of the cliffs. Also note that we can consider the cliffs to be a straight vertical line and so we have a right-angled triangle.

To understand the angle of depression let us now consider the same situation as above but instead our observer is standing on top of the cliffs looking down at the ship.



In this diagram α is the angle of depression.

DEFINITION: *Angle of depression*

The angle of depression is the angle formed by the line of sight and the horizontal plane for an object below the horizontal plane.

In our diagram the line of sight is from the top of the cliffs to the ship. The horizontal plane is from the top of the cliffs through P . Note that this is parallel to the line between the base of the cliffs and the ship. P lies directly above the ship. We can construct a vertical, perpendicular line to the horizontal plane at the point P .

Finally we can compare the angle of elevation and the angle of depression. In the following diagram the line from the base of the cliffs to the ship is parallel to the line from the top of the cliffs to P . The angle of elevation and the angle of depression are indicated. Notice that $\alpha = \theta$.

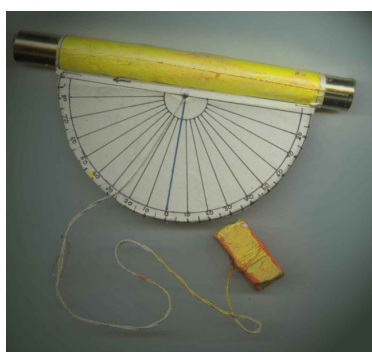
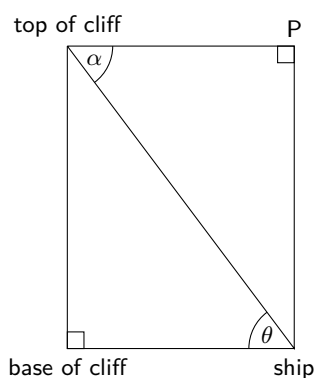


Figure 11.1: An inclinometer. Inclinometers can be used to measure angles of inclination and so can be used to determine the height of an object.

VISIT:

You can make an [inclinometer](#) to measure the angle of inclination of a tall building or tree. Once you have the angle of inclination you can determine the height of the building or tree.

NOTE:

In trigonometry the angle of inclination is the same as the angle of elevation.

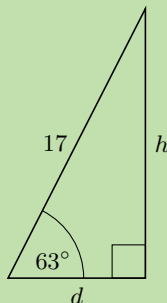
QUESTION

Mandla flies a kite on a 17 m string at an inclination of 63° .

1. What is the height, h , of the kite above the ground?
2. If Mandla's friend Sipho stands directly below the kite, calculate the distance, d , between the two friends.

SOLUTION

Step 1: Make a sketch and identify the opposite and adjacent sides and the hypotenuse



Step 2: Use given information and appropriate ratio to solve for h and d

1.

$$\begin{aligned}\sin 63^\circ &= \frac{\text{opposite}}{\text{hypotenuse}} \\ \sin 63^\circ &= \frac{h}{17} \\ \therefore h &= 17 \sin 63^\circ \\ &= 15,14711... \\ &\approx 15,15 \text{ m}\end{aligned}$$

2.

$$\begin{aligned}\cos 63^\circ &= \frac{\text{adjacent}}{\text{hypotenuse}} \\ \cos 63^\circ &= \frac{d}{17} \\ \therefore d &= 17 \cos 63^\circ \\ &= 7,7178... \\ &\approx 7,72 \text{ m}\end{aligned}$$

Note that the third side of the triangle can also be calculated using the theorem of Pythagoras: $d^2 = 17^2 - h^2$.

Step 3: Write the final answers

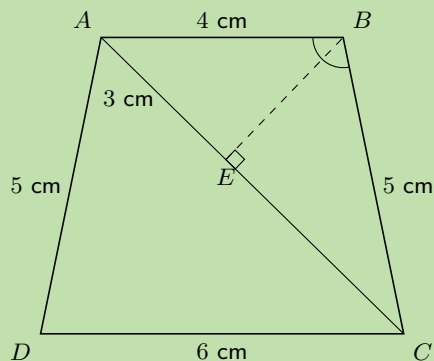
1. The kite is 15,15 m above the ground.
2. Mandla and Sipho are 7,72 m apart.

QUESTION

$ABCD$ is a trapezium with $AB = 4$ cm, $CD = 6$ cm, $BC = 5$ cm and $AD = 5$ cm. Point E on diagonal AC divides the diagonal such that $AE = 3$ cm. $\hat{BEC} = 90^\circ$. Find $\hat{ABC}$.

SOLUTION

Step 1: Draw the trapezium and label all given lengths on diagram. Indicate that $\hat{BEC} = 90^\circ$



We will use $\triangle ABE$ and $\triangle CBE$ to find $\hat{ABE}$ and $\hat{CBE}$. We can then add these two angles together to find $\hat{ABC}$.

Step 2: Find the first angle, $\hat{ABE}$

The hypotenuse and opposite side are given for both triangles, therefore we use the sin function.

In $\triangle ABE$:

$$\begin{aligned}\sin \hat{ABE} &= \frac{\text{opposite}}{\text{hypotenuse}} \\ &= \frac{3}{5} \\ \hat{ABE} &= 36,87^\circ \dots \\ &\approx 37^\circ\end{aligned}$$

Step 3: Use the theorem of Pythagoras to determine BE

In $\triangle ABE$:

$$\begin{aligned}BE^2 &= AB^2 - AE^2 \\ &= 5^2 - 3^2 \\ &= 16 \\ \therefore BE &= \sqrt{16}\text{cm} \\ &= 4\text{cm}\end{aligned}$$

Step 4: Find the second angle $\hat{CBE}$

In $\triangle CBE$:

$$\begin{aligned}\cos \hat{CBE} &= \frac{\text{adjacent}}{\text{hypotenuse}} \\ &= \frac{4}{5} \\ &= 0,8 \\ \hat{CBE} &= 36,87^\circ \dots \\ &\approx 37^\circ\end{aligned}$$

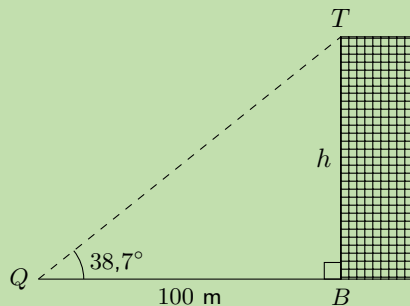
Step 5: Calculate the sum of the angles

$$\hat{ABC} = 48,6^\circ + 58,1^\circ = 106,7^\circ$$

Another application is using trigonometry to find the height of a building. We could use a tape measure lowered from the roof, but this is impractical (and dangerous) for tall buildings. It is much more sensible to use trigonometry.

Worked example 3: Finding the height of a building**QUESTION**

The given diagram shows a building of unknown height h . We start at point B and walk 100 m away from the building to point Q . Next we measure the angle of elevation from the ground to the top of the building, T , and find that the angle is $38,7^\circ$. Calculate the height of the building, correct to the nearest metre.

**SOLUTION****Step 1: Identify the opposite and adjacent sides and the hypotenuse**

We have a right-angled triangle and know the length of one side and an angle. We can therefore calculate the height of the building.

Step 2:

In $\triangle QTB$:

$$\begin{aligned}\tan 38,7^\circ &= \frac{\text{opposite}}{\text{adjacent}} \\ &= \frac{h}{100}\end{aligned}$$

Step 3: Rearrange and solve for h

$$\begin{aligned}h &= 100 \times \tan 38,7^\circ \\ &= 80,1151... \\ &\approx 80\end{aligned}$$

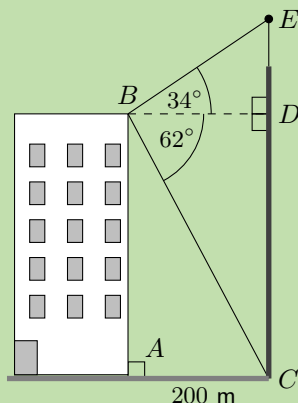
Step 4: Write final answer

The height of the building is 80 m.

QUESTION

A block of flats is 200 m away from a cellphone tower. Someone stands at B . They measure the angle from B to the top of the tower (E) to be 34° (the angle of elevation). They then measure the angle from B to the bottom of the tower (C) to be 62° (the angle of depression).

What is the height of the cellphone tower (correct to the nearest metre)?



Note: diagram not drawn to scale

SOLUTION

Step 1: To determine height CE , first calculate lengths DE and CD

$\triangle BDE$ and $\triangle BDC$ are both right-angled triangles. In each of the triangles, the length BD is known. Therefore we can calculate the sides of the triangles.

Step 2: Calculate CD

The length AC is given. $CABD$ is a rectangle so $BD = AC = 200$ m.

In $\triangle CBD$:

$$\begin{aligned}\tan \hat{CBD} &= \frac{CD}{BD} \\ \therefore CD &= BD \times \tan \hat{CBD} \\ &= 200 \times \tan 62^\circ \\ &= 376,1452... \\ &\approx 376 \text{ m}\end{aligned}$$

Step 3: Calculate DE

In $\triangle DBE$:

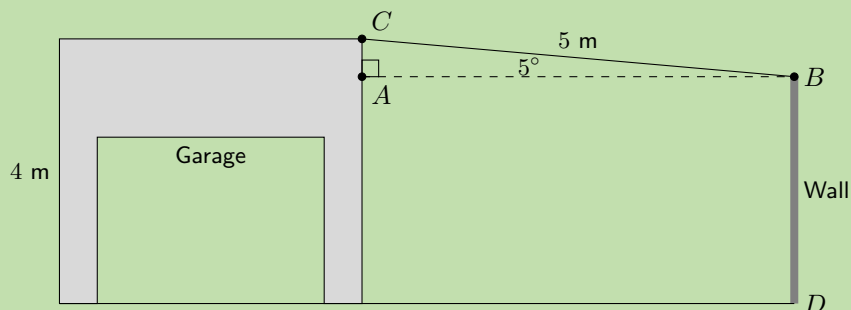
$$\begin{aligned}\tan \hat{DBE} &= \frac{DE}{BD} \\ \therefore DE &= BD \times \tan \hat{DBE} \\ &= 200 \times \tan 34^\circ \\ &= 134,9017... \\ &\approx 135 \text{ m}\end{aligned}$$

Step 4: Add the two heights to get the final answer

The height of the tower is: $CE = CD + DE = 135 \text{ m} + 376 \text{ m} = 511 \text{ m}$.

QUESTION

Mr Nkosi has a garage at his house and he decides to add a corrugated iron roof to the side of the garage. The garage is 4 m high, and his sheet for the roof is 5 m long. If the angle of the roof is 5° , how high must he build the wall BD ? Give the answer correct to 1 decimal place.



SOLUTION

Step 1: Identify opposite and adjacent sides and hypotenuse

$\triangle ABC$ is right-angled. The hypotenuse and an angle are known therefore we can calculate AC . The height of the wall BD is then the height of the garage minus AC .

$$\begin{aligned}\sin \hat{ABC} &= \frac{AC}{BC} \\ \therefore AC &= BC \times \sin \hat{ABC} \\ &= 5 \sin 5^\circ \\ &= 0,43577... \\ &\approx 0,4 \text{ m}\end{aligned}$$

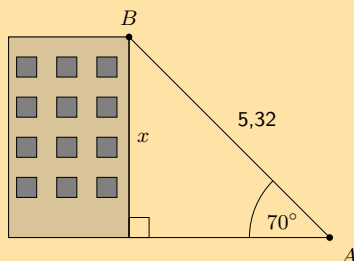
$$\begin{aligned}\therefore BD &= 4 \text{ m} - 0,4 \text{ m} \\ &= 3,6 \text{ m}\end{aligned}$$

Step 2: Write the final answer

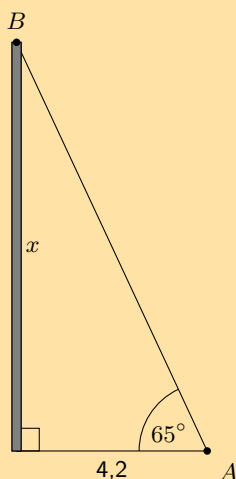
Mr Nkosi must build his wall to be 3,6 m high.

Exercise 11 – 1:

1. A person stands at point A , looking up at a bird sitting on the top of a building, point B . The height of the building is x meters, the line of sight distance from point A to the top of the building (point B) is 5,32 meters, and the angle of elevation to the top of the building is 70° . Calculate the height of the building (x) as shown in the diagram below:



2. A person stands at point A , looking up at a bird sitting on the top of a pole (point B). The height of the pole is x meters, point A is 4,2 meters away from the foot of the pole, and the angle of elevation to the top of the pole is 65° . Calculate the height of the pole (x), to the nearest metre.



3. A boy flying a kite is standing 30 m from a point directly under the kite. If the kite's string is 50 m long, find the angle of elevation of the kite.
4. What is the angle of elevation of the sun when a tree 7,15 m tall casts a shadow 10,1 m long?
5. From a distance of 300 m, Susan looks up at the top of a lighthouse. The angle of elevation is 5° . Determine the height of the lighthouse to the nearest metre.
6. A ladder of length 25 m is resting against a wall, the ladder makes an angle 37° to the wall. Find the distance between the wall and the base of the ladder to the nearest metre.

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2GPN 2. 2GPP 3. 2GPQ 4. 2GPR 5. 2GPS 6. 2GPT



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11.2 Chapter summary

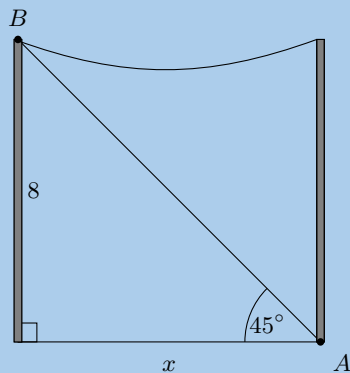
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▶ See presentation: 2GPV at www.everythingmaths.co.za

- We can define three trigonometric ratios for right-angled triangles: sine (sin), cosine (cos) and tangent (tan).
- Trigonometry is used to help us solve problems in two dimensions that involve right-angled triangles, such as finding the height of a building.
- The angle of elevation is the angle formed by the line of sight and the horizontal plane for an object above the horizontal plane.
- The angle of depression is the angle formed by the line of sight and the horizontal plane for an object below the horizontal plane.

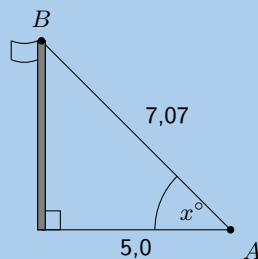
1. A ladder of length 15 m is resting against a wall, the base of the ladder is 5 m from the wall. Find the angle between the wall and the ladder.
2. Captain Jack is sailing towards a cliff with a height of 10 m.
 - a) The distance from the boat to the bottom of the cliff is 30 m. Calculate the angle of elevation from the boat to the top of the cliff (correct to the nearest degree).
 - b) If the boat sails 7 m closer to the cliff, what is the new angle of elevation from the boat to the top of the cliff?
3. Jim stands at point A at the base of a telephone pole, looking up at a bird sitting on the top of another telephone pole (point B). The height of each of the telephone poles is 8 meters, and the angle of elevation from A to the top of B is 45° .

Calculate the distance between the telephone poles (x) as shown in the diagram below:

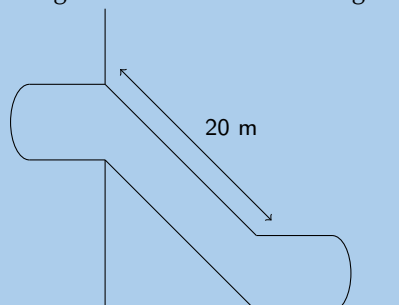


4. Alfred stands at point A , looking up at a flag on a pole (point B). Point A is 5,0 meters away from the bottom of the flag pole, the line of sight distance from point A to the top of the flag pole (point B) is 7,07 meters, and the angle of elevation to the top of the flag pole is x° .

Calculate the angle of elevation to the top of the flag pole (x) as shown in the diagram below:

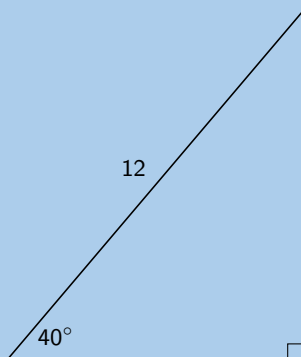


5. A rugby player is trying to kick a ball through the poles. The rugby crossbar is 3,4 m high. The ball is placed 24 m from the poles. What is the minimum angle he needs to launch the ball to get it over the bar?
6. The escalator at a mall slopes at an angle of 30° and is 20 m long.



Through what height would a person be lifted by travelling on the escalator?

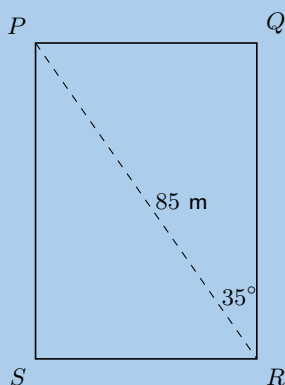
7. A ladder is 8 metres long. It is leaning against the wall of a house and reaches 6 metres up the wall.
 - a) Draw a sketch of the situation.
 - b) Calculate the angle which the ladder makes with the flat (level) ground.
8. Nandi is standing on level ground 70 metres away from a tall tower. From her position, the angle of elevation of the top of the tower is 38° .
 - a) Draw a sketch of the situation.
 - b) What is the height of the tower?
9. The top of a pole is anchored by a 12 m cable which makes an angle of 40° with the horizontal. What is the height of the pole?



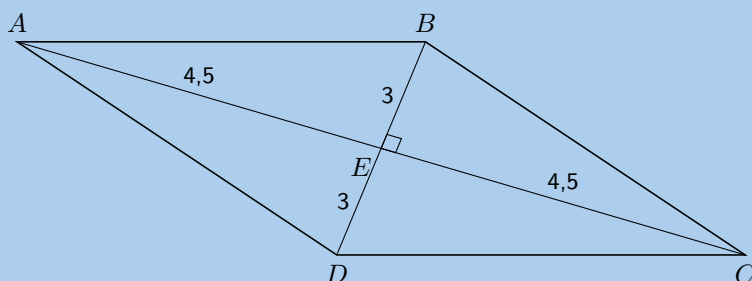
10. A ship's navigator observes a lighthouse on a cliff. According to the navigational charts the top of the lighthouse is 35 metres above sea level. She measures the angle of elevation of the top of the lighthouse to be $0,7^\circ$.

Ships have been advised to stay at least 4 km away from the shore. Is the ship safe?

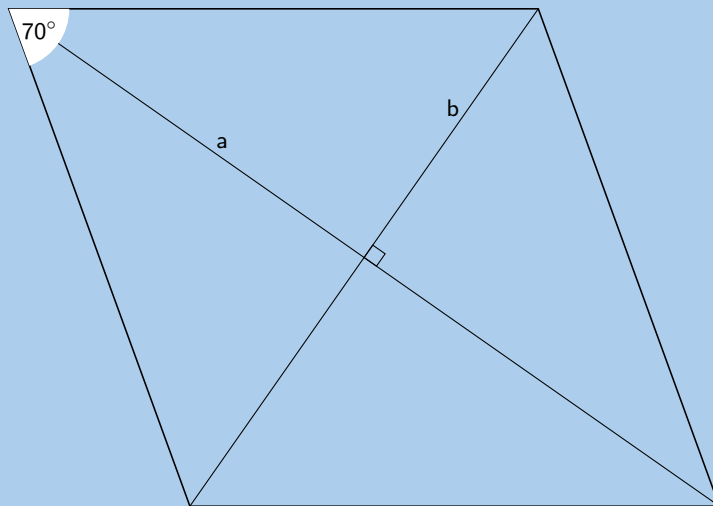
11. Determine the perimeter of rectangle $PQRS$:



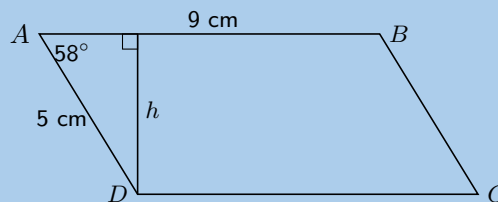
12. A rhombus has diagonals of lengths 6 cm and 9 cm. Calculate the sizes of its interior vertex angles.



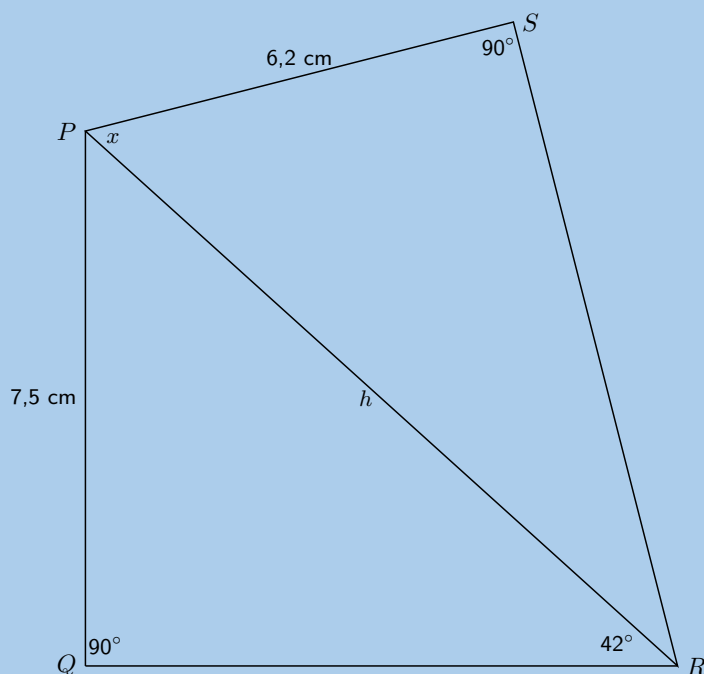
13. A rhombus has edge lengths of 7 cm. Its acute interior vertex angles are both 70° . Calculate the lengths of both of its diagonals.



14. A parallelogram has edge-lengths of 5 cm and 9 cm respectively, and an angle of 58° between them. Calculate the perpendicular distance between the two longer edges.



15. One of the angles of a rhombus with perimeter 20 cm is 30° .
- Find the lengths the sides of the rhombus.
 - Find the length of both diagonals.
16. Upright sticks and the shadows they cast can be used to judge the sun's altitude in the sky (the angle the sun makes with the horizontal) and the heights of objects.
- An upright stick, 1 metre tall, casts a shadow which is 1,35 metres long. What is the altitude of the sun?
 - At the same time, the shadow of a building is found to be 47 metres long. What is the height of the building?
17. The angle of elevation of a hot air balloon, climbing vertically, changes from 25 degrees at 11:00 am to 60 degrees at 11:02 am. The point of observation of the angle of elevation is situated 300 metres away from the take off point.
- Draw a sketch of the situation.
 - Calculate the increase in height between 11:00 am and 11:02 am.
18. When the top, T , of a mountain is viewed from point A , 2000 m from the ground, the angle of depression (a) is equal to 15° . When it is viewed from point B on the ground, the angle of elevation (b) is equal to 10° . If the points A and B are on the same vertical line, find the height, h , of the mountain. Round your answer to one decimal place.
19. The diagram below shows quadrilateral $PQRS$, with $PQ = 7,5$ cm, $PS = 6,2$ cm, angle $R = 42^\circ$ and angles S and Q are right angles.



a) Find PR , correct to 2 decimal places.

b) Find the size of the angle marked x , correct to one decimal place.

20. From a boat at sea (S), the angle of elevation of the top of a lighthouse PQ , on a cliff QR , is 27° . The lighthouse is 10 m high and the cliff top is 75 m above sea level. How far is the boat from the base of the cliff, to the nearest metre?

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

- | | | | | | |
|----------|----------|----------|----------|----------|----------|
| 1. 2GPW | 2. 2GPX | 3. 2GPY | 4. 2GPZ | 5. 2GQ2 | 6. 2GQ3 |
| 7. 2GQ4 | 8. 2GQ5 | 9. 2GQ6 | 10. 2GQ7 | 11. 2GQ8 | 12. 2GQ9 |
| 13. 2GQB | 14. 2GQC | 15. 2GQD | 16. 2GQF | 17. 2GQG | 18. 2GQH |
| 19. 2GQJ | 20. 2GQK | | | | |



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Euclidean geometry

12.1 *Proofs and conjectures*

404

12.2 *Chapter summary*

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Geometry (from the Greek “geo” = earth and “metria” = measure) arose as the field of knowledge dealing with spatial relationships. Geometry can be split into Euclidean geometry and analytical geometry. Analytical geometry deals with space and shape using algebra and a coordinate system. Euclidean geometry deals with space and shape using a system of logical deductions.

12.1 Proofs and conjectures

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We will now apply what we have learnt about geometry and the properties of polygons (in particular triangles and quadrilaterals) to prove some of these properties. We will also look at how we can prove a particular quadrilateral is one of the special quadrilaterals.

VISIT:

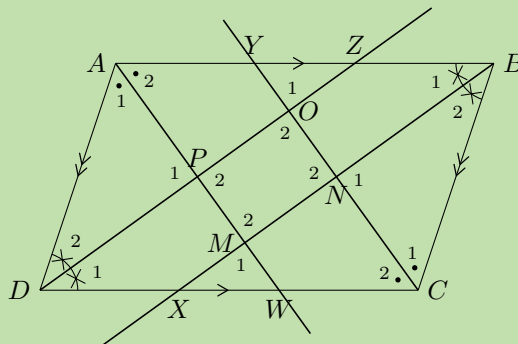
This video shows how to prove that the diagonals of a rhombus are perpendicular.

▶ See video: [2GQM](https://www.everythingmaths.co.za) at www.everythingmaths.co.za

Worked example 1: Proving a quadrilateral is a parallelogram

QUESTION

In parallelogram $ABCD$, the bisectors of the angles (AW , BX , CY and DZ) have been constructed. You are also given $AB = CD$, $AD = BC$, $AB \parallel CD$, $AD \parallel BC$, $\hat{A} = \hat{C}$, $\hat{B} = \hat{D}$. Prove that $MNOP$ is a parallelogram.



SOLUTION

Step 1: Use properties of the parallelogram $ABCD$ to fill in on the diagram all equal sides and angles.

Step 2: Prove that $\hat{M}_2 = \hat{O}_2$

In $\triangle CDZ$ and $\triangle ABX$,

$$\begin{aligned} \hat{DCZ} &= \hat{BAX} && \text{(given)} \\ \hat{D}_1 &= \hat{B}_1 && \text{(given)} \\ DC &= AB && \text{(given)} \\ \therefore \triangle CDZ &\equiv \triangle ABX && \text{(AAS)} \\ \therefore CZ &= AX \\ \text{and } \hat{CZD} &= \hat{AXB} \end{aligned}$$

In $\triangle XAM$ and $\triangle ZCO$

$$\begin{aligned}
 \hat{XAM} &= \hat{ZCO} && \text{(given: } \triangle CDZ \equiv \triangle ABX) \\
 \hat{AXM} &= \hat{CZO} && \text{(proved above)} \\
 AX &= CZ && \text{(proved above)} \\
 \therefore \triangle XAM &\equiv \triangle ZCO && \text{(AAS)} \\
 \therefore \hat{M}_1 &= \hat{O}_1 \\
 \text{but } \hat{M}_1 &= \hat{M}_2 && \text{(vert opp } \angle\text{s =)} \\
 \text{and } \hat{O}_1 &= \hat{O}_2 && \text{(vert opp } \angle\text{s =)} \\
 \therefore \hat{M}_2 &= \hat{O}_2
 \end{aligned}$$

Step 3: Similarly, we can show that $\hat{N}_2 = \hat{P}_2$

First show $\triangle ADW \equiv \triangle CBY$. Then show $\triangle PDW \equiv \triangle NBY$.

Step 4: Conclusion

Both pairs of opposite angles of $MNOP$ are equal. Therefore $MNOP$ is a parallelogram.

VISIT:

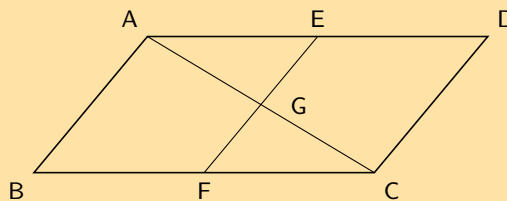
This video shows how to prove that the opposite angles of a parallelogram are equal.

▶ See video: 2GQN at www.everythingmaths.co.za

Exercise 12 – 1:

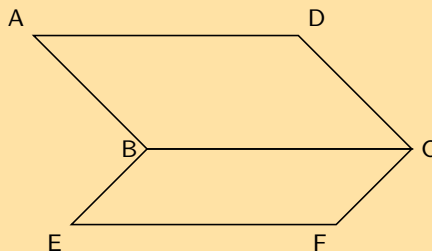
- In the diagram below, AC and EF bisect each other at G . E is the midpoint of AD , and F is the midpoint of BC .

a) Prove $AECF$ is a parallelogram.

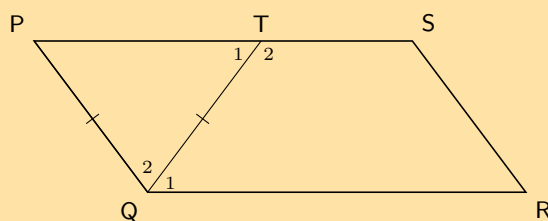


b) Prove $ABCD$ is a parallelogram.

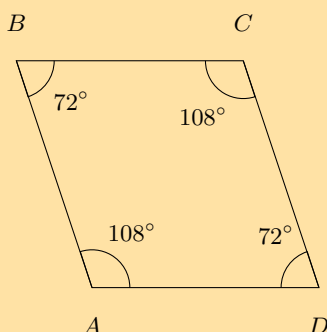
- Parallelogram $ABCD$ and $BEFC$ are shown below. Prove $AD = EF$.



- In the diagram below $PQRS$ is a parallelogram with $PQ = TQ$. Prove $\hat{Q}_1 = \hat{R}$

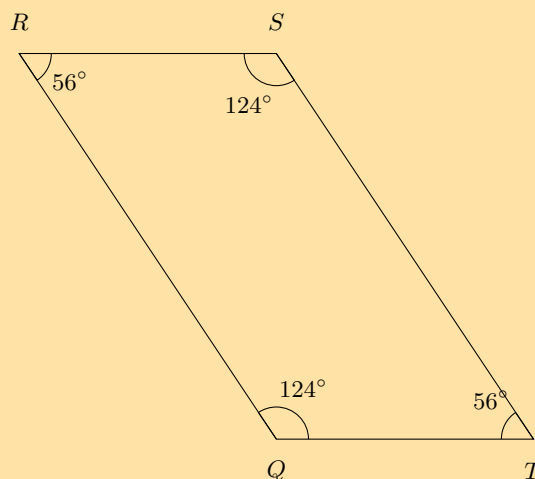


4. Study the quadrilateral $ABCD$ with opposite angles $\hat{A} = \hat{C} = 108^\circ$ and angles $\hat{B} = \hat{D} = 72^\circ$ carefully. Fill in the missing reasons and steps to prove that the quadrilateral $ABCD$ is a parallelogram.



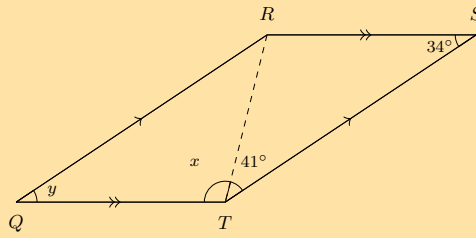
Steps	Reasons
?	given both $\angle$'s $= 108^\circ$
$\hat{A} = \hat{C}$	given both $\angle$'s $= 72^\circ$
$\hat{A} + \hat{B} + \hat{C} + \hat{D} = 360^\circ$	sum of $\angle$ s in quad
$\hat{B} + \hat{D} = 180^\circ$	given $108^\circ + 72^\circ = 180^\circ$
$\therefore AB \parallel DC$	?
$\therefore BC \parallel AD$	?
$\therefore ABCD$ is a parallelogram	opp sides of quad $\parallel$

5. Study the quadrilateral $QRST$ with opposite angles $Q = S = 124^\circ$ and angles $R = T = 56^\circ$ carefully. Fill in the missing reasons and steps to prove that the quadrilateral $QRST$ is a parallelogram.

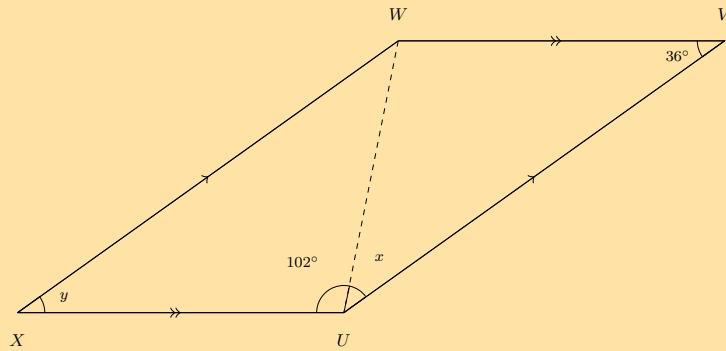


Steps	Reasons
$\hat{R} = \hat{T}$	given both $\angle$ s $= 124^\circ$
?	given both $\angle$ s $= 56^\circ$
$\hat{Q} + \hat{R} + \hat{S} + \hat{T} = 360^\circ$	?
$\hat{R} + \hat{T} = 180^\circ$	?
$\therefore QR \parallel TS$	co-int $\angle$ s; $QR \parallel TS$
$\therefore RS \parallel QT$	co-int $\angle$ s; $RS \parallel QT$
?	opp sides of quad $\parallel$

6. a) Quadrilateral $QRST$ with sides $QR \parallel TS$ and $QT \parallel RS$ is given. You are also given that: $\hat{Q} = y$ and $\hat{S} = 34^\circ$; $\hat{QTR} = x$ and $\hat{RTS} = 41^\circ$. Prove that $QRST$ is a parallelogram.



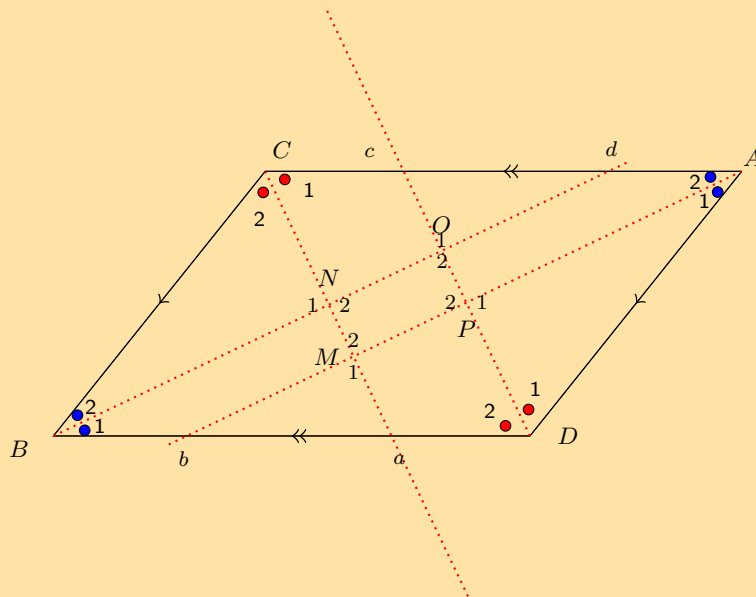
- b) Find the value of y .
 c) Find the value of x .
7. a) Quadrilateral $XWVU$ with sides $XW \parallel UV$ and $XU \parallel WV$ is given. Also given is $\hat{X} = y$ and $\hat{V} = 36^\circ$; $\hat{XUW} = 102^\circ$ and $\hat{WUV} = x$. Prove that $XWVU$ is a parallelogram.



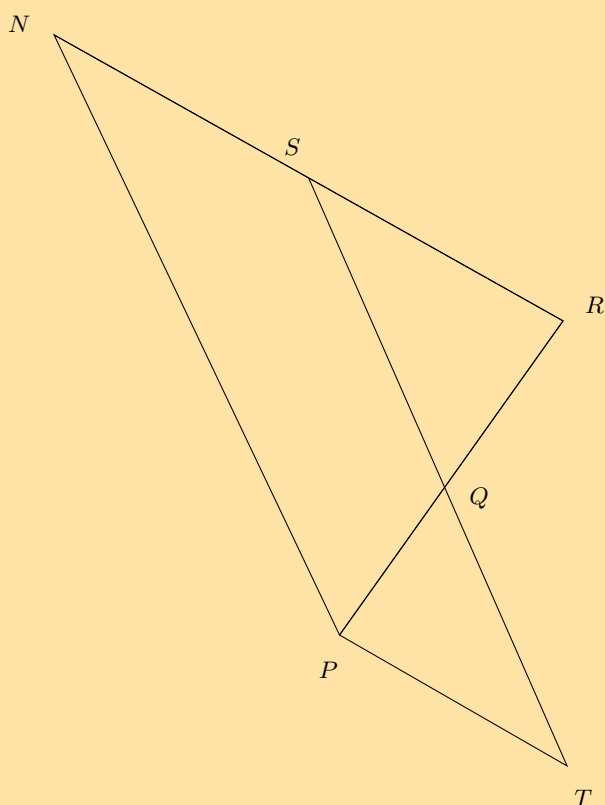
- b) Determine the value of y .
 c) Determine the value of x .
8. In parallelogram $ADBC$, the bisectors of the angles (A, D, B, C) have been constructed, indicated with the red lines below. You are also given $AD = CB$, $DB = AC$, $AD \parallel CB$, $DB \parallel AC$, $\hat{A} = \hat{B}$ and $\hat{D} = \hat{C}$.

Prove that the quadrilateral $MNOP$ is a parallelogram.

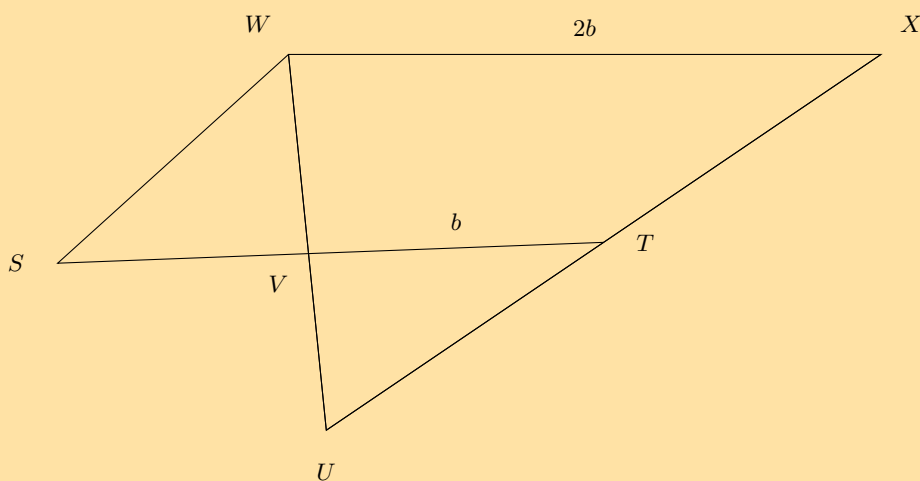
Note the diagram is drawn to scale.



9. Study the diagram below; it is not necessarily drawn to scale. Two triangles in the figure are congruent: $\triangle QRS \equiv \triangle QPT$. Additionally, $SN = SR$. You need to prove that $NPTS$ is a parallelogram.



10. Study the diagram below; it is not necessarily drawn to scale. Quadrilateral $XWST$ is a parallelogram and TV and XW have lengths b and $2b$, respectively, as shown. You need to prove that $\triangle TVU \equiv \triangle SVW$.



For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2GQP 2. 2GQQ 3. 2GQR 4. 2GQS 5. 2GQT 6. 2GQV
7. 2GQW 8. 2GQX 9. 2GQY 10. 2GQZ



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🔊 See presentation: 2GR2 at www.everythingmaths.co.za

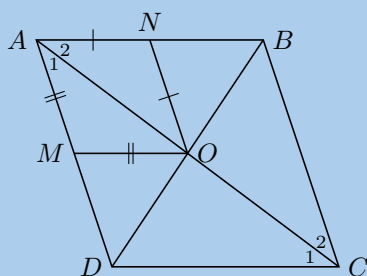
- A quadrilateral is a closed shape consisting of four straight line segments.
- A parallelogram is a quadrilateral with both pairs of opposite sides parallel.
 - Both pairs of opposite sides are equal in length.
 - Both pairs of opposite angles are equal.
 - Both diagonals bisect each other.
- A rectangle is a parallelogram that has all four angles equal to 90°
 - Both pairs of opposite sides are parallel.
 - Both pairs of opposite sides are equal in length.
 - The diagonals bisect each other.
 - The diagonals are equal in length.
 - All interior angles are equal to 90°
- A rhombus is a parallelogram that has all four sides equal in length.
 - Both pairs of opposite sides are parallel.
 - All sides are equal in length.
 - Both pairs of opposite angles are equal.
 - The diagonals bisect each other at 90°
 - The diagonals of a rhombus bisect both pairs of opposite angles.
- A square is a rhombus that has all four interior angles equal to 90°
 - Both pairs of opposite sides are parallel.
 - The diagonals bisect each other at 90°
 - All interior angles are equal to 90°
 - The diagonals are equal in length.
 - The diagonals bisect both pairs of interior opposite angles (i.e. all are 45°)
- A trapezium is a quadrilateral with one pair of opposite sides parallel.
- A kite is a quadrilateral with two pairs of adjacent sides equal.
 - One pair of opposite angles are equal (the angles are between unequal sides).
 - The diagonal between equal sides bisects the other diagonal.
 - The diagonal between equal sides bisects the interior angles.
 - The diagonals intersect at 90°
- The mid-point theorem: The line joining the mid-points of two sides of a triangle is parallel to the third side and equal to half the length of the third side.

VISIT:

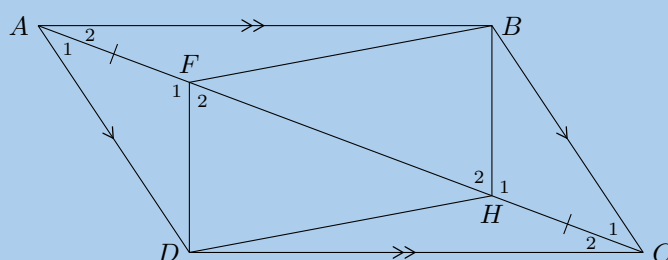
This video shows how to prove that the opposite sides of a parallelogram are equal.

🔊 See video: 2GR3 at www.everythingmaths.co.za

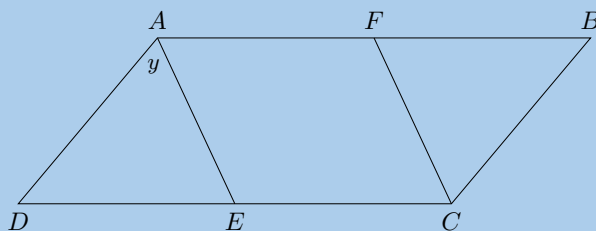
1. $ABCD$ is a rhombus with $AM = MO$ and $AN = NO$. Prove $ANOM$ is also a rhombus.



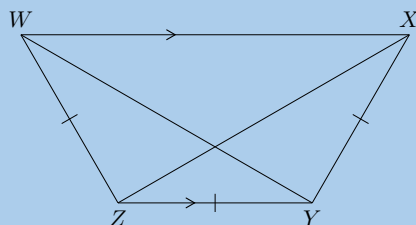
2. $ABCD$ is a parallelogram with diagonal AC . Given that $AF = HC$, show that:



- $\triangle AFD \equiv \triangle CHB$
 - $DF \parallel HB$
 - $DFBH$ is a parallelogram
3. Given parallelogram $ABCD$ with AE bisecting $\hat{A}$ and FC bisecting $\hat{C}$.

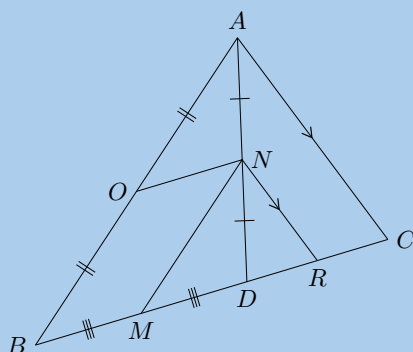


- Write all interior angles in terms of y .
 - Prove that $AFCE$ is a parallelogram.
4. Given that $WZ = ZY = YX$, $\hat{W} = \hat{X}$ and $WX \parallel ZY$, prove that:

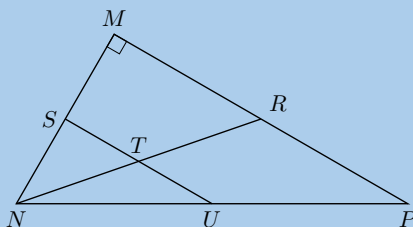


- XZ bisects $\hat{X}$
- $WY = XZ$

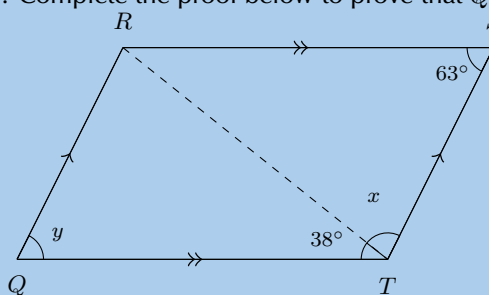
5. D is a point on BC , in $\triangle ABC$. N is the mid-point of AD . O is the mid-point of AB and M is the mid-point of BD . $NR \parallel AC$.



- Prove that $OBMN$ is a parallelogram.
 - Prove that $BC = 2MR$.
6. In $\triangle MNP$, $\hat{M} = 90^\circ$, S is the mid-point of MN and T is the mid-point of NR .



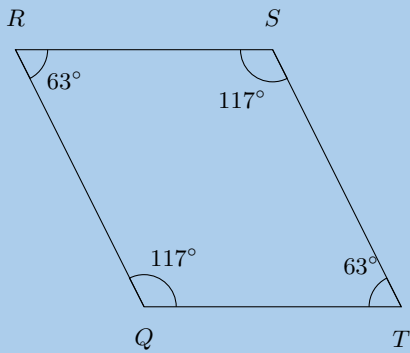
- Prove U is the mid-point of NP .
 - If $ST = 4$ cm and the area of $\triangle SNT$ is 6 cm^2 , calculate the area of $\triangle MNR$.
 - Prove that the area of $\triangle MNR$ will always be four times the area of $\triangle SNT$, let $ST = x$ units and $SN = y$ units.
7. a) Given quadrilateral $QRST$ with sides $QR \parallel TS$ and $QT \parallel RS$. Also given: $\hat{Q} = y$ and $\hat{S} = 63^\circ$; $\hat{QTR} = 38^\circ$ and $\hat{RTS} = x$. Complete the proof below to prove that $QRST$ is a parallelogram.



Steps	Reasons
$\hat{QTR} = \hat{TRS}$	alt $\angle$ s; $QT \parallel RS$
$\hat{STR} = \hat{QRT}$	alt $\angle$ s; $QR \parallel TS$
?	?
$\therefore \triangle QRT \equiv \triangle STR$	(AAS)
?	congruent triangles
$\hat{Q} = \hat{S}$	congruent triangles
$\therefore QRST$ is a parallelogram	?

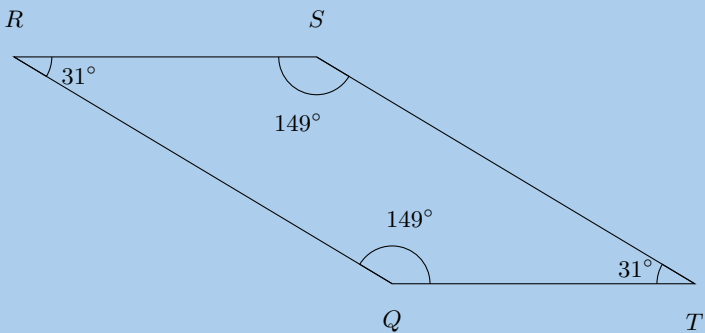
- Calculate the value of y .
- Calculate the value of x .

8. Study the quadrilateral $QRST$ with opposite angles $\hat{Q} = \hat{S} = 117^\circ$ and angles $\hat{R} = \hat{T} = 63^\circ$ carefully. Fill in the correct reasons or steps to prove that the quadrilateral $QRST$ is a parallelogram.



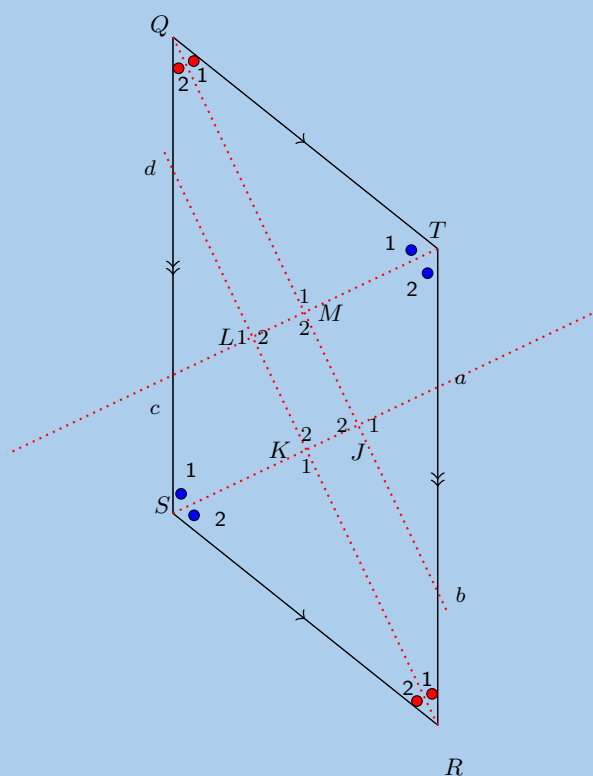
Steps	Reasons
?	given both $\angle s = 117^\circ$
$\hat{Q}\hat{R}\hat{S} = \hat{Q}\hat{T}\hat{S}$	given both $\angle s = 63^\circ$
?	sum of $\angle s$ in quad
$\hat{R}\hat{Q}\hat{T} + \hat{Q}\hat{T}\hat{S} = 180^\circ$	$117^\circ + 63^\circ = 180^\circ$
$\therefore QR \parallel TS$	co-int $\angle s$; $QR \parallel TS$
$\therefore RS \parallel QT$	?
$\therefore QRST$ is a parallelogram	?

9. Study the quadrilateral $QRST$ with $\hat{Q} = \hat{S} = 149^\circ$ and $\hat{R} = \hat{T} = 31^\circ$ carefully. Fill in the correct reasons or steps to prove that the quadrilateral $QRST$ is a parallelogram.

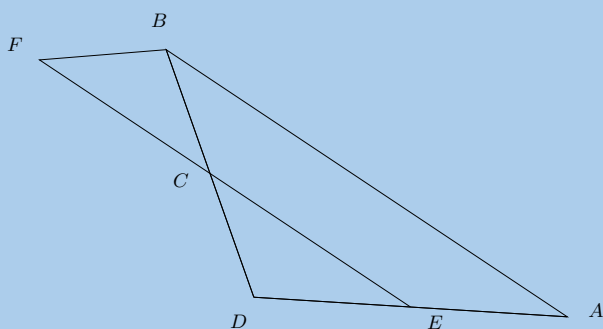


Steps	Reasons
$\hat{R}\hat{Q}\hat{T} = \hat{R}\hat{S}\hat{T}$	given both $\angle s = 149^\circ$
$\hat{Q}\hat{R}\hat{S} = \hat{Q}\hat{T}\hat{S}$	?
$\hat{Q} + \hat{R} + \hat{S} + \hat{T} = 360^\circ$	sum of $\angle s$ in quad
$\hat{R}\hat{Q}\hat{T} + \hat{Q}\hat{T}\hat{S} = 180^\circ$	?
?	co-int $\angle s$; $QR \parallel TS$
?	co-int $\angle s$; $RS \parallel QT$
$\therefore QRST$ is a parallelogram	opp. sides are parallel

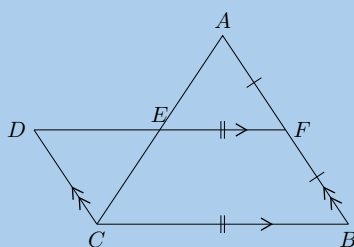
10. In parallelogram $QTRS$, the bisectors of the angles have been constructed, indicated with the red lines below. You are also given $QT = SR$, $TR = QS$, $QT \parallel SR$, $TR \parallel QS$, $\hat{Q} = \hat{R}$ and $\hat{T} = \hat{S}$. Prove that the quadrilateral $JKLM$ is a parallelogram.
Note the diagram is drawn to scale.



11. Study the diagram below; it is not necessarily drawn to scale. Two triangles in the figure are congruent: $\triangle CDE \equiv \triangle CBF$. Additionally, $EA = ED$. You need to prove that $ABFE$ is a parallelogram.

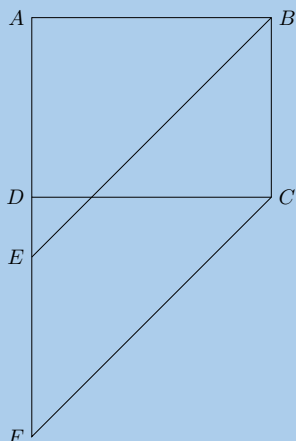


12. Given the following diagram:

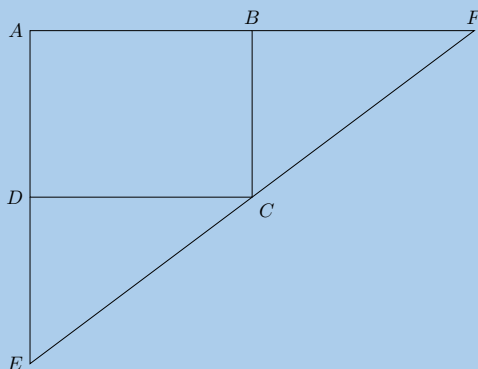


- Show that $BCDF$ is a parallelogram.
- Show that $ADCF$ is a parallelogram.
- Prove that $AE = EC$.

13. $ABCD$ is a parallelogram. $BEFC$ is a parallelogram. $ADEF$ is a straight line. Prove that $AE = DF$.



14. In the figure below $AB = BF$, $AD = DE$. $ABCD$ is a parallelogram. Prove EF is a straight line.



For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

- | | | | | | | |
|-------------------------|-------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1. 2GR4 | 2. 2GR5 | 3. 2GR6 | 4. 2GR7 | 5. 2GR8 | 6. 2GR9 | 7. 2GRB |
| 8. 2GRC | 9. 2GRD | 10. 2GRF | 11. 2GRG | 12. 2GRH | 13. 2GRJ | 14. 2GRK |



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Measurements

13.1	<i>Area of a polygon</i>	416
13.2	<i>Right prisms and cylinders</i>	420
13.3	<i>Right pyramids, right cones and spheres</i>	432
13.4	<i>The effect of multiplying a dimension by a factor of k</i>	451
13.5	<i>Chapter summary</i>	456

Knowing how to calculate the surface area and volume of an object can be useful in many contexts, particularly when we need to know how much a task is going to cost or how much material is needed to create an object. Some examples of this are calculating the surface area of a container, to help us work out the cost of the material, or calculating the volume of a dam, so we know how much water the dam can hold.

This chapter examines the surface areas and volumes of three dimensional objects, otherwise known as solids. In order to work with these objects, you need to know how to calculate the surface area and perimeter of two dimensional shapes.



Figure 13.1: A tennis court. The position of each of the lines is carefully calculated to ensure that the area of the rectangles is the same anywhere in the world.

VISIT:
To revise how to calculate the area and perimeter of squares and rectangles you can watch the video below.
▶ See video: [2GRM](#) at www.everythingmaths.co.za

13.1 Area of a polygon

EMA7K

DEFINITION: *Area*
Area is the two dimensional space inside the boundary of a flat object. It is measured in square units.

Name	Shape	Formula
Square		$\text{area (A)} = s^2$
Rectangle		$\text{area (A)} = b \times h$
Triangle		$\text{area (A)} = \frac{1}{2}b \times h$
Trapezium		$\text{area (A)} = \frac{1}{2}(a + b) \times h$
Parallelogram		$\text{area (A)} = b \times h$
Circle		$\text{area (A)} = \pi r^2$ (circumference = $2\pi r$)

DID YOU KNOW?

The acre and the hectare are two common measurements used for the area of land. One hectare is about 0,01 square kilometres and one acre is about 0,004 square kilometres.

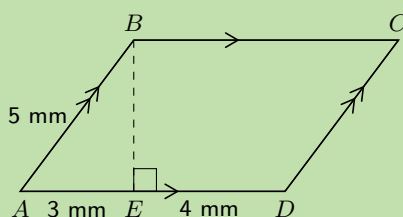
VISIT:

The video below shows some examples of calculations involving the area of a circle.

▶ See video: [2GRN](#) at www.everythingmaths.co.za

Worked example 1: Finding the area of a polygon**QUESTION**

Find the area of the following parallelogram:

**SOLUTION**

Step 1: Find the height BE

$$\begin{aligned} AB^2 &= BE^2 + AE^2 && \text{Pythagoras} \\ \therefore BE^2 &= AB^2 - AE^2 \\ &= 5^2 - 3^2 \\ &= 16 \\ \therefore BE &= 4 \text{ mm} \end{aligned}$$

Step 2: Find the area using the formula for a parallelogram

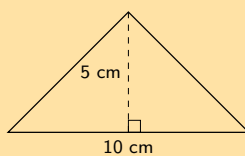
$$\begin{aligned} \text{area} &= b \times h \\ &= AD \times BE \\ &= 7 \times 4 \\ &= 28 \text{ mm}^2 \end{aligned}$$

VISIT:

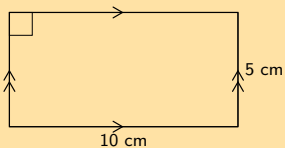
The following Phet simulation allows you to build different shapes and calculate the area and perimeter for the shapes: [Phet: area builder](#).

1. Find the area of each of the polygons below:

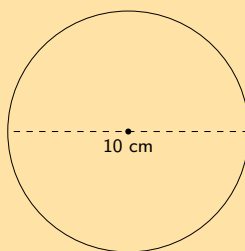
a)



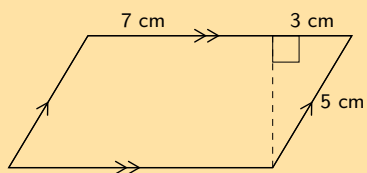
b)



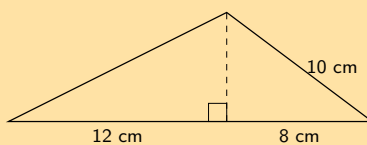
c)



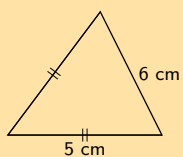
d)



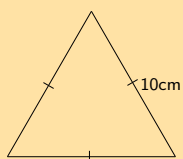
e)



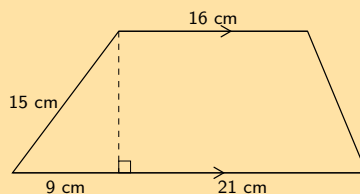
f)



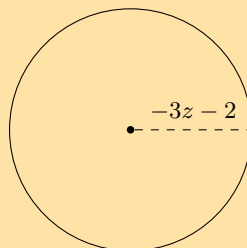
g)



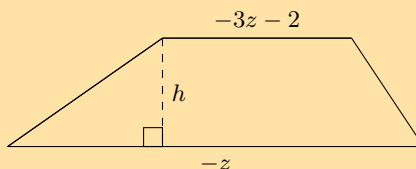
h)



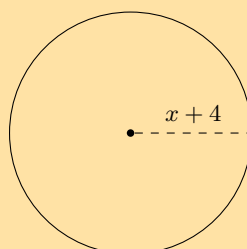
2. a) Find an expression for the area of this figure in terms of z and π . The circle has a radius of $-3z - 2$. Write your answer in expanded form (not factorised).



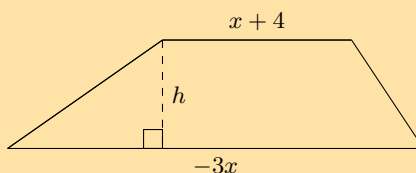
- b) Find an expression for the area of this figure in terms of z and h . The height of the figure is h , and two sides are labelled as $-3z - 2$ and $-z$. Write your answer in expanded form (not factorised).



3. a) Find an expression for the area of this figure in terms of x and π . The circle has a radius of $x + 4$. Write your answer in expanded form (not factorised).



- b) Find an expression for the area of this figure in terms of x and h . The height of the figure is h , and two sides are labelled as $x + 4$ and $-3x$. Write your answer in expanded form (not factorised).



For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

- 1a. 2GRP 1b. 2GRQ 1c. 2GRR 1d. 2GRS 1e. 2GRT 1f. 2GRV
1g. 2GRW 1h. 2GRX 2. 2GRY 3. 2GRZ



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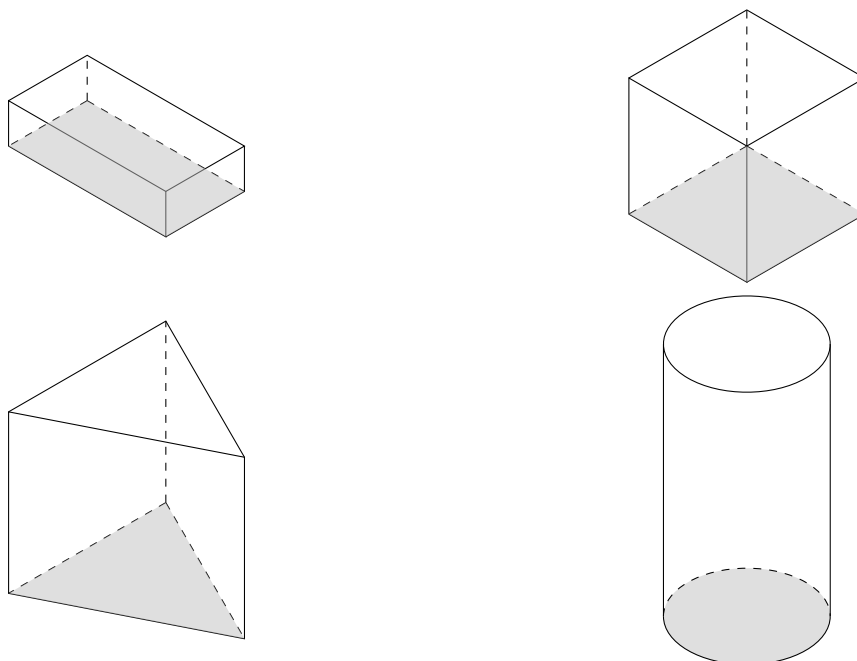


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DEFINITION: *Right prism*

A right prism is a geometric solid that has a polygon as its base and vertical faces perpendicular to the base. The base and top surface are the same shape and size. It is called a “right” prism because the angles between the base and faces are right angles.

A triangular prism has a triangle as its base, a rectangular prism has a rectangle as its base, and a cube is a rectangular prism with all its sides of equal length. A cylinder has a circle as its base. Examples of right prisms and a cylinder are given below: a rectangular prism, a cube and a triangular prism.



Surface area of prisms and cylinders

DEFINITION: *Surface area*

Surface area is the total area of the exposed or outer surfaces of a prism.

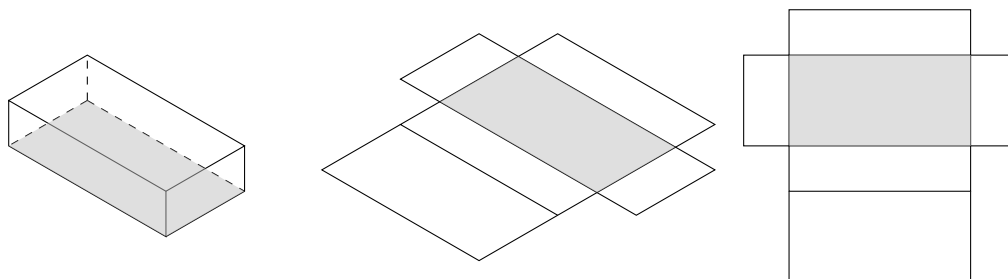
This is easier to understand if we imagine the prism to be a cardboard box that we can unfold. A solid that is unfolded like this is called a net. When a prism is unfolded into a net, we can clearly see each of its faces. In order to calculate the surface area of the prism, we can then simply calculate the area of each face, and add them all together.

For example, when a triangular prism is unfolded into a net, we can see that it has two faces that are triangles and three faces that are rectangles. To calculate the surface area of the prism, we find the area of each triangle and each rectangle, and add them together.

In the case of a cylinder the top and bottom faces are circles and the curved surface flattens into a rectangle with a length that is equal to the circumference of the circular base. To calculate the surface area we therefore find the area of the two circles and the rectangle and add them together.

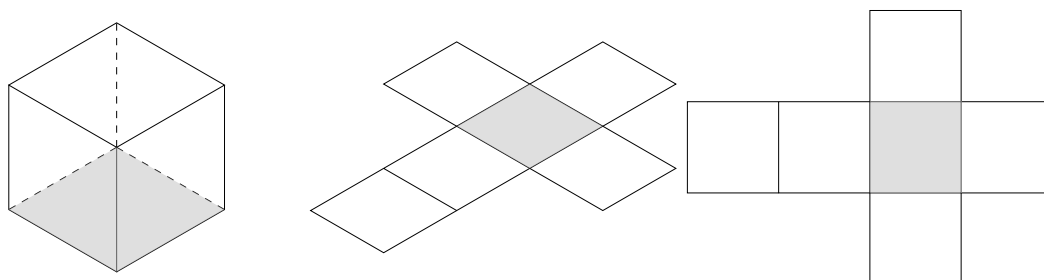
Below are examples of right prisms and a cylinder that have been unfolded into nets:

Rectangular prism



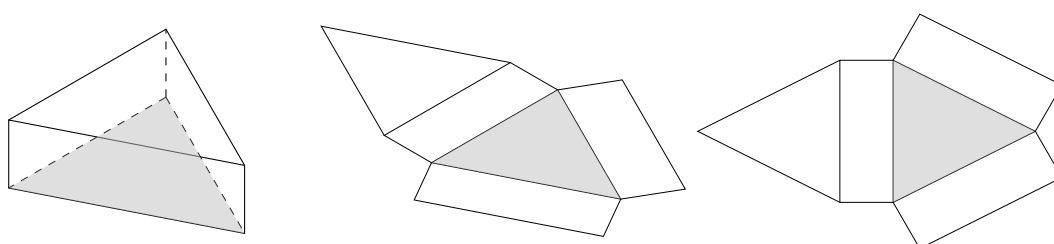
A rectangular prism unfolded into a net is made up of six rectangles.

Cube



A cube unfolded into a net is made up of six identical squares.

Triangular prism



A triangular prism unfolded into a net is made up of two triangles and three rectangles. The sum of the lengths of the rectangles is equal to the perimeter of the triangles.

Cylinder



A cylinder unfolded into a net is made up of two identical circles and a rectangle with a length equal to the circumference of the circles.

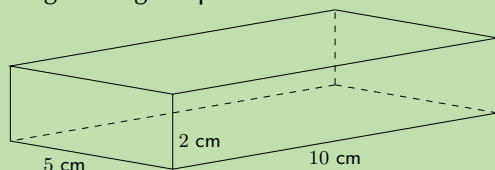
VISIT:

This video explains how we can unfold solids into nets.

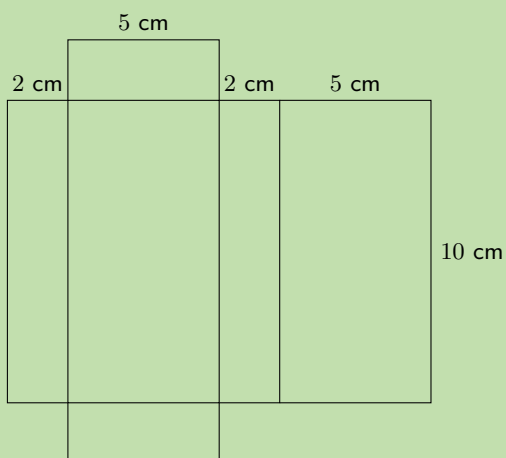
▶ See video: 2GS2 at www.everythingmaths.co.za

Worked example 2: Finding the surface area of a rectangular prism**QUESTION**

Find the surface area of the following rectangular prism:

**SOLUTION**

Step 1: Sketch and label the net of the prism



Step 2: Find the areas of the different shapes in the net

$$\begin{aligned}\text{large rectangle} &= \text{perimeter of small rectangle} \times \text{length} \\ &= (2 + 5 + 2 + 5) \times 10 \\ &= 14 \times 10 \\ &= 140 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}2 \times \text{small rectangle} &= 2(5 \times 2) \\ &= 2(10) \\ &= 20 \text{ cm}^2\end{aligned}$$

Step 3: Find the sum of the areas of the faces

$$\text{large rectangle} + 2 \times \text{small rectangle} = 140 + 20 = 160$$

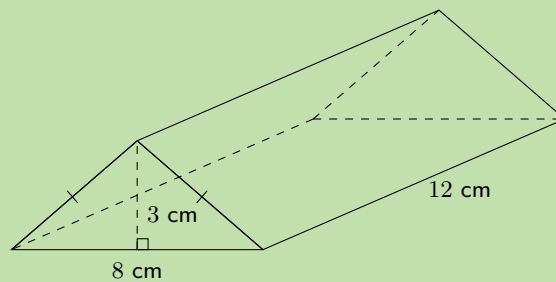
Step 4: Write the final answer

The surface area of the rectangular prism is 160 cm^2 .

Worked example 3: Finding the surface area of a triangular prism

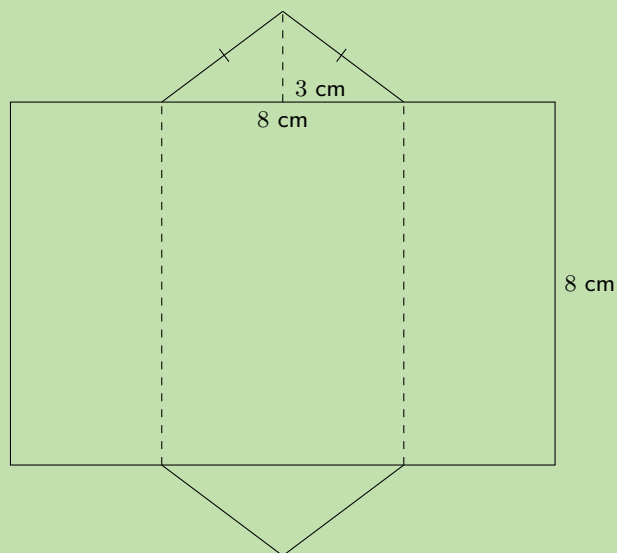
QUESTION

Find the surface area of the following triangular prism:



SOLUTION

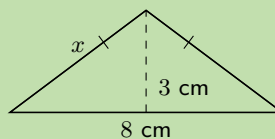
Step 1: Sketch and label the net of the prism



Step 2: Find the area of the different shapes in the net

To find the area of the rectangle, we need to calculate its length, which is equal to the perimeter of the triangles.

To find the perimeter of the triangle, we have to first find the length of its sides using the theorem of Pythagoras:



$$x^2 = 3^2 + \left(\frac{8}{2}\right)^2$$

$$x^2 = 3^2 + 4^2$$

$$= 25$$

$$\therefore x = 5 \text{ cm}$$

$$\therefore \text{perimeter of triangle} = 5 + 5 + 8$$

$$= 18 \text{ cm}$$

$$\therefore \text{area of large rectangle} = \text{perimeter of triangle} \times \text{length}$$

$$= 18 \times 12$$

$$= 216 \text{ cm}^2$$

$$\text{area of triangle} = \frac{1}{2}b \times h$$

$$= \frac{1}{2} \times 8 \times 3$$

$$= 12 \text{ cm}^2$$

Step 3: Find the sum of the areas of the faces

$$\text{surface area} = \text{area large rectangle} + (2 \times \text{area of triangle})$$

$$= 216 + 2(12)$$

$$= 240 \text{ cm}^2$$

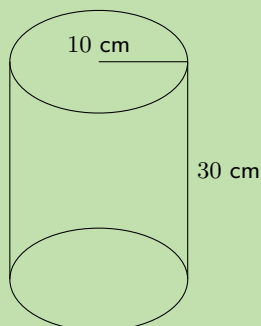
Step 4: Write the final answer

The surface area of the triangular prism is 240 cm^2 .

Worked example 4: Finding the surface area of a cylinder

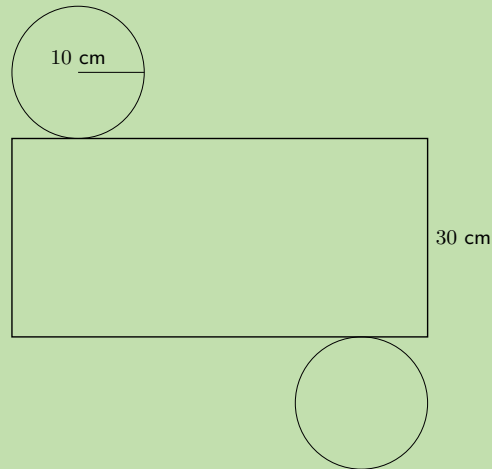
QUESTION

Find the surface area of the following cylinder (correct to 1 decimal place):



SOLUTION

Step 1: Sketch and label the net of the cylinder



Step 2: Find the area of the different shapes in the net

area of large rectangle = circumference of circle $\times$ length

$$= 2\pi r \times l$$

$$= 2\pi (10) \times 30$$

$$= 1884,9555... \text{ cm}^2$$

area of circle = πr^2

$$= \pi (10)^2$$

$$= 314,1592... \text{ cm}^2$$

surface area = area large rectangle + $(2 \times \text{area of circle})$

$$= 1884,9555... + 2 (314,1592...)$$

$$= 2513,3 \text{ cm}^2$$

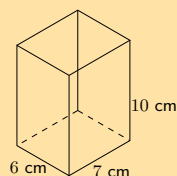
Step 3: Write the final answer

The surface area of the cylinder is $2513,3 \text{ cm}^2$.

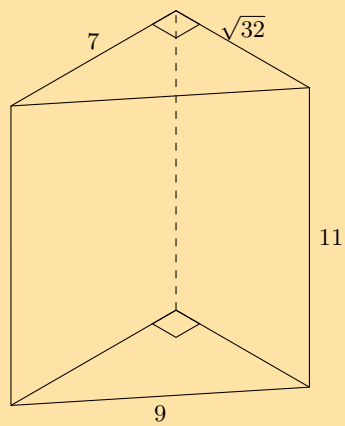
Exercise 13 – 2:

1. Calculate the surface area of the following prisms:

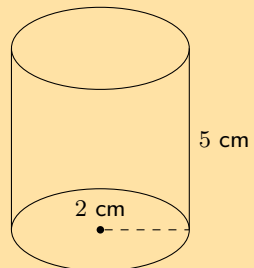
a)



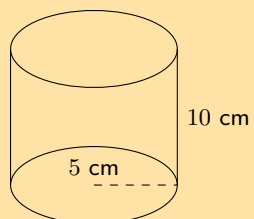
b)



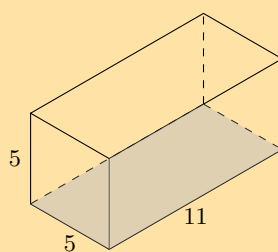
c)



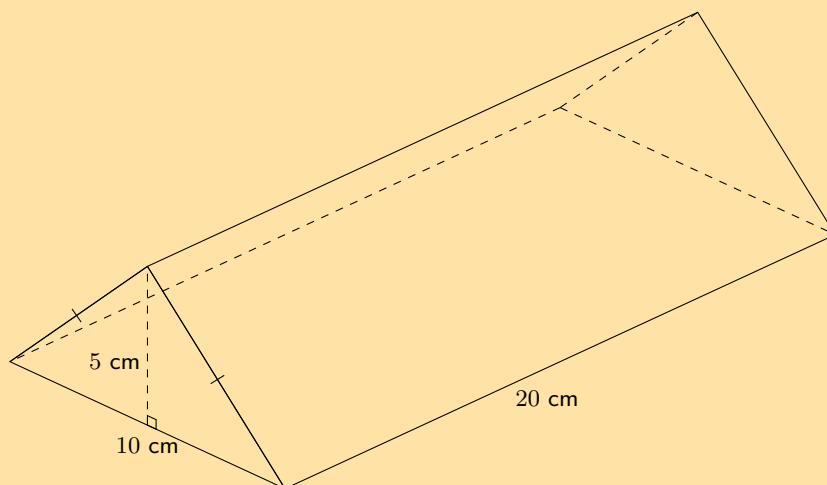
d)



e)

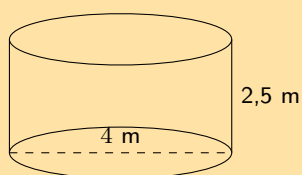


f)



2. If a litre of paint covers an area of 2 m^2 , how much paint does a painter need to cover:

- a rectangular swimming pool with dimensions $4 \text{ m} \times 3 \text{ m} \times 2,5 \text{ m}$ (the inside walls and floor only);
- the inside walls and floor of a circular reservoir with diameter 4 m and height $2,5 \text{ m}$.



For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1a. [2GS3](#) 1b. [2GS4](#) 1c. [2GS5](#) 1d. [2GS6](#) 1e. [2GS7](#) 1f. [2GS8](#) 2. [2GS9](#)



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Volume of prisms and cylinders

EMA7P

DEFINITION: Volume

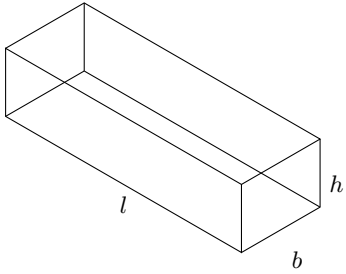
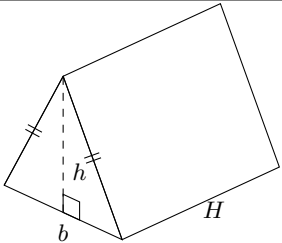
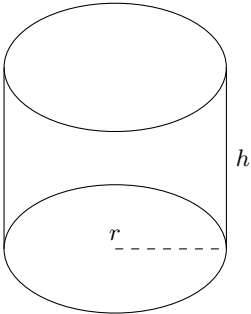
Volume is the three dimensional space occupied by an object, or the contents of an object. It is measured in cubic units.

The volume of right prisms and cylinders is simply calculated by multiplying the area of the base of the solid by the height of the solid.

VISIT:

The video below shows several examples of calculating the volume of a right prism.

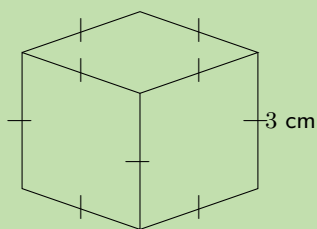
▶ See video: [2GSB](#) at www.everythingmaths.co.za

Rectangular prism		Volume = area of base $\times$ height = area of rectangle $\times$ height = $l \times b \times h$
Triangular prism		Volume = area of base $\times$ height = area of triangle $\times$ height = $(\frac{1}{2}b \times h) \times H$
Cylinder		Volume = area of base $\times$ height = area of circle $\times$ height = $\pi r^2 \times h$

Worked example 5: Finding the volume of a cube

QUESTION

Find the volume of the following cube:



SOLUTION

Step 1: Find the area of the base

$$\begin{aligned}
 \text{area of square} &= s^2 \\
 &= 3^2 \\
 &= 9 \text{ cm}^2
 \end{aligned}$$

Step 2: Multiply the area of the base by the height of the solid to find the volume

$$\begin{aligned}\text{volume} &= \text{area of base} \times \text{height} \\ &= 9 \times 3 \\ &= 27 \text{ cm}^3\end{aligned}$$

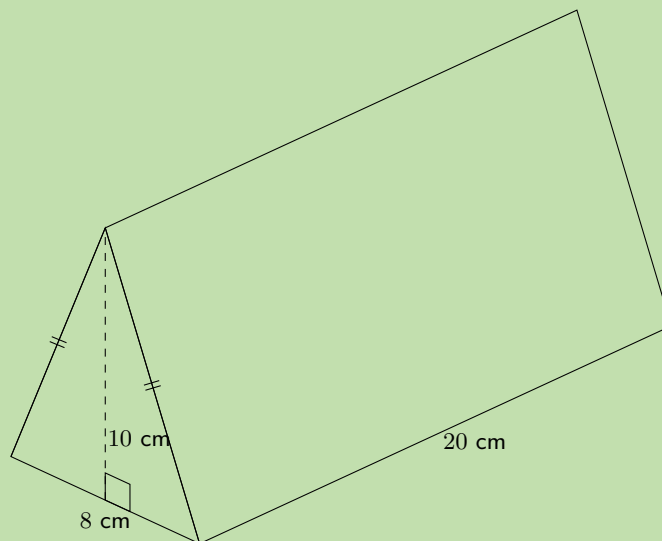
Step 3: Write the final answer

The volume of the cube is 27 cm^3 .

Worked example 6: Finding the volume of a triangular prism

QUESTION

Find the volume of the triangular prism:



SOLUTION

Step 1: Find the area of the base

$$\begin{aligned}\text{area of triangle} &= \frac{1}{2}b \times h \\ &= \left(\frac{1}{2} \times 8\right) \times 10 \\ &= 40 \text{ cm}^2\end{aligned}$$

Step 2: Multiply the area of the base by the height of the solid to find the volume

$$\begin{aligned}\text{volume} &= \text{area of base} \times \text{height} \\ &= \frac{1}{2}b \times h \times H \\ &= 40 \times 20 \\ &= 800 \text{ cm}^3\end{aligned}$$

Step 3: Write the final answer

The volume of the triangular prism is 800 cm^3 .

Worked example 7: Finding the volume of a cylinder

QUESTION

Find the volume of the following cylinder (correct to 1 decimal place):



SOLUTION

Step 1: Find the area of the base

$$\begin{aligned}\text{area of circle} &= \pi r^2 \\ &= \pi(4)^2 \\ &= 16\pi \text{ cm}^2\end{aligned}$$

Step 2: Multiply the area of the base by the height of the solid to find the volume

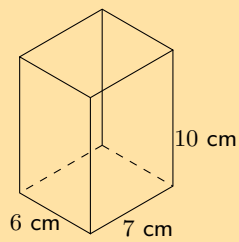
$$\begin{aligned}\text{volume} &= \text{area of base} \times \text{height} \\ &= \pi r^2 \times h \\ &= 16\pi \times 15 \\ &\approx 754,0 \text{ cm}^3\end{aligned}$$

Step 3: Write the final answer

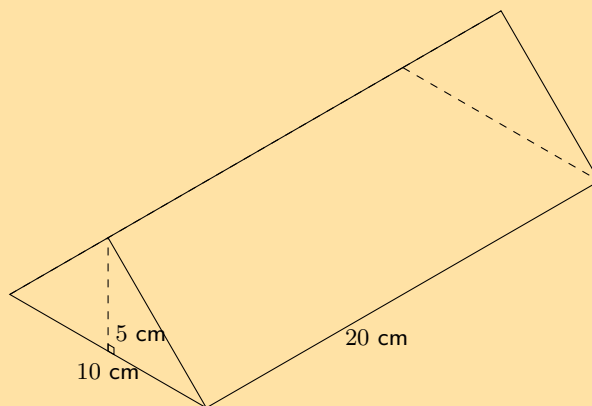
The volume of the cylinder is $754,0 \text{ cm}^3$.

1. Calculate the volumes of the following prisms (correct to 1 decimal place):

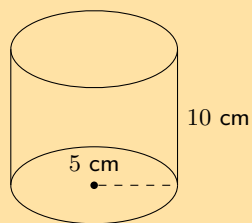
a)



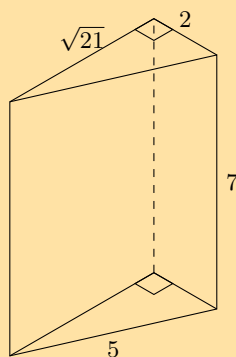
b)



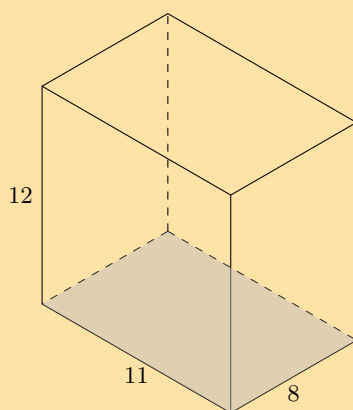
c)



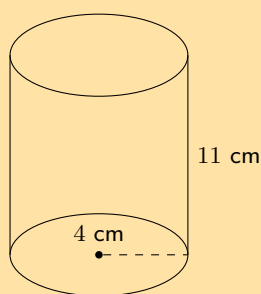
2. The figure here is a triangular prism. The height of the prism is 7 units; the triangles, which both contain right angles, have sides which are 2, $\sqrt{21}$ and 5 units long. Calculate the volume of the figure. Round to two decimal places if necessary.



3. The figure here is a rectangular prism. The height of the prism is 12 units; the other dimensions of the prism are 11 and 8 units. Find the volume of the figure.



4. The picture below shows a cylinder. The height of the cylinder is 11 units; the radius of the cylinder is $r = 4$ units. Determine the volume of the figure. Round your answer to two decimal places.



For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1a. [2GSC](#) 1b. [2GSD](#) 1c. [2GSF](#) 2. [2GSG](#) 3. [2GSH](#) 4. [2GSJ](#)



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13.3 Right pyramids, right cones and spheres

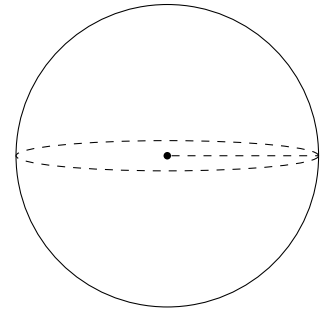
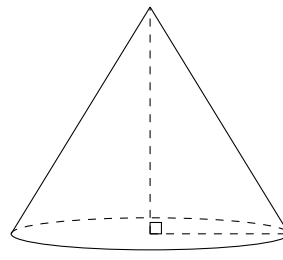
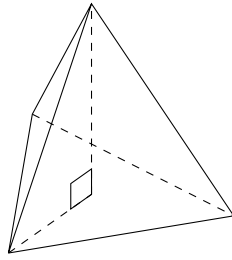
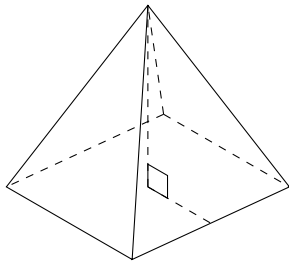
EMA7Q

DEFINITION: Pyramid

A pyramid is a geometric solid that has a polygon as its base and faces that converge at a point called the apex. In other words the faces are **not** perpendicular to the base.

The triangular pyramid and square pyramid take their names from the shape of their base. We call a pyramid a "right pyramid" if the line between the apex and the centre of the base is perpendicular to the base. Cones are similar to pyramids except that their bases are circles instead of polygons. Spheres are solids that are perfectly round and look the same from any direction.

Examples of a square pyramid, a triangular pyramid, a cone and a sphere:



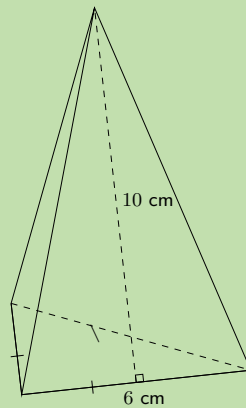
Surface area of pyramids, cones and spheres

EMA7R

Square pyramid		Surface area = area of base + area of triangular sides $= b^2 + 4 \left(\frac{1}{2} b h_s \right)$ $= b (b + 2 h_s)$
Triangular pyramid		Surface area = area of base + area of triangular sides $= \left(\frac{1}{2} b \times h_b \right) + 3 \left(\frac{1}{2} b \times h_s \right)$ $= \frac{1}{2} b (h_b + 3 h_s)$
Right cone		Surface area = area of base + area of walls $= \pi r^2 + \frac{1}{2} \times 2 \pi r h$ $= \pi r (r + h)$
Sphere		Surface area = $4 \pi r^2$

QUESTION

Find the surface area of the following triangular pyramid (correct to one decimal place):

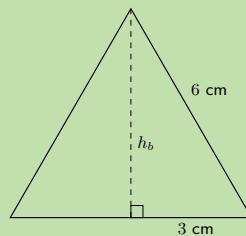


SOLUTION

Step 1: Find the area of the base

$$\text{area of base triangle} = \frac{1}{2}bh_b$$

To find the height of the base triangle (h_b) we use the theorem of Pythagoras:



$$\begin{aligned} 6^2 &= 3^2 + h_b^2 \\ \therefore h_b &= \sqrt{6^2 - 3^2} \\ &= 3\sqrt{3} \\ \therefore \text{area of base triangle} &= \frac{1}{2} \times 6 \times 3\sqrt{3} \\ &= 9\sqrt{3} \text{ cm}^2 \end{aligned}$$

Step 2: Find the area of the sides

$$\begin{aligned} \text{area of sides} &= 3 \left(\frac{1}{2} \times b \times h_s \right) \\ &= 3 \left(\frac{1}{2} \times 6 \times 10 \right) \\ &= 90 \text{ cm}^2 \end{aligned}$$

Step 3: Find the sum of the areas

$$9\sqrt{3} + 90 = 105,6 \text{ cm}^2$$

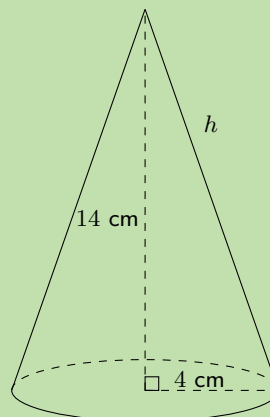
Step 4: Write the final answer

The surface area of the triangular pyramid is $105,6 \text{ cm}^2$.

Worked example 9: Finding the surface area of a cone

QUESTION

Find the surface area of the following cone (correct to 1 decimal place):



SOLUTION

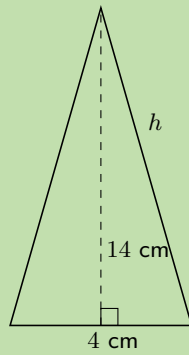
Step 1: Find the area of the base

$$\begin{aligned}\text{area of base circle} &= \pi r^2 \\ &= \pi \times 4^2 \\ &= 16\pi\end{aligned}$$

Step 2: Find the area of the walls

$$\text{area of sides} = \pi r h$$

To find the slant height, h , we use the theorem of Pythagoras:



$$\begin{aligned} h^2 &= 4^2 + 14^2 \\ \therefore h &= \sqrt{4^2 + 14^2} \\ &= 2\sqrt{53} \text{ cm} \end{aligned}$$

$$\begin{aligned} \text{area of walls} &= \frac{1}{2} 2\pi r h \\ &= \pi (4) (2\sqrt{53}) \\ &= 8\pi\sqrt{53} \text{ cm}^2 \end{aligned}$$

Step 3: Find the sum of the areas

$$\begin{aligned} \text{total surface area} &= 16\pi + 8\pi\sqrt{53} \\ &= 233,2 \text{ cm}^2 \end{aligned}$$

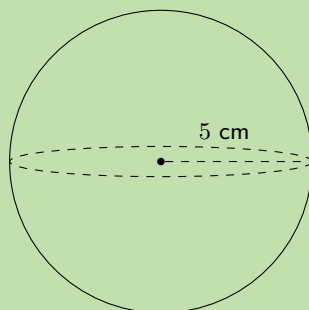
Step 4: Write the final answer

The surface area of the cone is $233,2 \text{ cm}^2$.

Worked example 10: Finding the surface area of a sphere

QUESTION

Find the surface area of the following sphere (correct to 1 decimal place):



SOLUTION

$$\begin{aligned}\text{surface area of sphere} &= 4\pi r^2 \\ &= 4\pi(5)^2 \\ &= 100\pi \\ &= 314,2 \text{ cm}^2\end{aligned}$$

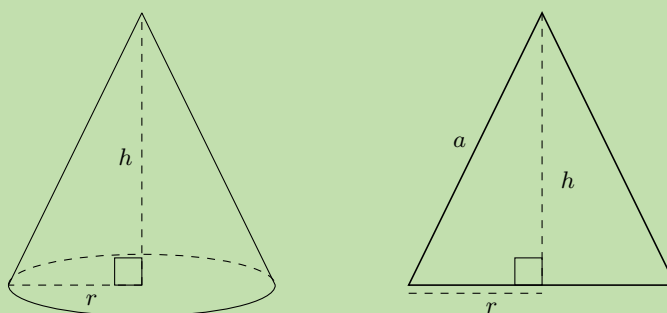
Worked example 11: Examining the surface area of a cone

QUESTION

If a cone has a height of h and a base of radius r , show that the surface area is: $\pi r^2 + \pi r\sqrt{r^2 + h^2}$

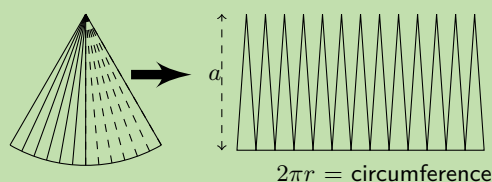
SOLUTION

Step 1: Sketch and label the cone



Step 2: Identify the faces that make up the cone

The cone has two faces: the base and the walls. The base is a circle of radius r and the walls can be opened out to a sector of a circle:



This curved surface can be cut into many thin triangles with height close to a (where a is the slant height). The area of these triangles or sectors can be summed as follows:

$$\begin{aligned}\text{Area of sector} &= \frac{1}{2} \times \text{base} \times \text{height (of a small triangle)} \\ &= \frac{1}{2} \times 2\pi r \times a \\ &= \pi r a\end{aligned}$$

Step 3: Calculate a

a can be calculated using the theorem of Pythagoras:

$$a = \sqrt{r^2 + h^2}$$

Step 4: Calculate the area of the circular base (A_b)

$$A_b = \pi r^2$$

Step 5: Calculate the area of the curved walls (A_w)

$$\begin{aligned} A_w &= \pi r a \\ &= \pi r \sqrt{r^2 + h^2} \end{aligned}$$

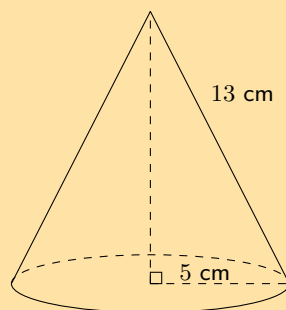
Step 6: Find the sum of the areas A

$$\begin{aligned} A &= A_b + A_w \\ &= \pi r^2 + \pi r \sqrt{r^2 + h^2} \\ &= \pi r \left(r + \sqrt{r^2 + h^2} \right) \end{aligned}$$

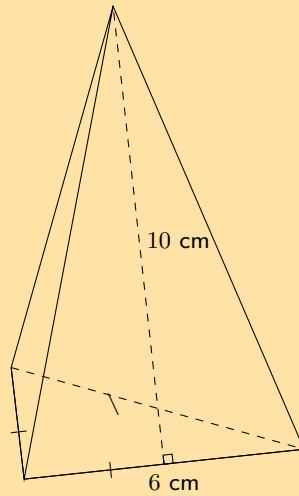
Exercise 13 – 4:

1. Find the total surface area of the following objects (correct to 1 decimal place if necessary):

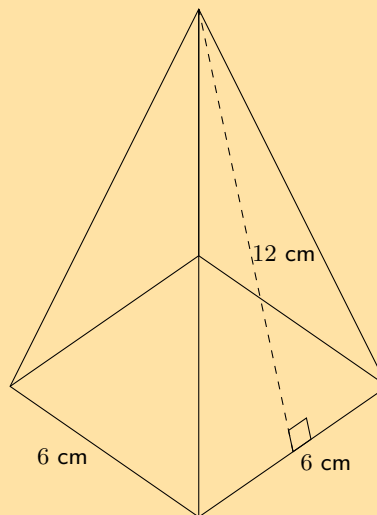
a)



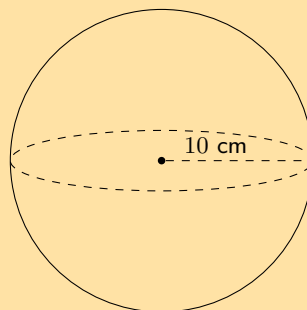
b)



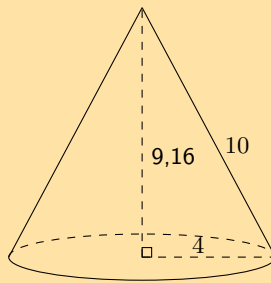
c)



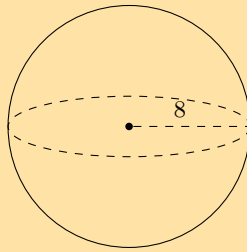
d)



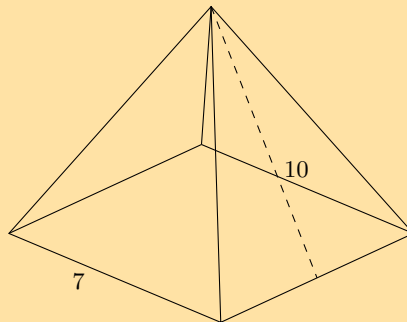
2. The figure here is a cone. The vertical height of the cone is $H = 9,16$ units and the slant height of the cone is $h = 10$ units; the radius of the cone is shown, $r = 4$ units. Calculate the surface area of the figure. Round your answer to two decimal places.



3. The figure here is a sphere. The radius of the sphere is $r = 8$ units. Calculate the surface area of the figure. Round your answer to two decimal places.



4. The figure here shows a pyramid with a square base. The sides of the base are each 7 units long. The vertical height of the pyramid is 9,36 units, and the slant height of the pyramid is 10 units. Determine the surface area of the pyramid.



For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

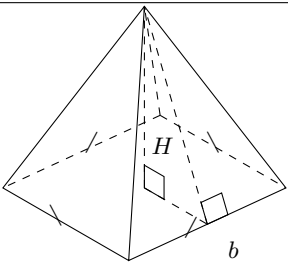
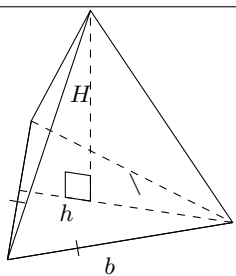
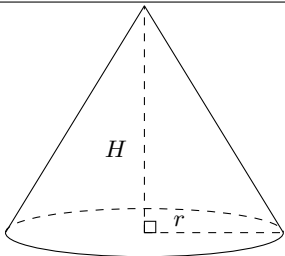
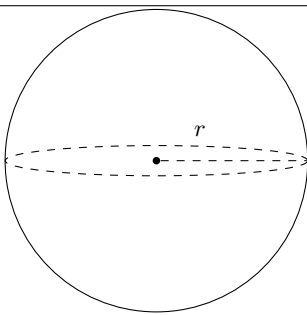
1a. [2GSK](#) 1b. [2GSM](#) 1c. [2GSN](#) 1d. [2GSP](#) 2. [2GSQ](#) 3. [2GSR](#) 4. [2GSS](#)



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Square pyramid		Volume $= \frac{1}{3} \times \text{area of base} \times \text{height of pyramid}$ $= \frac{1}{3} \times b^2 \times H$
Triangular pyramid		Volume $= \frac{1}{3} \times \text{area of base} \times \text{height of pyramid}$ $= \frac{1}{3} \times \frac{1}{2}bh \times H$
Right cone		Volume $= \frac{1}{3} \times \text{area of base} \times \text{height of cone}$ $= \frac{1}{3} \times \pi r^2 \times H$
Sphere		Volume $= \frac{4}{3}\pi r^3$

VISIT:

This video gives an example of calculating the volume of a sphere.

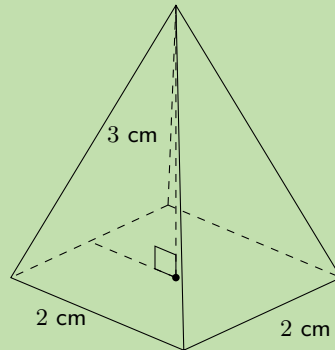
▶ See video: 2GST at www.everythingmaths.co.za

Worked example 12: Finding the volume of a square pyramid**QUESTION**

Find the volume of a square pyramid with a height of 3 cm and a side length of 2 cm.

SOLUTION

Step 1: Sketch and label the pyramid



Step 2: Select the correct formula and substitute the given values

We are given $b = 2$ and $H = 3$, therefore:

$$\begin{aligned} V &= \frac{1}{3} \times b^2 \times H \\ V &= \frac{1}{3} \times 2^2 \times 3 \\ &= \frac{1}{3} \times 12 \\ &= 4 \text{ cm}^3 \end{aligned}$$

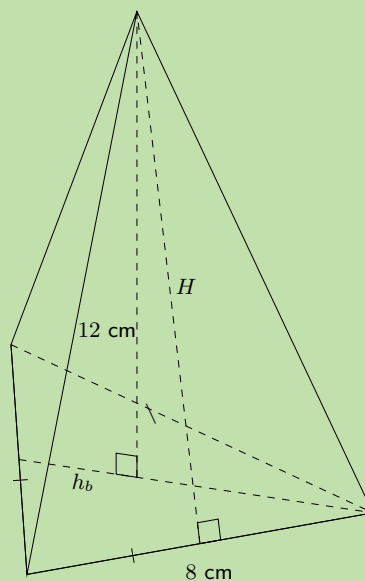
Step 3: Write the final answer

The volume of the square pyramid is 4 cm^3 .

Worked example 13: Finding the volume of a triangular pyramid

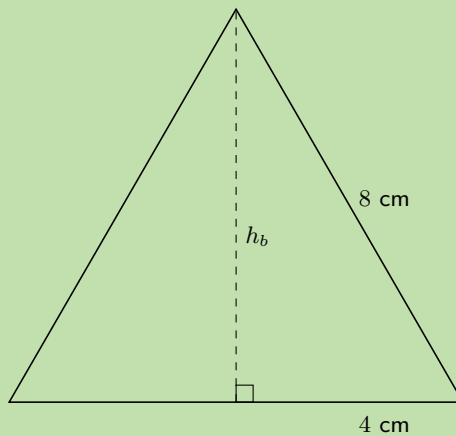
QUESTION

Find the volume of the following triangular pyramid (correct to 1 decimal place):



SOLUTION

Step 1: Sketch the base triangle and calculate its area



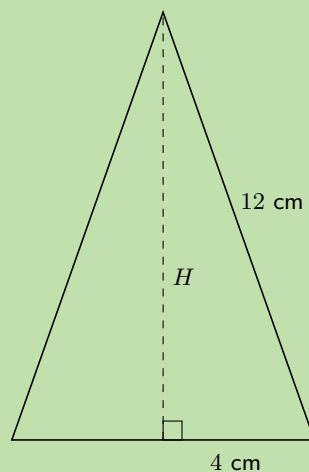
The height of the base triangle (h_b) is:

$$\begin{aligned}8^2 &= 4^2 + h_b^2 \\ \therefore h_b &= \sqrt{8^2 - 4^2} \\ &= 4\sqrt{3} \text{ cm}\end{aligned}$$

The area of the base triangle is:

$$\begin{aligned}\text{area of base triangle} &= \frac{1}{2}b \times h_b \\ &= \frac{1}{2} \times 4 \times 4\sqrt{3} \\ &= 8\sqrt{3} \text{ cm}^2\end{aligned}$$

Step 2: Sketch the side triangle and calculate pyramid height H



$$12^2 = 2\sqrt{3}^2 + H^2$$

$$H^2 = 130$$

$$\therefore H = \sqrt{130} \text{ cm}$$

Step 3: Calculate the volume of the pyramid

$$V = \frac{1}{3} \times \frac{1}{2}bh_b \times H$$

$$= \frac{1}{3} \times 16\sqrt{3} \times \sqrt{130}$$

$$= 105,3 \text{ cm}^3$$

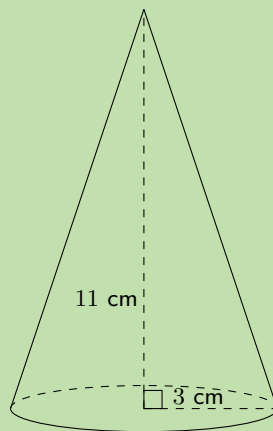
Step 4: Write the final answer

The volume of the triangular pyramid is $105,3 \text{ cm}^3$.

Worked example 14: Finding the volume of a cone

QUESTION

Find the volume of the following cone (correct to 1 decimal place):



SOLUTION

Step 1: Find the area of the base

$$\text{area of circle} = \pi r^2$$

$$= \pi \times 3^2$$

$$= 9\pi \text{ cm}^2$$

Step 2: Calculate the volume

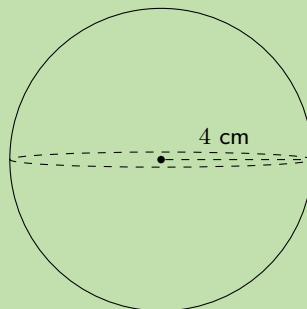
$$\begin{aligned}V &= \frac{1}{3} \times \pi r^2 \times H \\&= \frac{1}{3} \times 9\pi \times 11 \\&= 103,7 \text{ cm}^3\end{aligned}$$

Step 3: Write the final answer

The volume of the cone is $103,7 \text{ cm}^3$.

Worked example 15: Finding the volume of a sphere**QUESTION**

Find the volume of the following sphere (correct to 1 decimal place):

**SOLUTION****Step 1: Use the formula to find the volume**

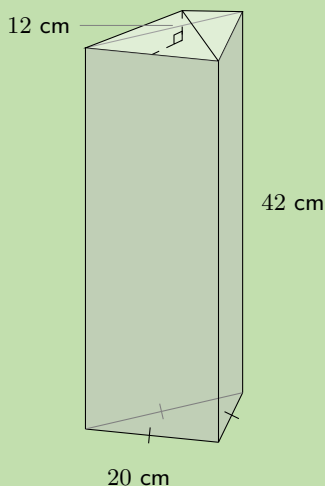
$$\begin{aligned}\text{volume} &= \frac{4}{3}\pi r^3 \\&= \frac{4}{3}\pi(4)^3 \\&= 268,1 \text{ cm}^3\end{aligned}$$

Step 2: Write the final answer

The volume of the sphere is $268,1 \text{ cm}^3$.

QUESTION

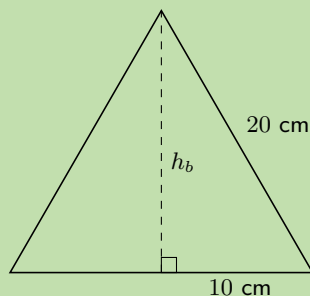
A triangular pyramid is placed on top of a triangular prism, as shown below. The base of the prism is an equilateral triangle of side length 20 cm and the height of the prism is 42 cm. The pyramid has a height of 12 cm. Calculate the total volume of the object.



SOLUTION

Step 1: Calculate the volume of the prism

First find the height of the base triangle using the theorem of Pythagoras:



$$\begin{aligned} 20^2 &= 10^2 + h_b^2 \\ \therefore h_b &= \sqrt{20^2 - 10^2} \\ &= 10\sqrt{3} \text{ cm} \end{aligned}$$

Next find the area of the base triangle:

$$\begin{aligned} \text{area of base triangle} &= \frac{1}{2} \times 20 \times 10\sqrt{3} \\ &= 100\sqrt{3} \text{ cm}^2 \end{aligned}$$

Now we can find the volume of the prism:

$$\begin{aligned}\therefore \text{volume of prism} &= \text{area of base triangle} \times \text{height of prism} \\ &= 100\sqrt{3} \times 42 \\ &= 4200\sqrt{3} \text{ cm}^3\end{aligned}$$

Step 2: Calculate the volume of the pyramid

The area of the base triangle is equal to the area of the base of the pyramid.

$$\begin{aligned}\therefore \text{volume of pyramid} &= \frac{1}{3} (\text{area of base}) \times H \\ &= \frac{1}{3} \times 100\sqrt{3} \times 12 \\ &= 400\sqrt{3} \text{ cm}^3\end{aligned}$$

Step 3: Calculate the total volume

$$\begin{aligned}\text{total volume} &= 4200\sqrt{3} + 400\sqrt{3} \\ &= 4600\sqrt{3} \\ &= 7967,4 \text{ cm}^3\end{aligned}$$

Therefore the total volume of the object is $7967,4 \text{ cm}^3$.

Worked example 17: Finding the surface area of a complex object

QUESTION

With the same complex object as in the previous example, you are given the additional information that the slant height $h_s = 13,3 \text{ cm}$. Now calculate the total surface area of the object.

SOLUTION

Step 1: Calculate the surface area of each exposed face of the pyramid

$$\begin{aligned}\text{area of one pyramid face} &= \frac{1}{2} b \times h_s \\ &= \frac{1}{2} \times 20 \times 13,3 \\ &= 133 \text{ cm}^2\end{aligned}$$

Because the base triangle is equilateral, each face has the same base, and therefore the same surface area. Therefore the surface area for each face of the pyramid is 133 cm^2 .

Step 2: Calculate the surface area of each side of the prism

Each side of the prism is a rectangle with base $b = 20 \text{ cm}$ and height $h_p = 42 \text{ cm}$.

$$\begin{aligned}
 \text{area of one prism side} &= b \times h_p \\
 &= 20 \times 42 \\
 &= 840 \text{ cm}^2
 \end{aligned}$$

Because the base triangle is equilateral, each side of the prism has the same area. Therefore the surface area for each side of the prism is 840 cm^2 .

Step 3: Calculate the total surface area of the object

$$\begin{aligned}
 \text{total surface area} &= \text{area of base of prism} + \text{area of sides of prism} + \text{area of exposed faces of pyramid} \\
 &= (100\sqrt{3}) + 3(840) + 3(133) \\
 &= 3092,2 \text{ cm}^2
 \end{aligned}$$

Therefore the total surface area (of the exposed faces) of the object is $3092,2 \text{ cm}^2$.

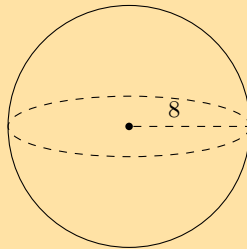
VISIT:

This video shows an example of calculating the volume of a complex object.

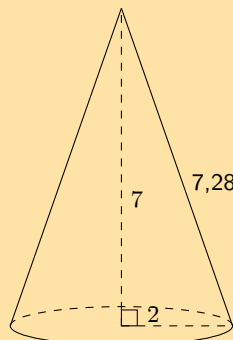
▶ See video: 2GSV at www.everythingmaths.co.za

Exercise 13 – 5:

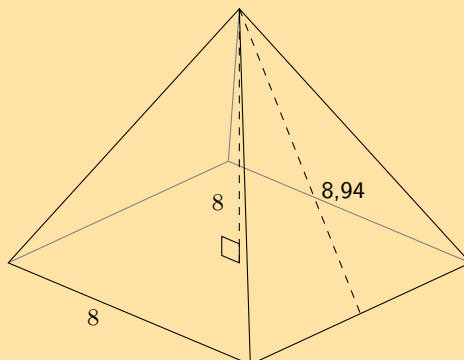
1. The figure below shows a sphere. The radius of the sphere is $r = 8$ units. Determine the volume of the figure. Round your answer to two decimal places.



2. The figure here is a cone. The vertical height of the cone is $H = 7$ units and the slant height is $h = 7,28$ units; the radius of the cone is shown, $r = 2$ units. Calculate the volume of the figure. Round your answer to two decimal places.

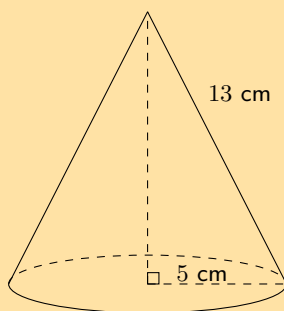


3. The figure here is a pyramid with a square base. The vertical height of the pyramid is $H = 8$ units and the slant height is $h = 8,94$ units; each side of the base of the pyramid is $b = 8$ units. Round your answer to two decimal places.

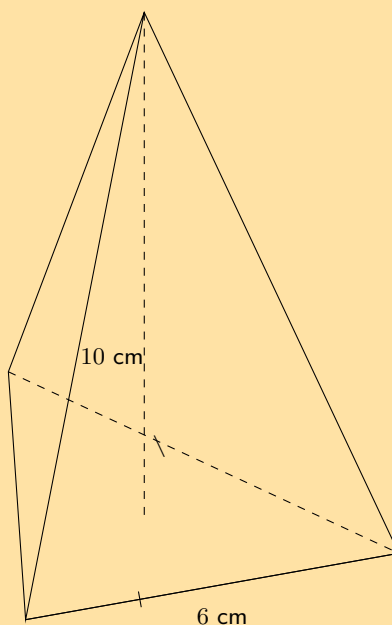


4. Find the volume of the following objects (round off to 1 decimal place if needed):

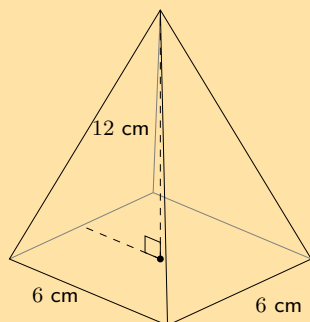
a)



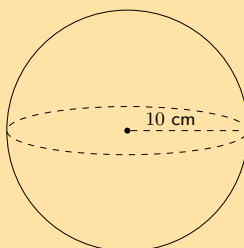
b)



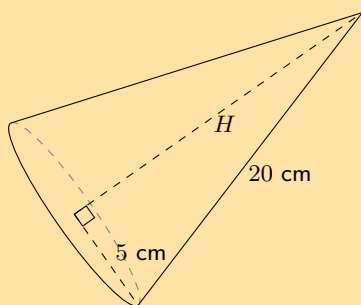
c)



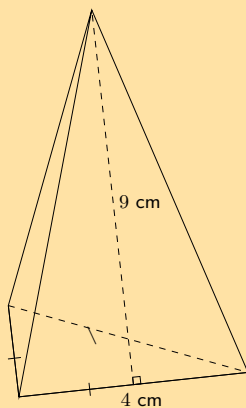
d)



5. Find the surface area and volume of the cone shown here. Round your answers to the nearest integer.

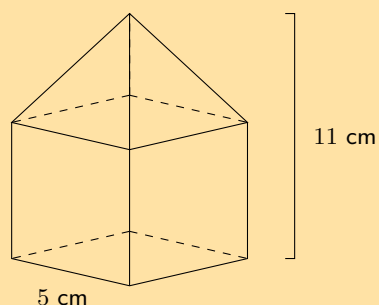


6. Calculate the following properties for the pyramid shown below. Round your answers to two decimal places.



- a) Surface area
- b) Volume

7. The solid below is made up of a cube and a square pyramid. Find its volume and surface area (correct to 1 decimal place):



For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. [2GSW](#) 2. [2GSX](#) 3. [2GSY](#) 4a. [2GSZ](#) 4b. [2GT2](#) 4c. [2GT3](#)
4d. [2GT4](#) 5. [2GT5](#) 6. [2GT6](#) 7. [2GT7](#)



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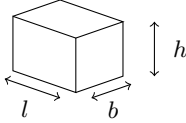
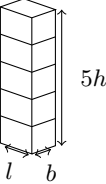
13.4 The effect of multiplying a dimension by a factor of k

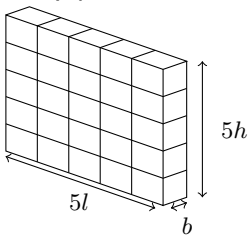
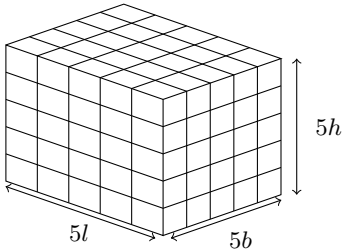
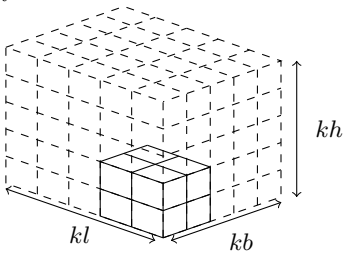
EMA7T

When one or more of the dimensions of a prism or cylinder is multiplied by a constant, the surface area and volume will change. The new surface area and volume can be calculated by using the formulae from the preceding section.

It is possible to see a relationship between the change in dimensions and the resulting change in surface area and volume. These relationships make it simpler to calculate the new volume or surface area of an object when its dimensions are scaled up or down.

Consider a rectangular prism of dimensions l , b and h . Below we multiply one, two and three of its dimensions by a constant factor of 5 and calculate the new volume and surface area.

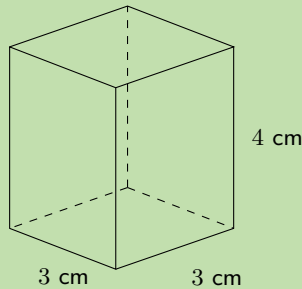
Dimensions	Volume	Surface
Original dimensions 	$V = l \times b \times h$ $= lbh$	$A = 2[(l \times h) + (l \times b) + (b \times h)]$ $= 2(lh + lb + bh)$
Multiply one dimension by 5 	$V_1 = l \times b \times 5h$ $= 5(lbh)$ $= 5V$	$A_1 = 2[(l \times 5h) + (l \times b) + (b \times 5h)]$ $= 2(5lh + lb + 5bh)$

Dimensions	Volume	Surface
Multiply two dimensions by 5 	$V_2 = 5l \times b \times 5h$ $= 5.5(lbh)$ $= 5^2 \times V$	$A_2 = 2[(5l \times 5h) + (5l \times b) + (b \times 5h)]$ $= 2 \times 5(5lh + lb + bh)$
Multiply all three dimensions by 5 	$V_3 = 5l \times 5b \times 5h$ $= 5^3(lbh)$ $= 5^3V$	$A_3 = 2[(5l \times 5h) + (5l \times 5b) + (5b \times 5h)]$ $= 2(5^2lh + 5^2lb + 5^2bh)$ $= 5^2 \times 2(lh + lb + bh)$ $= 5^2A$
Multiply all three dimensions by k 	$V_k = kl \times kb \times kh$ $= k^3(lbh)$ $= k^3V$	$A_k = 2[(kl \times kh) + (kl \times kb) + (kb \times kh)]$ $= k^2 \times 2(lh + lb + bh)$ $= k^2A$

Worked example 18: Calculating the new dimensions of a rectangular prism

QUESTION

Consider a rectangular prism with a height of 4 cm and base lengths of 3 cm.



1. Calculate the surface area and volume.
2. Calculate the new surface area (A_n) and volume (V_n) if the base lengths are multiplied by a constant factor of 3.
3. Express the new surface area and volume as a factor of the original surface area and volume.

SOLUTION

Step 1: Calculate the original volume and surface area

$$\begin{aligned}V &= l \times b \times h \\&= 3 \times 3 \times 4 \\&= 36 \text{ cm}^3\end{aligned}$$

$$\begin{aligned}A &= 2[(l \times h) + (l \times b) + (b \times h)] \\&= 2[(3 \times 4) + (3 \times 3) + (3 \times 4)] \\&= 66 \text{ cm}^2\end{aligned}$$

Step 2: Calculate the new volume and surface area

Two of the dimensions are multiplied by a factor of 3.

$$\begin{aligned}V_n &= 3l \times 3b \times h \\&= 3(3) \times 3(3) \times 4 \\&= 324 \text{ cm}^3\end{aligned}$$

$$\begin{aligned}A_n &= 2[(3l \times h) + (3l \times 3b) + (3b \times h)] \\&= 2[(3(3) \times 4) + (3(3) \times 3(3)) + (3(3) \times 4)] \\&= 306 \text{ cm}^2\end{aligned}$$

Step 3: Express the new dimensions as a factor of the original dimensions

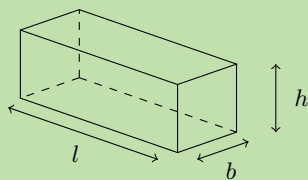
$$\begin{aligned}V &= 36 \\V_n &= 324 \\\frac{V_n}{V} &= \frac{324}{36} \\&= 9 \\\therefore V_n &= 9V \\&= 3^2V\end{aligned}$$

$$\begin{aligned}A &= 66 \\A_n &= 306 \\\frac{A_n}{A} &= \frac{306}{66} \\\therefore A_n &= \frac{306}{66}A \\&= \frac{51}{11}A\end{aligned}$$

Worked example 19: Multiplying the dimensions of a rectangular prism by k

QUESTION

Prove that if the height of a rectangular prism with dimensions l , b and h is multiplied by a constant value of k , the volume will also increase by a factor k .



SOLUTION

Step 1: Calculate the original volume

We are given the original dimensions l , b and h and so the original volume is $V = l \times b \times h$.

Step 2: Calculate the new volume

The new dimensions are l , b , and kh and so the new volume is:

$$\begin{aligned}V_n &= l \times b \times (kh) \\&= k(lbh) \\&= kV\end{aligned}$$

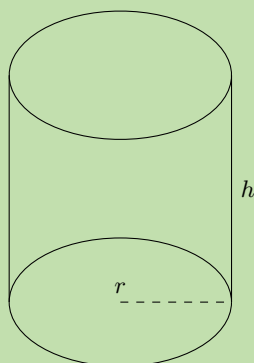
Step 3: Write the final answer

If the height of a rectangular prism is multiplied by a constant k , then the volume also increases by a factor of k .

Worked example 20: Multiplying the dimensions of a cylinder by k

QUESTION

Consider a cylinder with a radius of r and a height of h . Calculate the new volume and surface area (expressed in terms of r and h) if the radius is multiplied by a constant factor of k .



SOLUTION

Step 1: Calculate the original volume and surface area

$$V = \pi r^2 \times h$$

$$A = \pi r^2 + 2\pi rh$$

Step 2: Calculate the new volume and surface area

The new dimensions are kr and h .

$$V_n = \pi(kr)^2 \times h$$

$$= \pi k^2 r^2 \times h$$

$$= k^2 \times \pi r^2 h$$

$$= k^2 V$$

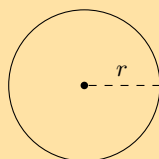
$$A_n = \pi(kr)^2 + 2\pi(kr)h$$

$$= \pi k^2 r^2 + 2\pi krh$$

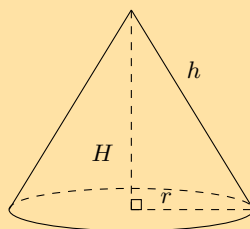
$$= k^2 (\pi r^2) + k (2\pi rh)$$

Exercise 13 – 6:

1. If the length of the radius of a circle is a third of its original size, what will the area of the new circle be?

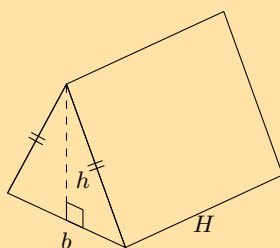


2. If the length of the base's radius and height of a cone is doubled, what will the surface area of the new cone be?



3. If the height of a prism is doubled, how much will its volume increase by?
4. Describe the change in the volume of a rectangular prism if the:
- length and breadth increase by a constant factor of 3.
 - length, breadth and height are multiplied by a constant factor of 3.

5. If the length of each side of a triangular prism is quadrupled, what will the volume of the new triangular prism be?



6. Given a prism with a volume of 493 cm^3 and a surface area of 6007 cm^2 , find the new surface area and volume for a prism if all dimensions are increased by a constant factor of 4.

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1. 2GT8 2. 2GT9 3. 2GTB 4a. 2GTC 4b. 2GTD 5. 2GTF 6. 2GTC



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13.5 Chapter summary

EMA7V

📺 See presentation: 2GTH at www.everythingmaths.co.za

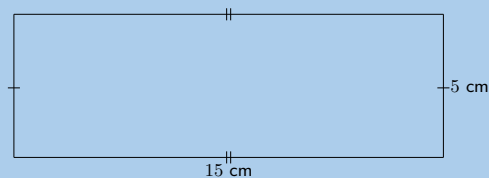
- Area is the two dimensional space inside the boundary of a flat object. It is measured in square units.
- Area formulae:
 - square: s^2
 - rectangle: $b \times h$
 - triangle: $\frac{1}{2}b \times h$
 - trapezium: $\frac{1}{2}(a + b) \times h$
 - parallelogram: $b \times h$
 - circle: πr^2
- A right prism is a geometric solid that has a polygon as its base and vertical sides perpendicular to the base. The base and top surface are the same shape and size. It is called a “right” prism because the angles between the base and sides are right angles.
- A triangular prism has a triangle as its base, a rectangular prism has a rectangle as its base, and a cube is a rectangular prism with all its sides of equal length. A cylinder is another type of right prism which has a circle as its base.
- Surface area is the total area of the exposed or outer surfaces of a prism.
- A net is the unfolded “plan” of a solid.
- Volume is the three dimensional space occupied by an object, or the contents of an object. It is measured in cubic units.
- Volume formulae for prisms and cylinders:
 - Volume of a rectangular prism: $l \times b \times h$
 - Volume of a triangular prism: $(\frac{1}{2}b \times h) \times H$
 - Volume of a square prism or cube: s^3
 - Volume of a cylinder: $\pi r^2 \times h$

- A pyramid is a geometric solid that has a polygon as its base and sides that converge at a point called the apex. The sides are not perpendicular to the base.
- The triangular pyramid and square pyramid take their names from the shape of their base. We call a pyramid a “right pyramid” if the line between the apex and the centre of the base is perpendicular to the base. Cones are similar to pyramids except that their bases are circles instead of polygons. Spheres are solids that are perfectly round and look the same from any direction.
- Surface area formulae for right pyramids, right cones and spheres:
 - square pyramid: $b(b + 2h)$
 - triangular pyramid: $\frac{1}{2}b(h_b + 3h_s)$
 - right cone: $\pi r(r + h_s)$
 - sphere: $4\pi r^2$
- Volume formulae for right pyramids, right cones and spheres:
 - square pyramid: $\frac{1}{3} \times b^2 \times H$
 - triangular pyramid: $\frac{1}{3} \times \frac{1}{2}bh \times H$
 - right cone: $\frac{1}{3} \times \pi r^2 \times H$
 - sphere: $\frac{4}{3}\pi r^3$
- Multiplying one or more dimensions of a prism or cylinder by a constant k affects the surface area and volume.

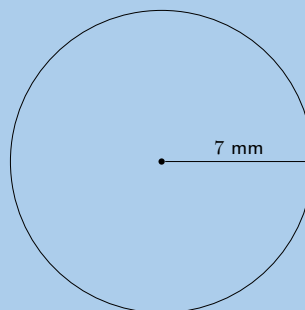
End of chapter Exercise 13 – 7:

1. Find the area of each of the shapes shown. Round your answer to two decimal places if necessary.

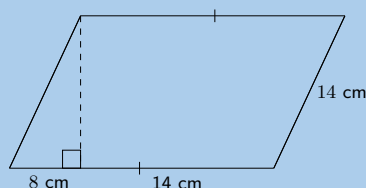
a)



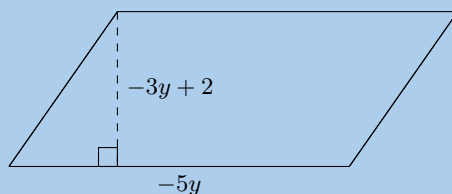
b)



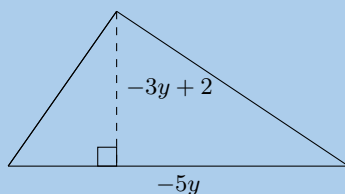
c)



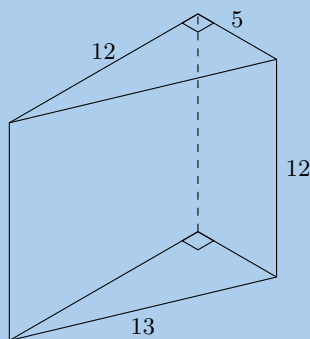
2. a) Find an expression for the area of this figure in terms of y . The dimensions of the figure are labelled $-5y$ and $-3y + 2$. Write your answer in expanded form (not factorised).



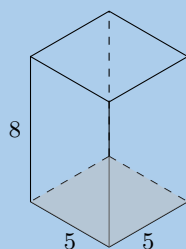
- b) Find an expression for the area of this figure in terms of y . The figure has dimensions of $-5y$ and $-3y + 2$, as labelled. Write your answer in expanded form (not factorised).



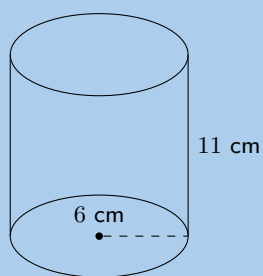
3. The figure here is a triangular prism. The height of the prism is 12 units; the triangles, which are both right triangles, have sides which are 5, 12 and 13 units long. Find the surface area of the figure.



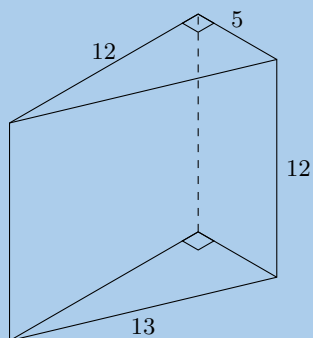
4. The figure here is a rectangular prism. The height of the prism is 5 units; the other dimensions of the prism are 8 and 5 units. Find the surface area of the figure.



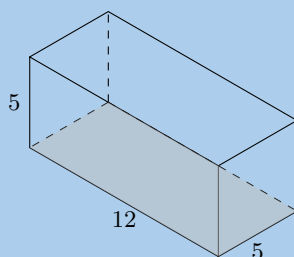
5. A cylinder is shown below. The height of the cylinder is 11 cm; the radius of the cylinder is $r = 6$ cm, as shown. Find the surface area of the figure. Round your answer to two decimal places.



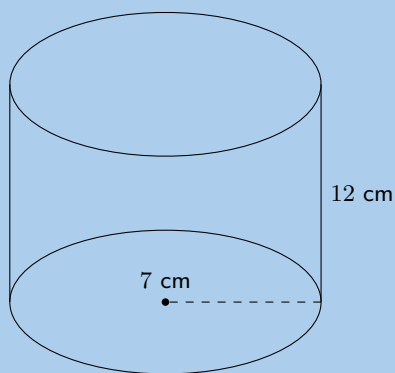
6. The figure here is a triangular prism. The height of the prism is 12 units; the triangles, which both contain right angles, have sides which are 5, 12 and 13 units long. Determine the volume of the figure.



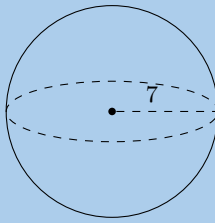
7. The figure here is a rectangular prism. The height of the prism is 5 units; the other dimensions of the prism are 12 and 5 units. Calculate the volume of the figure.



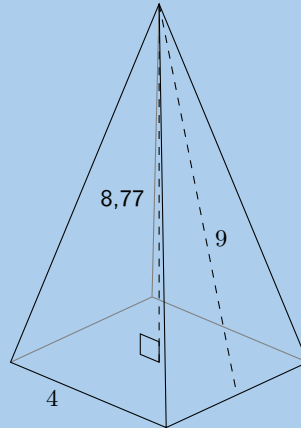
8. The picture below shows a cylinder. The height of the cylinder is 12 cm; the radius of the cylinder is $r = 7$ cm. Calculate the volume of the figure. Round your answer two decimal places.



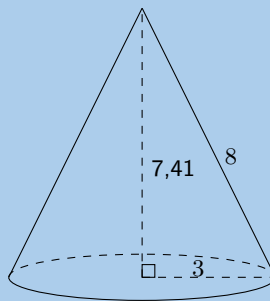
9. The figure here is a sphere. The radius of the sphere is $r = 7$ units. Find the surface area of the figure. Round your answer two decimal places.



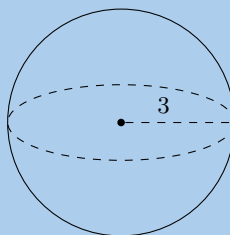
10. The figure here shows a pyramid with a square base. The sides of the base are each 4 units long. The vertical height of the pyramid is 8,77 units, and the slant height of the pyramid is 9 units. Determine the surface area of the pyramid.



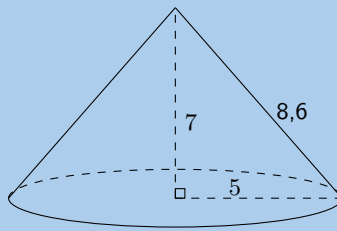
11. The figure here is a cone. The vertical height of the cone is $H = 7,41$ units and the slant height of the cone is $h = 8$ units; the radius of the cone is shown, $r = 3$ units. Find the surface area of the figure. Round your answer two decimal places.



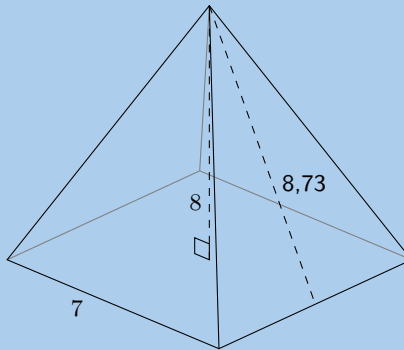
12. The figure below shows a sphere. The radius of the sphere is $r = 3$ units. Determine the volume of the figure. Round your answer to two decimal places.



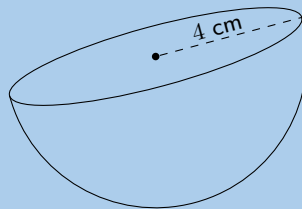
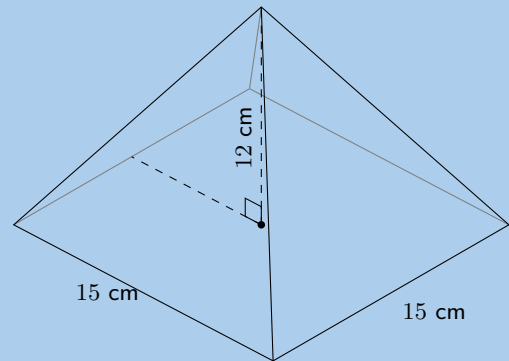
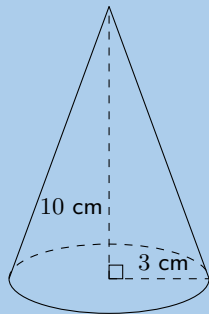
13. The figure here is a cone. The vertical height of the cone is $H = 7$ units and the slant height is $h = 8,60$ units; the radius of the cone is shown, $r = 5$ units. Find the volume of the figure. Round your answer to two decimal places.



14. The figure here is a pyramid with a square base. The vertical height of the pyramid is $H = 8$ units and the slant height is $h = 8,73$ units; each side of the base of the pyramid is $b = 7$ units. Find the volume of the figure. Round your answer to two decimal places.

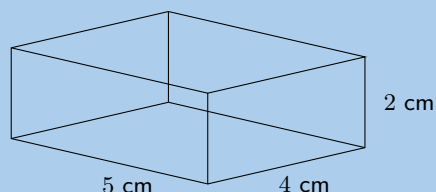
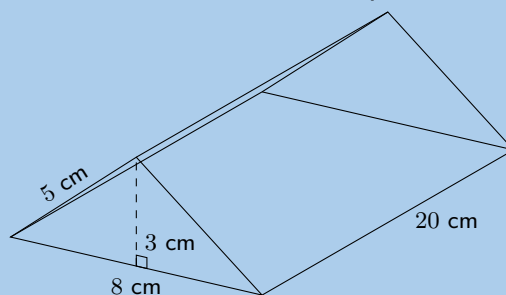
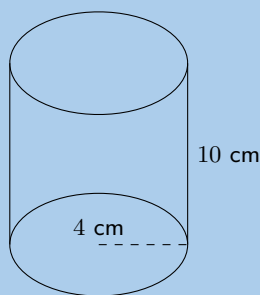


15. Consider the solids below:

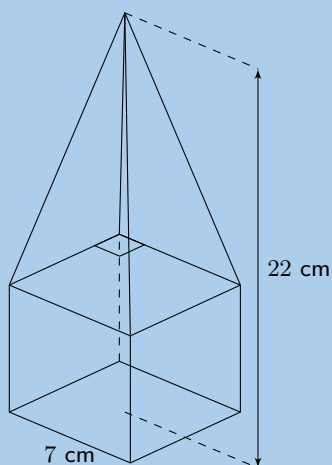


- Calculate the surface area of each solid.
 - Calculate the volume of each solid.
- If the length of each side of a square is a quarter of its original size, what will the area of the new square be?
 - If the length of each side of a square pyramid is a third of its original size, what will the surface area of the new square pyramid be?
 - If the length of the base's radius and the height of a cylinder is halved, what will the volume of the new cylinder be?

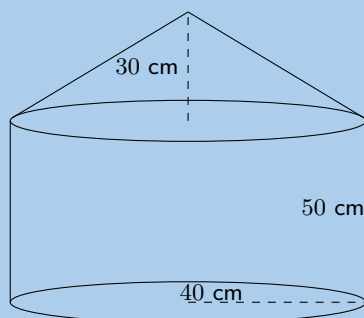
19. Consider the solids below and answer the questions that follow (correct to 1 decimal place, if necessary):



- Calculate the surface area of each solid.
 - Calculate volume of each solid.
 - If each dimension of the solids is increased by a factor of 3, calculate the new surface area of each solid.
 - If each dimension of the solids is increased by a factor of 3, calculate the new volume of each solid.
20. The solid below is made of a cube and a square pyramid. Answer the following:

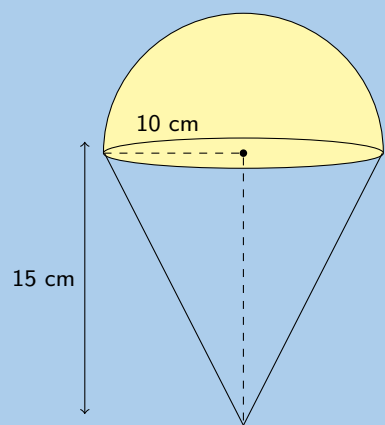


- Find the surface area of the solid shown. Give your answers to two decimal places.
 - Now determine the volume of the shape. Give your answer to the nearest integer value.
21. Calculate the volume and surface area of the solid below (correct to 1 decimal place):

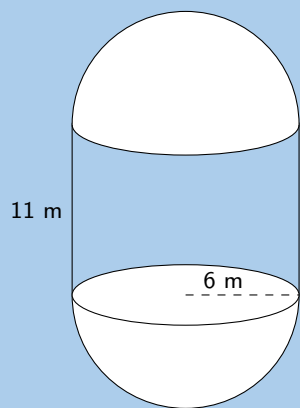


22. Find the volume and surface areas of the following composite shapes.

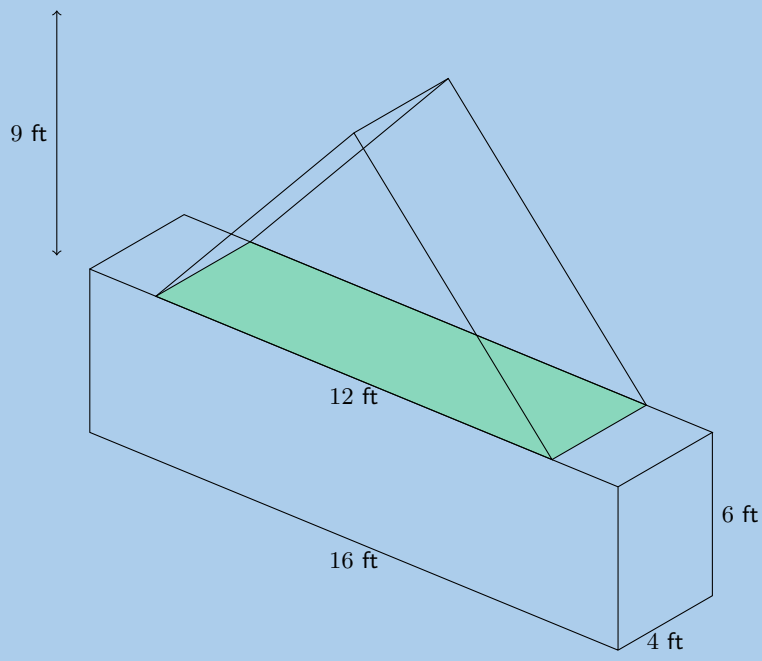
a)



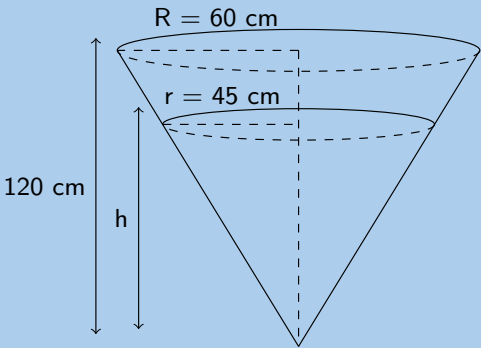
b)



c)

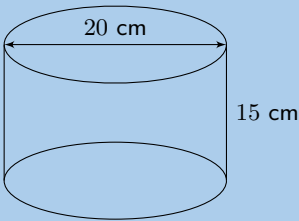


23. An ice-cream cone (right cone) has a radius of 3 cm and a height of 12 cm. A half scoop of ice-cream (hemisphere) is placed on top of the cone. If the ice-cream melts, will it fit into the cone? Show all your working.
24. A receptacle filled with petrol has the shape of an inverted right circular cone of height 120 cm and base radius of 60 cm. A certain amount of fuel is siphoned out of the receptacle leaving a depth of h cm.

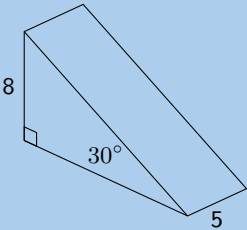


- a) Show that $h = 90$ cm.
- b) Determine the volume of fuel that has been siphoned out. Express your answer in litres if $1 \text{ l} = 1000 \text{ cm}^3$
25. Find the **volume** and **surface** area of the following prisms.

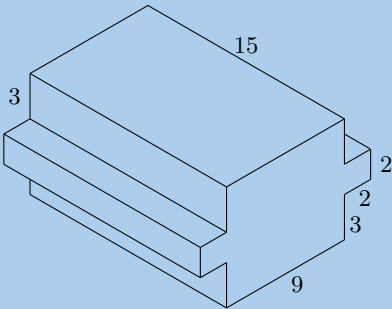
a)



b)

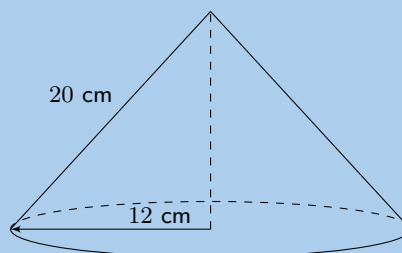


c)

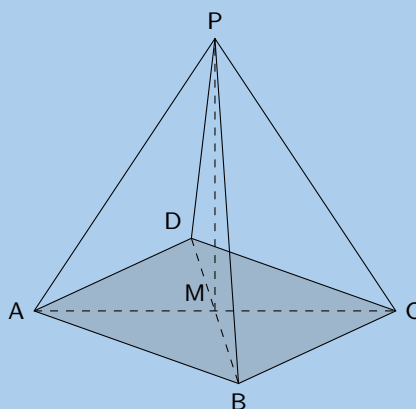


26. Determine the volume of the following:

a)

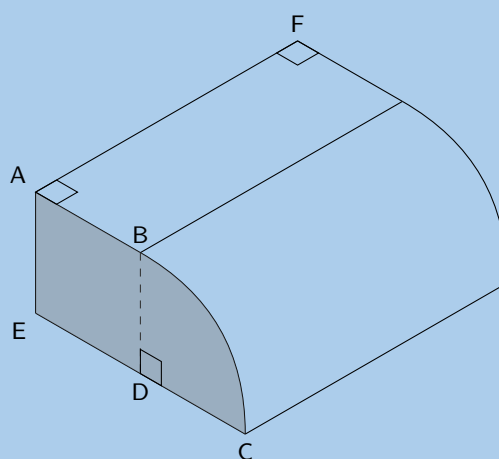


b) $ABCD$ is a square, $AC = 12$ cm, $AP = 10$ cm.



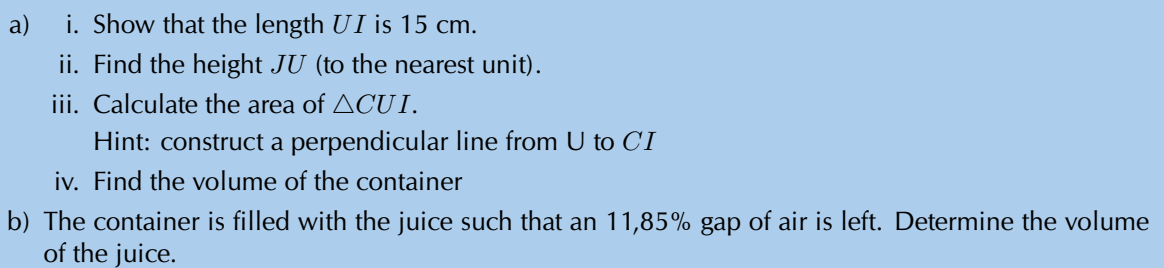
27. The prism alongside has the following dimensions:

$AB = 4$ units, $EC = 8$ units, $AF = 10$ units. BC is an arc of a circle with centre D . $AB \parallel EC$.

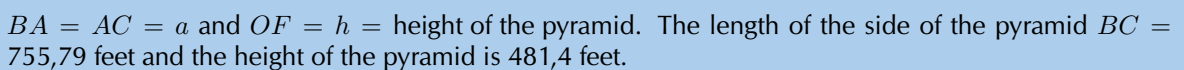


- Explain why BD , the radius of the arc BC , is 4 units.
- Calculate the area of the shaded surface.
- Find the volume of the prism.

28. A coldrink container is made in the shape of a pyramid with an isosceles triangular base. This is known as a tetrahedron. The angle of elevation of the top of the container is $33,557^\circ$. $CI = 7$ cm; $JI = 18$ cm.



This is a square-based pyramid and O is the centre of the square.



- Determine the area of the base of the pyramid in terms of a .
- Calculate $AF (= s)$ to 5 decimal places.
- From your calculation in question (b) determine $\frac{s}{a}$.
- Determine the volume and surface area of the pyramid.

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

- | | | | | | |
|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|--------------------------|
| 1. 2GTJ | 2. 2GTK | 3. 2GTM | 4. 2GTN | 5. 2GTP | 6. 2GTQ |
| 7. 2GTR | 8. 2GTS | 9. 2GTT | 10. 2GTV | 11. 2GTW | 12. 2GTX |
| 13. 2GTY | 14. 2GTZ | 15a. 2GV2 | 15b. 2GV3 | 16. 2GV4 | 17. 2GV5 |
| 18. 2GV6 | 19a. 2GV7 | 19b. 2GV8 | 19c. 2GV9 | 19d. 2GVB | 20. 2GVC |
| 21. 2GVD | 22a. 2GVF | 22b. 2GVG | 22c. 2GVH | 23. 2GVJ | 24. 2GVK |
| 25a. 2GVM | 25b. 2GVN | 25c. 2GVP | 26a. 2GVQ | 26b. 2GVR | 27. 2GVS |
| 28. 2GVT | 29. 2GVV | | | | |



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Probability

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We use probability to describe uncertain events. When you accidentally drop a slice of bread, you don't know if it's going to fall with the buttered side facing upwards or downwards. When your favourite sports team plays a game, you don't know whether they will win or not. When the weatherman says that there is a 40% chance of rain tomorrow, you may or may not end up getting wet. Uncertainty presents itself to some degree in every event that occurs around us and in every decision that we make.

We will see in this chapter that all of these uncertainties can be described using the rules of probability theory and that we can make definite conclusions about uncertain processes.

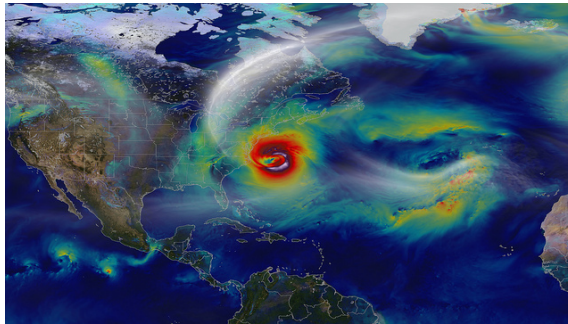


Figure 14.1: Tracking a superstorm. Meteorologists use computer software to help them track storms and predict the weather.

We'll use three examples of uncertain processes to help you understand the meanings of the different words used in probability theory: tossing a coin, rolling dice, and a soccer match.

VISIT:

The following video introduces the concepts used in probability.

▶ See video: 2GVW at www.everythingmaths.co.za

DEFINITION: Experiment

An experiment refers to an uncertain process.

DEFINITION: Outcome

An outcome of an experiment is a single result of that experiment.

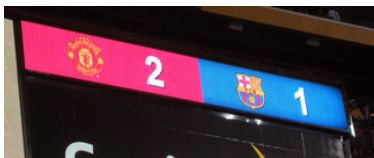
Experiment 1: A coin is tossed and it lands with either heads (H) or tails (T) facing upwards. An example outcome of tossing a coin is that it lands with heads facing up:



Experiment 2: Two dice are rolled and the total number of dots added up. An example outcome of rolling two dice:



Experiment 3: Two teams play a soccer match and we are interested in the final score. An example outcome of a soccer match:

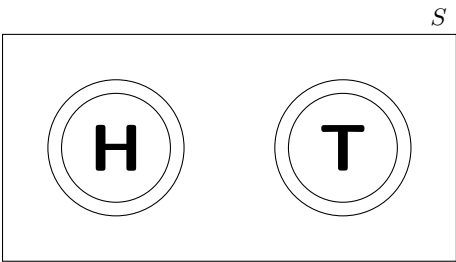


DEFINITION: *Sample space*

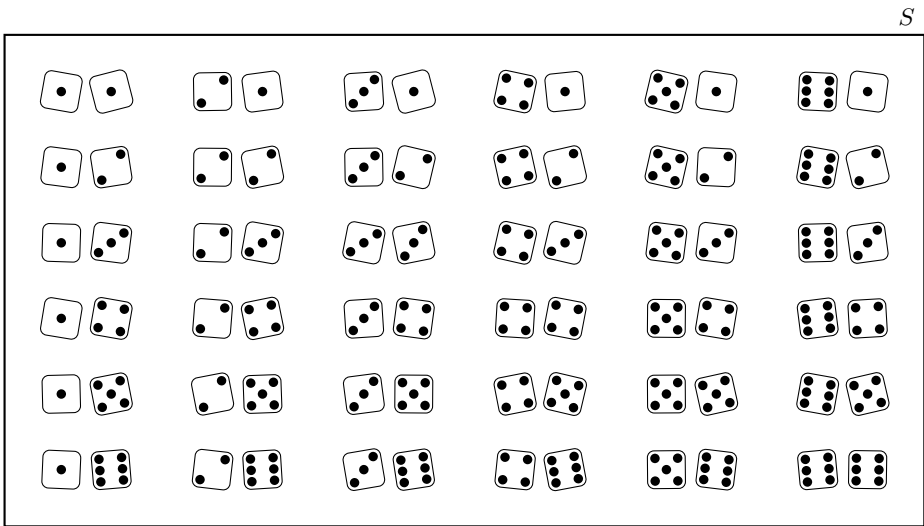
The sample space of an experiment is the set of all possible outcomes of that experiment. The sample space is denoted with the symbol S and the size of the sample space (the total number of possible outcomes) is denoted with $n(S)$

Even though we are usually interested in the outcome of an experiment, we also need to know what the other outcomes could have been. Let’s have a look at the sample spaces of each of our three experiments.

Experiment 1: Since a coin can land in one of only two ways (we will ignore the possibility that the coin lands on its edge), the sample space is the set $S = \{H; T\}$. The size of the sample space of the coin toss is $n(S) = 2$:



Experiment 2: Each of the dice can land on a number from 1 to 6. In this experiment the sample space of all possible outcomes is every possible combination of the 6 numbers on the first die with the 6 numbers on the second die. This gives a total of $n(S) = 6 \times 6 = 36$ possible outcomes. The figure below shows all of the outcomes in the sample space of rolling two dice:



Experiment 3: Each soccer team can get an integer score from 0 upwards. Usually we don't expect a score to go much higher than 5 goals, but there is no reason why this cannot happen. So the sample space of this experiment consists of all possible combinations of two non-negative integers. The figure below shows all of the possibilities. Since we do not limit the score of a team, this sample space is infinitely large:

	<i>S</i>				
	0 – 0	1 – 0	2 – 0	3 – 0	...
	0 – 1	1 – 1	2 – 1	3 – 1	...
	0 – 2	1 – 2	2 – 2	3 – 2	...
	0 – 3	1 – 3	2 – 3	3 – 3	
	⋮	⋮	⋮		⋱

NOTE:
When we represent a sample space containing real numbers we can either write out all the outcomes in the sample space: $\{1; 2; 3; 4; 5; 6; 7; 8; 9; 10\}$ or we can represent the sample space as: $\{n : n \in \mathbb{Z}, 1 \leq n \leq 10\}$.

DEFINITION: Event

An event is a specific set of outcomes of an experiment that you are interested in. An event is denoted with the letter E and the number of outcomes in the event with $n(E)$.

Experiment 1: Let us say that we would like the coin to land heads up. Here the event contains a single outcome: $E = \{H\}$. The size of the event set is $n(E) = 1$.

Experiment 2: Let us say that we are interested in the sum of the dice being 8. In this case the event set is:

$$E = \{(\begin{smallmatrix} \bullet & \bullet \\ \bullet & \bullet \end{smallmatrix}); (\begin{smallmatrix} \bullet & \bullet \\ \bullet & \bullet \end{smallmatrix}); (\begin{smallmatrix} \bullet & \bullet \\ \bullet & \bullet \end{smallmatrix}); (\begin{smallmatrix} \bullet & \bullet \\ \bullet & \bullet \end{smallmatrix}); (\begin{smallmatrix} \bullet & \bullet \\ \bullet & \bullet \end{smallmatrix})\}$$

since it contains all of the possible ways to get 8 dots with 2 dice. The size of the event set is $n(E) = 5$.

Experiment 3: We would like to know whether the first team will win. For this event to happen the first score must be greater than the second.

$$E = \{(1; 0); (2; 0); (2; 1); (3; 0); (3; 1); (3; 2); \dots\}.$$

This event set is infinitely large.

14.1 Theoretical probability

EMA7W

DEFINITION: Probability

A probability is a real number between 0 and 1 that describes how likely it is that an event will occur.

We can describe probabilities in three ways:

1. As a real number between 0 and 1. For example 0,75.
2. As a percentage. For example 0,75 can be written as 75%.
3. As a fraction. For example 0,75 can also be written as $\frac{3}{4}$.

We note the following about probabilities:

- A probability of 0 means that an event will never occur.
- A probability of 1 means that an event will always occur.
- A probability of 0,5 means that an event will occur half the time, or 1 time out of every 2.

When all of the possible outcomes of an experiment have an equal chance of occurring, we can compute the exact theoretical probability of an event. The probability of an event is the ratio between the number of outcomes in the event set and the number of possible outcomes in the sample space.

$$P(E) = \frac{n(E)}{n(S)}$$

VISIT:

The following video shows an example of calculating the theoretical probabilities of an event.

▶ See video: 2GVX at www.everythingmaths.co.za

Worked example 1: Theoretical probabilities

QUESTION

What is the theoretical probability of each of the events in the first two of our three experiments?

SOLUTION

Step 1: Write down the value of $n(S)$

Experiment 1 (coin): $n(S) = 2$

Experiment 2 (dice): $n(S) = 36$

Step 2: Write down the size of the event set

Experiment 1: $n(E) = 1$

Experiment 2: $n(E) = 5$

Step 3: Compute the theoretical probability

Experiment 1:

$$P(E) = \frac{n(E)}{n(S)} = \frac{1}{2} = 0,5$$

Experiment 2:

$$P(E) = \frac{n(E)}{n(S)} = \frac{5}{36} = 0,13\dot{8}$$

Note that we do not consider the theoretical probability of the third experiment. The third experiment is different from the first two in an important way, namely that all possible outcomes (all final scores) are not

equally likely. For example, we know that a soccer score of 1–1 is quite common, while a score of 11–15 is very, very rare. Because all outcomes are not equally likely, we cannot use the ratio between $n(E)$ and $n(S)$ to compute the theoretical probability of a team winning.

Exercise 14 – 1:

1. A learner wants to understand the term “event”. So the learner rolls 2 dice hoping to get a total of 8. Which of the following is the most appropriate example of the term “event”?
 - event set = $\{(4; 4)\}$
 - event set = $\{(2; 6); (3; 5); (4; 4); (5; 3); (6; 2)\}$
 - event set = $\{(2; 6); (6; 2)\}$
2. A learner wants to understand the term “sample space”. So the learner rolls a die. Which of the following is the most appropriate example of the term “sample space”?
 - $\{1; 2; 3; 4; 5; 6\}$
 - $\{H; T\}$
 - $\{1; 3; 5\}$
3. A learner finds a 6 sided die and then rolls the die once on a table. What is the probability that the die lands on either 1 or 2?
Write your answer as a simplified fraction.
4. A learner finds a textbook that has 100 pages. He then selects one page from the textbook. What is the probability that the page has an odd page number?
Write your answer as a decimal (correct to 2 decimal places).
5. Even numbers from 2 to 100 are written on cards. What is the probability of selecting a multiple of 5, if a card is drawn at random?
6. A bag contains 6 red balls, 3 blue balls, 2 green balls and 1 white ball. A ball is picked at random. Determine the probability that it is:
 - a) red
 - b) blue or white
 - c) not green
 - d) not green or red
7. A playing card is selected randomly from a pack of 52 cards. Determine the probability that it is:
 - a) the 2 of hearts
 - b) a red card
 - c) a picture card
 - d) an ace
 - e) a number less than 4

For more exercises, visit www.everythingmaths.co.za and click on ‘Practise Maths’.

1. [2GVY](#) 2. [2GVZ](#) 3. [2GW2](#) 4. [2GW3](#) 5. [2GW4](#) 6. [2GW5](#) 7. [2GW6](#)



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DEFINITION: *Relative frequency*

The relative frequency of an event is defined as the number of times that the event occurs during experimental trials, divided by the total number of trials conducted.

The relative frequency is not a theoretical quantity, but an experimental one. We have to repeat an experiment a number of times and count how many times the outcome of the trial is in the event set. Because it is experimental, it is possible to get a different relative frequency every time that we repeat an experiment.

VISIT:

The following video explains the concept of relative frequency using the throw of a dice.

► See video: [2GW7](https://www.everythingmaths.co.za/2GW7) at www.everythingmaths.co.za

Worked example 2: Relative frequency and theoretical probability

QUESTION

We toss a coin 30 times and observe the outcomes. The results of the trials are shown in the table below.

trial	1	2	3	4	5	6	7	8	9	10
outcome	H	T	T	T	H	T	H	H	H	T
trial	11	12	13	14	15	16	17	18	19	20
outcome	H	T	T	H	T	T	T	H	T	T
trial	21	22	23	24	25	26	27	28	29	30
outcome	H	H	H	T	H	T	H	T	T	T

What is the relative frequency of observing heads after each trial and how does it compare to the theoretical probability of observing heads?

SOLUTION

Step 1: Count the number of positive outcomes

A positive outcome is when the outcome is in our event set. The table below shows a running count (after each trial t) of the number of positive outcomes p we have observed. For example, after $t = 20$ trials we have observed heads 8 times and tails 12 times and so the positive outcome count is $p = 8$.

t	1	2	3	4	5	6	7	8	9	10
p	1	1	1	1	2	2	3	4	5	5
t	11	12	13	14	15	16	17	18	19	20
p	6	6	6	7	7	7	7	8	8	8
t	21	22	23	24	25	26	27	28	29	30
p	9	10	11	11	12	12	13	13	13	13

Step 2: Compute the relative frequency

Since the relative frequency is defined as the ratio between the number of positive trials and the total number of trials,

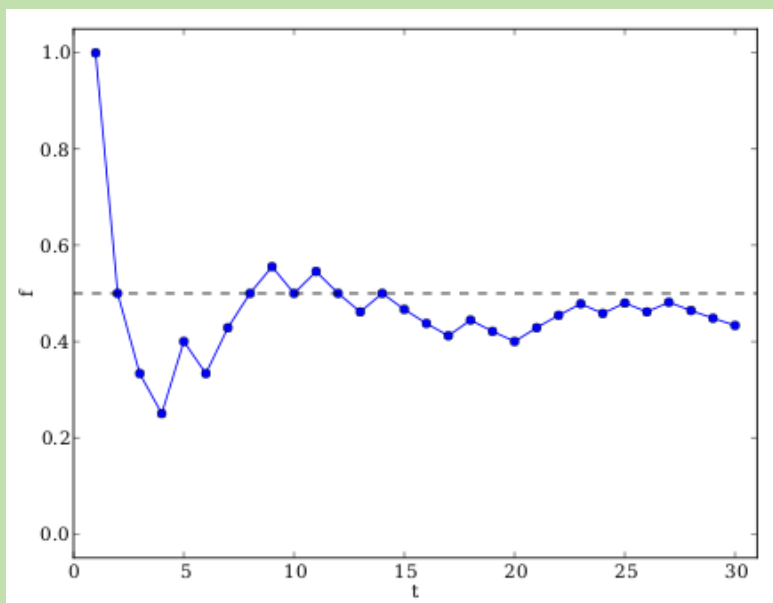
$$f = \frac{p}{t}$$

The relative frequency of observing heads, f , after having completed t coin tosses is:

t	1	2	3	4	5	6	7	8	9	10
f	1,00	0,50	0,33	0,25	0,40	0,33	0,43	0,50	0,56	0,50
t	11	12	13	14	15	16	17	18	19	20
f	0,55	0,50	0,46	0,50	0,47	0,44	0,41	0,44	0,42	0,40
t	21	22	23	24	25	26	27	28	29	30
f	0,43	0,45	0,48	0,46	0,48	0,46	0,48	0,46	0,45	0,43

From the last entry in this table we can now easily read the relative frequency after 30 trials, namely $\frac{13}{30} = 0,43$. The relative frequency is close to the theoretical probability of 0,5. In general, the relative frequency of an event tends to get closer to the theoretical probability of the event as we perform more trials.

A much better way to summarise the table of relative frequencies is in a graph:



The graph above is the plot of the relative frequency of observing heads, f , after having completed t coin tosses. It was generated from the table of numbers above by plotting the number of trials that have been completed, t , on the x -axis and the relative frequency, f , on the y -axis. In the beginning (after a small number of trials) the relative frequency fluctuates a lot around the theoretical probability at 0,5, which is shown with a dashed line. As the number of trials increases, the relative frequency fluctuates less and gets closer to the theoretical probability.

Worked example 3: Relative frequency and theoretical probability

QUESTION

While watching 10 soccer games where Team 1 plays against Team 2, we record the following final scores:

Trial	1	2	3	4	5	6	7	8	9	10
Team 1	2	0	1	1	1	1	1	0	5	3
Team 2	0	2	2	2	2	1	1	0	0	0

What is the relative frequency of Team 1 winning?

SOLUTION

Step 1:

In this experiment, each trial takes the form of Team 1 playing a soccer match against Team 2.

Step 2: Count the number of positive outcomes

We are interested in the event where Team 1 wins. From the table above we see that this happens 3 times.

Step 3: Compute the relative frequency

The total number of trials is 10. This means that the relative frequency of the event is

$$\frac{3}{10} = 0,3$$

It is important to understand the difference between the theoretical probability of an event and the observed relative frequency of the event in experimental trials. The theoretical probability is a number that we can compute if we have enough information about the experiment. If each possible outcome in the sample space is equally likely, we can count the number of outcomes in the event set and the number of outcomes in the sample space to compute the theoretical probability.

The relative frequency depends on the sequence of outcomes that we observe while doing a statistical experiment. The relative frequency can be different every time we redo the experiment. The more trials we run during an experiment, the closer the observed relative frequency of an event will get to the theoretical probability of the event.

So why do we need statistical experiments if we have theoretical probabilities? In some cases, like our soccer experiment, it is difficult or impossible to compute the theoretical probability of an event. Since we do not know exactly how likely it is that one soccer team will score goals against another, we can never compute the theoretical probability of events in soccer. In such cases we can still use the relative frequency to estimate the theoretical probability, by running experiments and counting the number of positive outcomes.

VISIT:

You can use this Phet simulation on [probability](#) to do some experiments with dropping a ball through a triangular grid.

Exercise 14 – 2:

1. A die is tossed 44 times and lands 5 times on the number 3.
What is the relative frequency of observing the die land on the number 3? Write your answer correct to 2 decimal places.
2. A coin is tossed 30 times and lands 17 times on heads.
What is the relative frequency of observing the coin land on heads? Write your answer correct to 2 decimal places.
3. A die is tossed 27 times and lands 6 times on the number 6.
What is the relative frequency of observing the die land on the number 6? Write your answer correct to 2 decimal places.

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2GW8 2. 2GW9 3. 2GWB



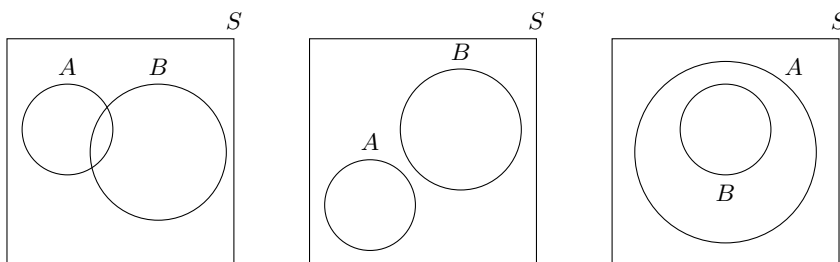
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A Venn diagram is a graphical way of representing the relationships between sets. In each Venn diagram a set is represented by a closed curve. The region inside the curve represents the elements that belong to the set, while the region outside the curve represents the elements that are excluded from the set.

Venn diagrams are helpful for thinking about probability since we deal with different sets. Consider two events, A and B , in a sample space S . The diagram below shows the possible ways in which the event sets can overlap, represented using Venn diagrams:



The sets are represented using a rectangle for S and circles for each of A and B . In the first diagram the two events overlap partially. In the second diagram the two events do not overlap at all. In the third diagram one event is fully contained in the other. Note that events will always appear inside the sample space since the sample space contains all possible outcomes of the experiment.

VISIT:

This video shows how to draw a Venn diagram using a deck of cards as the sample space.

▶ See video: [2GWC](https://www.everythingmaths.co.za) at www.everythingmaths.co.za

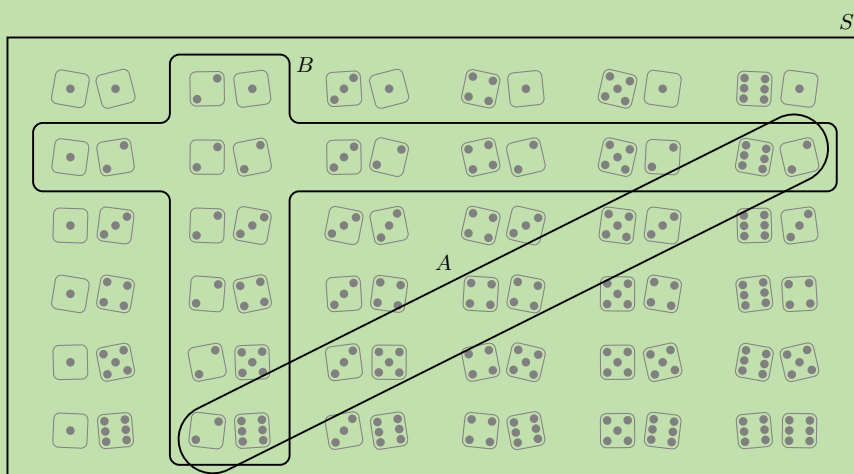
Worked example 4: Venn diagrams

QUESTION

Represent the sample space of two rolled dice and the following two events using a Venn diagram:

- Event A: the sum of the dice equals 8
- Event B: at least one of the dice shows a 2

SOLUTION



QUESTION

Consider the set of diamonds removed from a deck of cards. A random card is selected from the set of diamonds.

- Write down the sample space, S , for the experiment.
- What is the value of $n(S)$?
- Consider the following two events:
 - P : An even diamond is chosen
 - R : A royal diamond is chosen

Represent the sample space S and events P and R using a Venn diagram.

SOLUTION

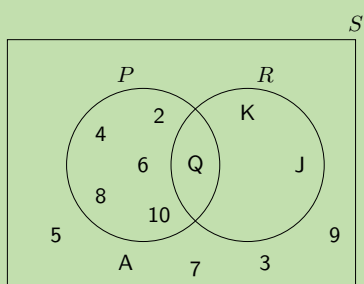
Step 1: Write down the sample space S

$$S = \{A; 2; 3; 4; 5; 6; 7; 8; 9; 10; J; Q; K\}$$

Step 2: Write down the value of $n(S)$

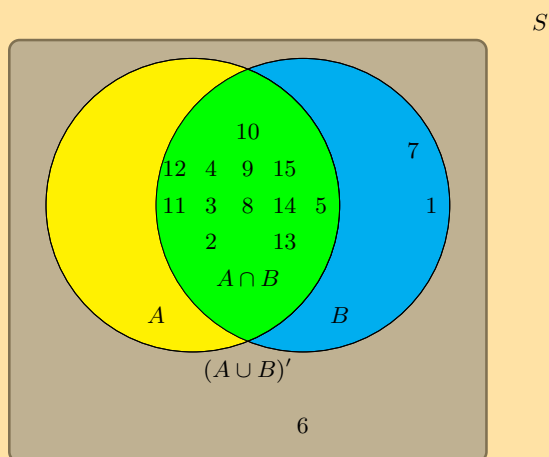
$$n(S) = 13$$

Step 3: Draw the Venn diagram



Exercise 14 – 3:

1. A group of learners are given the following Venn diagram:

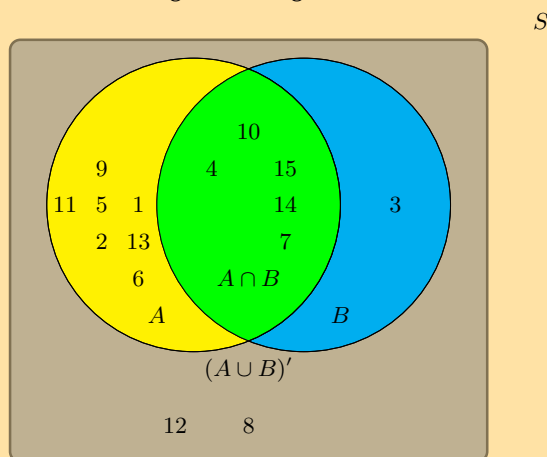


The sample space can be described as $\{n : n \in \mathbb{Z}, 1 \leq n \leq 15\}$.

They are asked to identify the event set of B . They get stuck, and you offer to help them find it. Which of the following sets best describes the event set of B ?

- $\{2; 3; 4; 5; 8; 9; 10; 11; 12; 13; 14; 15\}$
- $\{1; 2; 3; 4; 5; 7; 8; 9; 10; 11; 12; 14; 15\}$
- $\{1; 6; 7\}$
- $\{6\}$

2. A group of learners are given the following Venn diagram:



The sample space can be described as $\{n : n \in \mathbb{Z}, 1 \leq n \leq 15\}$.

They are asked to identify the event set of A . They get stuck, and you offer to help them find it. Which of the following sets best describes the event set of A ?

- $\{1; 2; 3; 4; 5; 6; 7; 8; 9; 10; 11; 12; 14; 15\}$
- $\{3; 8; 12\}$
- $\{3; 4; 7; 10; 14; 15\}$
- $\{1; 2; 4; 5; 6; 7; 9; 10; 11; 13; 14; 15\}$
- $\{4; 7; 10; 14; 15\}$

3. Pieces of paper labelled with the numbers 1 to 12 are placed in a box and the box is shaken. One piece of paper is taken out and then replaced.

- What is the sample space, S ?
- Write down the set A , representing the event of taking a piece of paper labelled with a factor of 12.
- Write down the set B , representing the event of taking a piece of paper labelled with a prime number.
- Represent A , B and S by means of a Venn diagram.
- Find:
 - $n(S)$
 - $n(A)$
 - $n(B)$

4. Let S denote the set of whole numbers from 1 to 16, X denote the set of even numbers from 1 to 16 and Y denote the set of prime numbers from 1 to 16. Draw a Venn diagram depicting S , X and Y .

5. There are 71 Grade 10 learners at school. All of these take some combination of Maths, Geography and History. The number who take Geography is 41, those who take History is 36, and 30 take Maths. The number who take Maths and History is 16; the number who take Geography and History is 6, and there are 8 who take Maths only and 16 who take History only.

- Draw a Venn diagram to illustrate all this information.
- How many learners take Maths and Geography but not History?
- How many learners take Geography only?
- How many learners take all three subjects?

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2GWD 2. 2GWF 3. 2GWC 4. 2GWH 5. 2GWJ



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14.4 Union and intersection

EMA7Z

DEFINITION: Union

The union of two sets is a new set that contains all of the elements that are in at least one of the two sets. The union is written as $A \cup B$ or "A or B".

DEFINITION: Intersection

The intersection of two sets is a new set that contains all of the elements that are in both sets. The intersection is written as $A \cap B$ or "A and B".

The figure below shows the union and intersection for different configurations of two events in a sample space, using Venn diagrams.

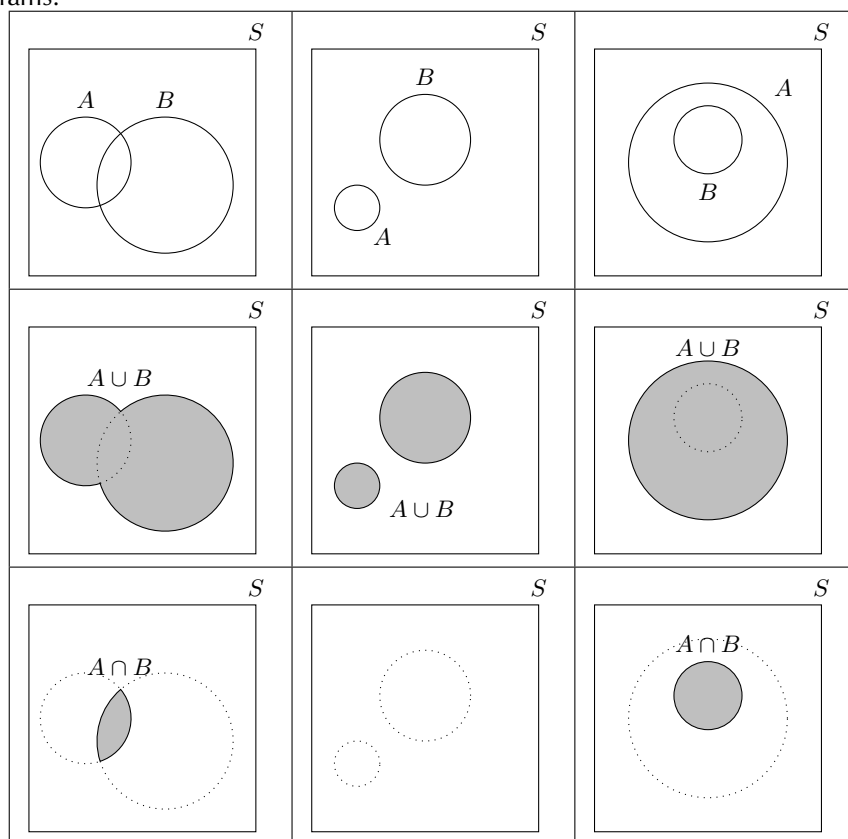
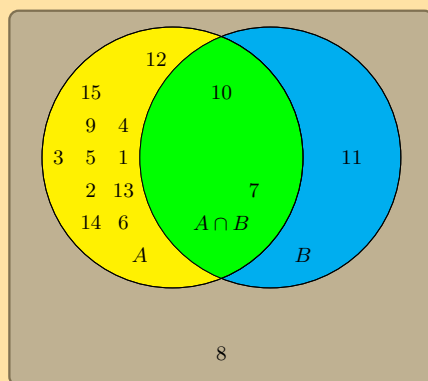


Figure 14.2: The unions and intersections of different events. Note that in the middle column the intersection, $A \cap B$, is empty since the two sets do not overlap. In the final column the union, $A \cup B$, is equal to A and the intersection, $A \cap B$, is equal to B since B is fully contained in A.

Exercise 14 – 4:

1. A group of learners are given the following Venn diagram:



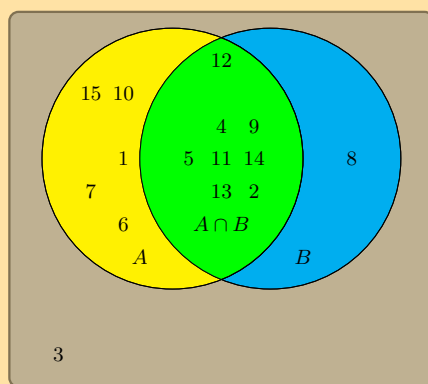
The sample space can be described as $\{n : n \in \mathbb{Z}, 1 \leq n \leq 15\}$.

They are asked to identify the event set of the intersection between event set A and event set B , also written as $A \cap B$. They get stuck, and you offer to help them find it.

Which set best describes the event set of $A \cap B$?

- $\{7; 10; 11\}$
- $\{1; 2; 3; 4; 5; 6; 7; 9; 10; 11\}$
- $\{1; 2; 3; 4; 5; 6; 7; 9; 10\}$
- $\{7; 10\}$

2. A group of learners are given the following Venn diagram:



The sample space can be described as $\{n : n \in \mathbb{Z}, 1 \leq n \leq 15\}$

They are asked to identify the event set of the union between event set A and event set B , also written as $A \cup B$. They get stuck, and you offer to help them find it.

Which set best describes the event set of $A \cup B$?

- $\{1; 6; 7; 10; 15\}$
- $\{1; 2; 4; 5; 6; 7; 8; 9; 10; 11; 12; 13; 14; 15\}$
- $\{2; 4; 5; 9; 10; 11; 12; 13; 14\}$
- $\{3\}$

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'. 1. 2GWK 2. 2GWM



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By definition, the sample space contains all possible outcomes of an experiment. So we know that the probability of observing an outcome from the sample space is 1.

$$P(S) = 1$$

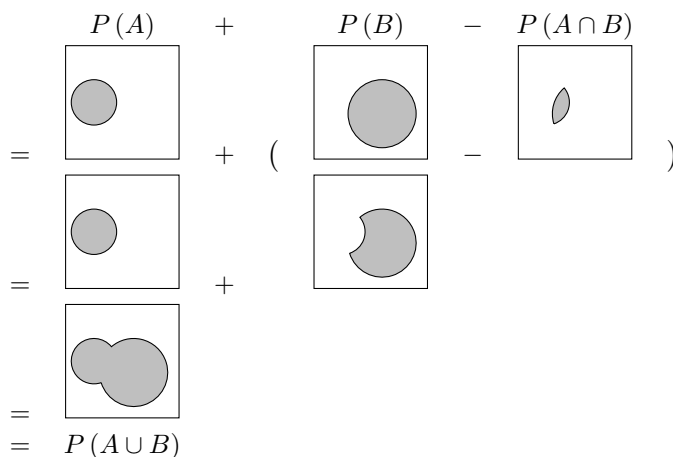
We can calculate the probability of the union of two events using:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

We will prove this identity using the Venn diagrams given above.

For each of the 4 terms in the union and intersection identity, we can draw the Venn diagram and then add and subtract the different diagrams. The area of a region represents its probability.

We will do this for the first column of the Venn diagram figure given previously. You should also try it for the other columns.



VISIT:

This video gives an example of how we can add probabilities together.

▶ See video: [2GWN](https://www.youtube.com/watch?v=2GWN) at www.everythingmaths.co.za

Worked example 6: Union and intersection of events

QUESTION

Relate the probabilities of events A and B from Example 4 (two rolled dice) and show that they satisfy the identity:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

SOLUTION

Step 1: Write down the probabilities of the two events, their union and their intersection

From the Venn diagram in Example 4, we can count the number of outcomes in each event. To get the probability of an event, we divide the size of the event by the size of the sample space, which is $n(S) = 36$.

$$\begin{aligned}
 P(A) &= \frac{n(A)}{n(S)} = \frac{5}{36} \\
 P(B) &= \frac{n(B)}{n(S)} = \frac{11}{36} \\
 P(A \cap B) &= \frac{n(A \cap B)}{n(S)} = \frac{2}{36} \\
 P(A \cup B) &= \frac{n(A \cup B)}{n(S)} = \frac{14}{36}
 \end{aligned}$$

Step 2: Write down and check the identity

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\text{RHS} = \frac{14}{36}$$

$$\text{LHS} = \frac{5}{36} + \frac{11}{36} - \frac{2}{36}$$

$$= \frac{5}{36} + \frac{9}{36}$$

$$= \frac{14}{36}$$

$$\therefore \text{RHS} = \text{LHS}$$

Exercise 14 – 5:

1. A group of learners is given the following event sets:

Event Set A	1	2	5	6
---------------	---	---	---	---

Event Set B	3
---------------	---

Event Set $A \cap B$	empty
----------------------	-------

The sample space can be described as $\{n : n \in \mathbb{Z}, 1 \leq n \leq 6\}$.

They are asked to calculate the value of $P(A \cup B)$. They get stuck, and you offer to calculate it for them. Give your answer as a decimal number, rounded to two decimal places.

2. A group of learners is given the following event sets:

Event Set A	1	2	6
---------------	---	---	---

Event Set B	1	5
---------------	---	---

Event Set $A \cup B$	1	2	5	6
----------------------	---	---	---	---

The sample space can be described as $\{n : n \in \mathbb{Z}, 1 \leq n \leq 6\}$.

They are asked to calculate the value of $P(A \cap B)$. They get stuck, and you offer to calculate it for them. Give your answer as a decimal number, rounded to two decimal value.

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'. 1. 2GWP 2. 2GWQ



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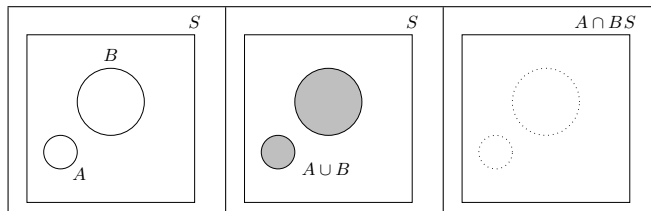


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DEFINITION: *Mutually exclusive events*

Two events are called mutually exclusive if they cannot occur at the same time. Whenever an outcome of an experiment is in the first event it cannot also be in the second event, and vice versa.

Another way of saying this is that the two event sets, A and B , cannot have any elements in common, or $P(A \cap B) = \emptyset$ (where $\emptyset$ denotes the empty set). We have already seen the Venn diagrams of mutually exclusive events in the middle column of the Venn diagrams provided earlier.



From this figure you can see that the intersection has no elements. You can also see that the probability of the union is the sum of the probabilities of the events.

$$P(A \cup B) = P(A) + P(B)$$

This relationship is true for mutually exclusive events only.

Worked example 7: Mutually exclusive events

QUESTION

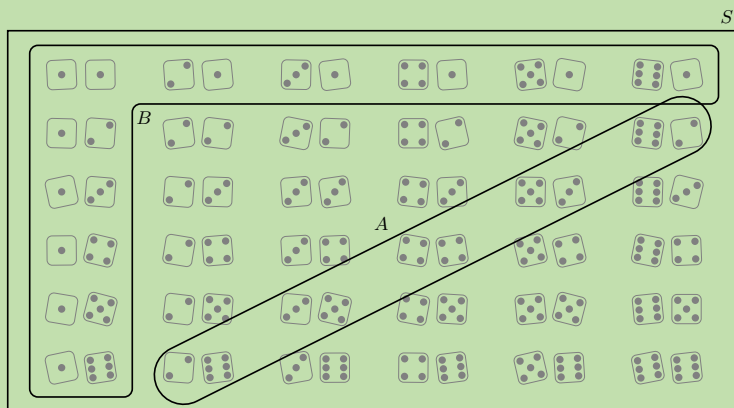
We roll two dice and are interested in the following two events:

- A : The sum of the dice equals 8
- B : At least one of the dice shows a 1

Show that the events are mutually exclusive.

SOLUTION

Step 1: Draw the sample space and the two events



Step 2: Determine the intersection

From the above figure we notice that there are no elements in common in A and B . Therefore the events are mutually exclusive.

Exercise 14 – 6:

State whether the following events are mutually exclusive or not.

1. A fridge contains orange juice, apple juice and grape juice. A cooldrink is chosen at random from the fridge. Event A: the cooldrink is orange juice. Event B: the cooldrink is apple juice.
2. A packet of cupcakes contains chocolate cupcakes, vanilla cupcakes and red velvet cupcakes. A cupcake is chosen at random from the packet. Event A: the cupcake is red velvet. Event B: the cupcake is vanilla.
3. A card is chosen at random from a deck of cards. Event A: the card is a red card. Event B: the card is a picture card.
4. A cricket team plays a game. Event A: they win the game. Event B: they lose the game.

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. [2GWR](#) 2. [2GWS](#) 3. [2GWT](#) 4. [2GWV](#)



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14.7 Complementary events

EMA84

DEFINITION: Complementary set

The complement of a set, A , is a new set that contains all of the elements that are not in A . We write the complement of A as A' , or sometimes not (A) .

For an experiment with sample space S and an event A we can derive some identities for complementary events. Since every element in A is not in A' , we know that complementary events are mutually exclusive.

$$A \cap A' = \emptyset$$

Since every element in the sample space is either in A or in A' , the union of complementary events covers the sample space.

$$A \cup A' = S$$

From the previous two identities, we also know that the probabilities of complementary events sum to 1.

$$P(A) + P(A') = P(A \cup A') = P(S) = 1$$

Worked example 8: Reasoning with Venn diagrams

QUESTION

In a survey 70 people were questioned about which product they use: A or B or both. The report of the survey shows that 25 people use product A, 35 people use product B and 15 people use neither. Use a Venn diagram to work out how many people:

1. use product A only
2. use product B only
3. use both product A and product B

SOLUTION

Step 1: Summarise the sizes of the sample space, the event sets, their union and their intersection

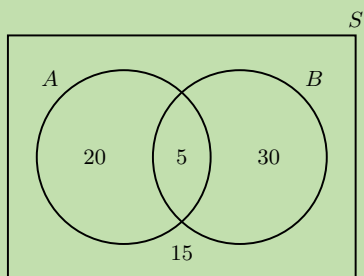
- We are told that 70 people were questioned, so the size of the sample space is $n(S) = 70$.
- We are told that 25 people use product A, so $n(A) = 25$.
- We are told that 35 people use product B, so $n(B) = 35$.
- We are told that 15 people use neither product. This means that $70 - 15 = 55$ people use at least one of the two products, so $n(A \cup B) = 55$.
- We are not told how many people use both products, so we have to work out the size of the intersection, $A \cap B$, by using the identity for the union of two events:

$$\begin{aligned}P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ \frac{n(A \cup B)}{n(S)} &= \frac{n(A)}{n(S)} + \frac{n(B)}{n(S)} - \frac{n(A \cap B)}{n(S)} \\ \frac{55}{70} &= \frac{25}{70} + \frac{35}{70} - \frac{n(A \cap B)}{70} \\ \therefore n(A \cap B) &= 25 + 35 - 55 \\ &= 5\end{aligned}$$

Step 2: Determine whether the events are mutually exclusive

Since the intersection of the events, $A \cap B$, is not empty, the events are not mutually exclusive. This means that their circles should overlap in the Venn diagram.

Step 3: Draw the Venn diagram and fill in the numbers

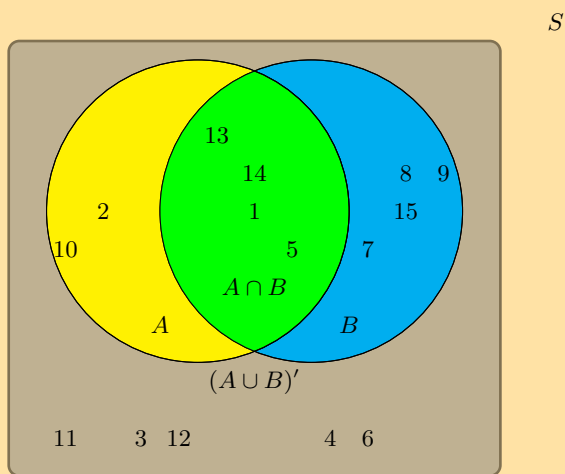


Step 4: Read off the answers

1. 20 people use product A only.
2. 30 people use product B only.
3. 5 people use both products.

Exercise 14 – 7:

1. A group of learners are given the following Venn diagram:



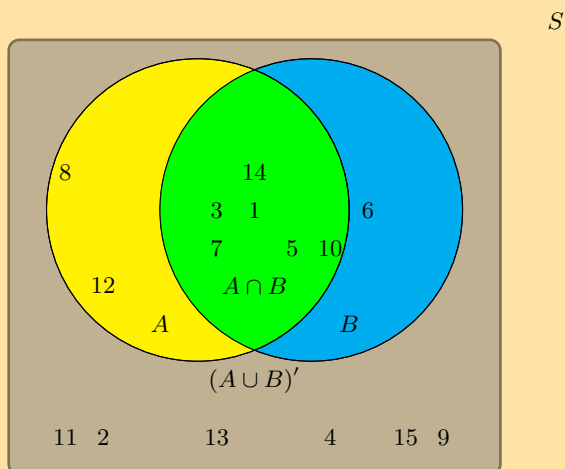
The sample space can be described as $\{n : n \in \mathbb{Z}, 1 \leq n \leq 15\}$.

They are asked to identify the complementary event set of B , also known as B' . They get stuck, and you offer to help them find it.

Which of the following sets best describes the event set of B' ?

- $\{1; 5; 13; 14\}$
- $\{2; 3; 4; 6; 10; 11; 12\}$
- $\{3; 4; 6; 11; 12\}$

2. A group of learners are given the following Venn diagram:



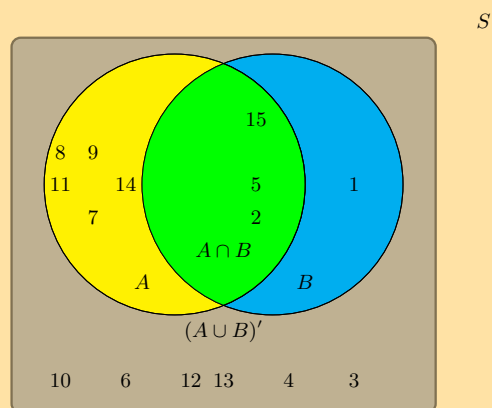
The sample space can be described as $\{n : n \in \mathbb{Z}, 1 \leq n \leq 15\}$.

They are asked to identify the complementary event set of $(A \cup B)$, also known as $(A \cup B)'$. They get stuck, and you offer to help them find it.

Which of the following sets best describes the event set of $(A \cup B)'$?

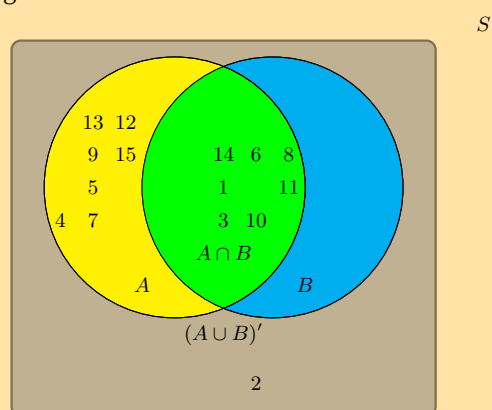
- $\{2; 4; 9; 11; 13; 15\}$
- $\{1; 3; 5; 6; 7; 8; 10; 12; 14\}$
- $\{6; 8; 12\}$

3. Given the following Venn diagram:



The sample space can be described as $\{n : n \in \mathbb{Z}, 1 \leq n \leq 15\}$. Are $(A \cup B)'$ and $A \cup B$ mutually exclusive?

4. Given the following Venn diagram:



The sample space can be described as $\{n : n \in \mathbb{Z}, 1 \leq n \leq 15\}$. Are A' and B' mutually exclusive?

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

1. 2GWW 2. 2GWX 3. 2GWY 4. 2GWZ



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14.8 Chapter summary

EMA85

► See presentation: 2GX2 at www.everythingmaths.co.za

- An experiment refers to an uncertain process.
- An outcome of an experiment is a single result of that experiment.
- The sample space of an experiment is the set of all possible outcomes of that experiment. The sample space is denoted with the symbol S and the size of the sample space (the total number of possible outcomes) is denoted with $n(S)$.
- An event is a specific set of outcomes of an experiment that you are interested in. An event is denoted with the letter E and the number of outcomes in the event with $n(E)$.
- A probability is a real number between 0 and 1 that describes how likely it is that an event will occur.

- A probability of 0 means that an event will never occur.
- A probability of 1 means that an event will always occur.
- A probability of 0,5 means that an event will occur half the time, or 1 time out of every 2.
- A probability can also be written as a percentage or as a fraction.
- When all of the possible outcomes of an experiment have an equal chance of occurring, we can compute the exact theoretical probability of an event. The probability of an event is the ratio between the number of outcomes in the event set and the number of possible outcomes in the sample space.

$$P(E) = \frac{n(E)}{n(S)}$$

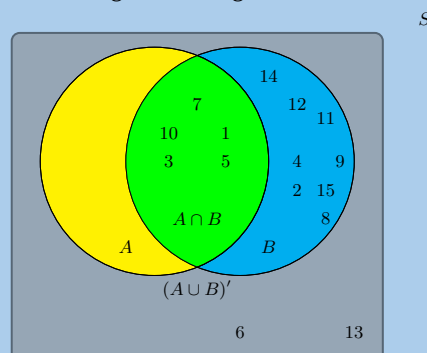
- The relative frequency of an event is defined as the number of times that the event occurs during experimental trials, divided by the total number of trials conducted.

$$f = \frac{\text{number of positive trials}}{\text{number of trials}} = \frac{p}{n}$$

- The union of two sets is a new set that contains all of the elements that are in at least one of the two sets. The union is written as $A \cup B$ or A or B .
- The intersection of two sets is a new set that contains all of the elements that are in both sets. The intersection is written as $A \cap B$ or A and B .
- The probability of observing an outcome from the sample space is 1: $P(S) = 1$.
- The probability of the union of two events is calculated using: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- Mutually exclusive events are two events that cannot occur at the same time. Whenever an outcome of an experiment is in the first event, it can not also be in the second event.
- The complement of a set, A , is a different set that contains all of the elements that are not in A . We write the complement of A as A' or “not (A)”.
- Complementary events are mutually exclusive: $A \cap A' = \emptyset$.
- Complementary events cover the sample space: $A \cup A' = S$
- Probabilities of complementary events sum to 1: $P(A) + P(A') = P(A \cup A') = P(S) = 1$.

End of chapter Exercise 14 – 8:

1. A learner wants to understand the term “outcome”. So the learner rolls a die. Which of the following is the most appropriate example of the term “outcome”?
 - A teacher walks into the class room.
 - The die lands on the number 5.
 - The clock strikes 3 pm.
2. A group of learners are given the following Venn diagram:

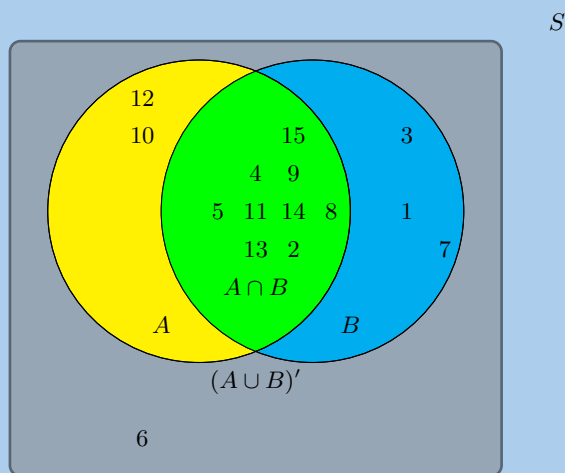


The sample space can be described as $\{n : n \in \mathbb{Z}, 1 \leq n \leq 15\}$.

They are asked to identify the event set of B . They get stuck, and you offer to help them find it. Which of the following sets best describes the event set of B ?

- $\{6; 13\}$
- $\{1; 3; 5; 7; 10\}$
- $\{2; 4; 6; 8; 9; 11; 12; 13; 14; 15\}$
- $\{1; 2; 3; 4; 5; 7; 8; 9; 10; 11; 12; 14; 15\}$

3. A group of learners are given the following Venn diagram:

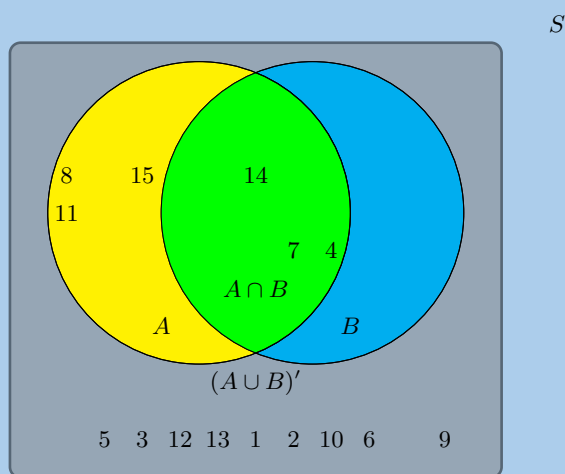


The sample space can be described as $\{n : n \in \mathbb{Z}, 1 \leq n \leq 15\}$.

They are asked to identify the event set of the union between event set A and event set B , also written as $A \cup B$. They get stuck, and you offer to help them find it.

Write down the event set that best describes $A \cup B$.

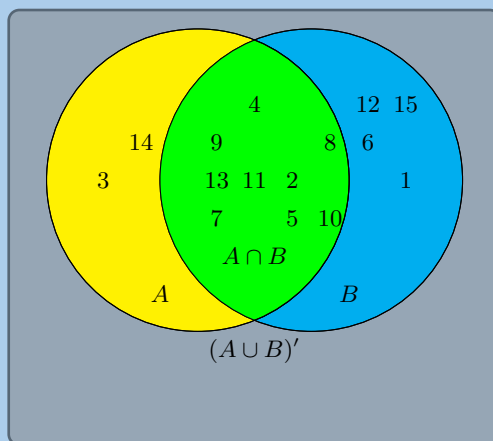
4. Given the following Venn diagram:



The sample space can be described as $\{n : n \in \mathbb{Z}, 1 \leq n \leq 15\}$.

Are $A \cup B$ and $(A \cup B)'$ mutually exclusive?

5. A group of learners are given the following Venn diagram:



The sample space can be described as $\{n : n \in \mathbb{Z}, 1 \leq n \leq 15\}$.

They are asked to identify the complementary event set of $(A \cap B)$, also known as $(A \cap B)'$. They get stuck, and you offer to help them find it.

Write down the set that best describes the event set of $(A \cap B)'$.

6. A learner finds a deck of 52 cards and then takes one card from the deck. What is the probability that the card is a king?
Write your answer as a decimal (correct to 2 decimal places).
7. A die is tossed 21 times and lands 2 times on the number 3.
What is the relative frequency of observing the die land on the number 3? Write your answer correct to 2 decimal places.
8. A coin is tossed 44 times and lands 22 times on heads.
What is the relative frequency of observing the coin land on heads? Write your answer correct to 2 decimal places.
9. A group of 45 children were asked if they eat Frosties, Strawberry Pops or both. 31 children said they eat both and 6 said they only eat Frosties. What is the probability that a child chosen at random will eat only Strawberry Pops?
10. In a group of 42 learners, all but 3 had a packet of chips or a cooldrink or both. If 23 had a packet of chips and 7 of these also had a cooldrink, what is the probability that one learner chosen at random has:
 - a) both chips and cooldrink
 - b) only cooldrink
11. A box contains coloured blocks. The number of each colour is given in the following table.

Colour	Purple	Orange	White	Pink
Number of blocks	24	32	41	19

A block is selected randomly. What is the probability that the block will be:

- a) purple
 - b) purple or white
 - c) pink and orange
 - d) not orange
12. A small nursery school has a class with children of various ages. The table gives the number of children of each age in the class.

Age	3 years old	4 years old	5 years old
Male	2	7	6
Female	6	5	4

If a child is selected at random what is the probability that the child will be:

- a) a female
- b) a 4 year old male
- c) aged 3 or 4
- d) aged 3 and 4
- e) not 5
- f) either 3 or female

13. Fiona has 85 labelled discs, which are numbered from 1 to 85. If a disc is selected at random what is the probability that the disc number:

- a) ends with 5
- b) is a multiple of 3
- c) is a multiple of 6
- d) is number 65
- e) is not a multiple of 5
- f) is a multiple of 3 or 4
- g) is a multiple of 2 and 6
- h) is number 1

14. Use a Venn diagram to work out the following probabilities for a die being rolled:

- a) a multiple of 5 and an odd number
- b) a number that is neither a multiple of 5 nor an odd number
- c) a number which is not a multiple of 5, but is odd

15. A packet has yellow sweets and pink sweets. The probability of taking out a pink sweet is $\frac{7}{12}$. What is the probability of taking out a yellow sweet?

16. In a car park with 300 cars, there are 190 Opels. What is the probability that the first car to leave the car park is:

- a) an Opel
- b) not an Opel

17. Nezi has 18 loose socks in a drawer. Eight of these are plain orange and two are plain pink. The remaining socks are neither orange nor pink. Calculate the probability that the first sock taken out at random is:

- a) orange
- b) not orange
- c) pink
- d) not pink
- e) orange or pink
- f) neither orange nor pink

18. A plate contains 9 shortbread cookies, 4 ginger biscuits, 11 chocolate chip cookies and 18 Jambos. If a biscuit is selected at random, what is the probability that:

- a) it is either a ginger biscuit or a Jambo
- b) it is not a shortbread cookie

19. 280 tickets were sold at a raffle. Jabulile bought 15 tickets. What is the probability that Jabulile:
- wins the prize
 - does not win the prize
20. A group of children were surveyed to see how many had red hair and brown eyes. 44 children had red hair but not brown eyes, 14 children had brown eyes and red hair, 5 children had brown eyes but not red hair and 40 children did not have brown eyes or red hair.
- How many children were in the school?
 - What is the probability that a child chosen at random has:
 - brown eyes
 - red hair
 - A child with brown eyes is chosen randomly. What is the probability that this child will have red hair?
21. A jar has purple sweets, blue sweets and green sweets in it. The probability that a sweet chosen at random will be purple is $\frac{1}{7}$ and the probability that it will be green is $\frac{3}{5}$.
- If I choose a sweet at random what is the probability that it will be:
 - purple or blue
 - green
 - purple
 - If there are 70 sweets in the jar how many purple ones are there?
 - $\frac{2}{5}$ of the purple sweets in (b) have streaks on them and the rest do not. How many purple sweets have streaks?
22. Box A contains 3 cards numbered 1, 2 and 3.
Box B contains 2 cards numbered 1 and 2.
One card is removed at random from each box.
Find the probability that:
- the sum of the numbers is 4.
 - the sum of the two numbers is a prime number.
 - the product of the two numbers is at least 3.
 - the sum is equal to the product.
23. A card is drawn at random from an ordinary pack of 52 playing cards.
- Find the probability that the card drawn is:
 - the three of diamonds
 - the three of diamonds or any heart
 - a diamond or a three
 - The card drawn is the three of diamonds. It is placed on the table and a second card is drawn. What is the probability that the second card drawn is not a diamond.
24. A group of learners is given the following event sets:

Event Set A	3	4
-------------	---	---

Event Set B	2	4	5
-------------	---	---	---

Event Set $A \cup B$	2	3	4	5
----------------------	---	---	---	---

The sample space can be described as $\{n : n \in \mathbb{Z}, 1 \leq n \leq 6\}$

They are asked to calculate the value of $P(A \cap B)$. They get stuck, and you offer to calculate it for them. Give your answer as a decimal number, rounded to two decimal value.

25. For each of the following, draw a Venn diagram to represent the situation and find an example to illustrate the situation.
- a sample space in which there are two events that are not mutually exclusive
 - a sample space in which there are two events that are complementary
26. Use a Venn diagram to prove that the probability of either event A or B occurring (A and B are not mutually exclusive) is given by:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

27. All the clubs are taken out of a pack of cards. The remaining cards are then shuffled and one card chosen. After being chosen, the card is replaced before the next card is chosen.
- What is the sample space?
 - Find a set to represent the event, P , of drawing a picture card.
 - Find a set for the event, N , of drawing a numbered card.
 - Represent the above events in a Venn diagram.
 - What description of the sets P and N is suitable? (Hint: Find any elements of P in N and of N in P .)
28. A survey was conducted at Mutende Primary School to establish how many of the 650 learners buy vetkoek and how many buy sweets during break. The following was found:
- 50 learners bought nothing
 - 400 learners bought vetkoek
 - 300 learners bought sweets
- Represent this information with a Venn diagram.
 - If a learner is chosen randomly, calculate the probability that this learner buys:
 - sweets only
 - vetkoek only
 - neither vetkoek nor sweets
 - vetkoek and sweets
 - vetkoek or sweets
29. In a survey at Lwandani Secondary School, 80 people were questioned to find out how many read the Sowetan, how many read the Daily Sun and how many read both. The survey revealed that 45 read the Daily Sun, 30 read the Sowetan and 10 read neither. Use a Venn diagram to find the percentage of people that read:
- only the Daily Sun
 - only the Sowetan
 - both the Daily Sun and the Sowetan
30. In a class there are
- 8 learners who play football and hockey
 - 7 learners who do not play football or hockey
 - 13 learners who play hockey
 - 19 learners who play football
- How many learners are there in the class?
31. Of 36 people, 17 have an interest in reading magazines and 12 have an interest in reading books, 6 have an interest in reading both magazines and books.

- Represent the information in a Venn diagram.
- How many people have no interest in reading magazines or books?
- If a person is chosen at random from the group, find the probability that the person will:
 - have an interest in reading magazines and books.
 - have an interest in reading books only.
 - not have any interest in reading books.

32. 30 learners were surveyed and the following information was revealed from this group:

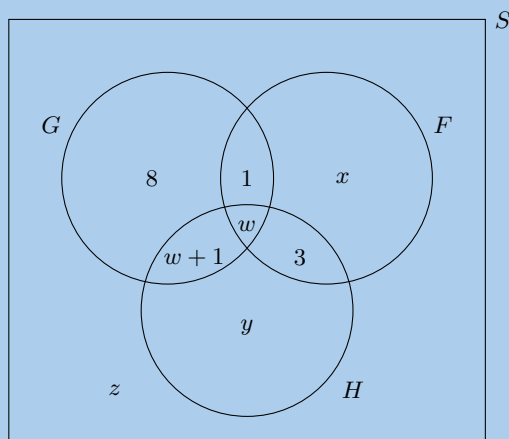
- 18 learners take Geography
- 10 learners take French
- 6 learners take History, but take neither Geography nor French.

In addition the following Venn Diagram has been filled in below:

Let G be the event that a learner takes Geography.

Let F be the event that a learner takes French.

Let H be the event that a learner takes History.



- From the information above, determine the values of w , x , y and z .
- Determine the probability that a learner chosen at random from this group:
 - takes only Geography.
 - takes French and History, but not Geography.

For more exercises, visit www.everythingmaths.co.za and click on 'Practise Maths'.

- | | | | | | | | |
|---------------------------|---------------------------|--------------------------|--------------------------|--------------------------|--------------------------|---------------------------|--------------------------|
| 1. 2GX3 | 2. 2GX4 | 3. 2GX5 | 4. 2GX6 | 5. 2GX7 | 6. 2GX8 | 7. 2GX9 | 8. 2GXB |
| 9. 2GXC | 10. 2GXD | 11. 2GXF | 12. 2GXF | 13. 2GXH | 14. 2GXJ | 15. 2G XK | 16. 2GXM |
| 17. 2GXN | 18. 2GXP | 19. 2GXQ | 20. 2GXR | 21. 2GXS | 22. 2GXT | 23. 2GXV | 24. 2GXW |
| 25a. 2GXX | 25b. 2GXY | 26. 2GXZ | 27. 2GY2 | 28. 2GY3 | 29. 2GY4 | 30. 2GY5 | 31. 2GY6 |
| 32. 2GY7 | | | | | | | |



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1 Algebraic expressions

Exercise 1 – 1:

- | | | | |
|----|--|--|---------------------------|
| 1. | a) $\mathbb{Z}$ | b) rational | b) $\frac{3}{25}$ |
| | b) (ii) | c) rational | c) $\frac{29}{50}$ |
| 2. | a) between the rectangle and $\mathbb{Z}$ | d) irrational | d) $\frac{2589}{10\,000}$ |
| | b) (ii) | 6. | a) rational |
| 3. | a) real | b) rational | 10. |
| | b) real | c) irrational | a) $0,\dot{1}$ |
| | c) non-real | d) rational | b) $0,\overline{12}$ |
| | d) undefined | 7. | c) $0,\overline{123}$ |
| | e) non-real | a) 7 and 11 | d) $0,1141\overline{45}$ |
| | f) real | b) $-\sqrt{8}$; $3,3231089\dots$; $3 + \sqrt{2}$; π | 11. |
| 4. | a) rational | c) $\sqrt{-1}$ | a) $0,\dot{5}$ |
| | b) irrational | d) -3 ; 0 ; $-8\frac{4}{5}$; $\frac{22}{7}$; 7 ; $1,\overline{34}$ | b) $0,\dot{5}$ |
| | c) rational, an integer, a whole number and a natural number | 9. | c) $0,2\dot{1}$ |
| | d) irrational | a) $9\frac{7}{10}$; 11 | d) $0,\dot{6}$ |
| | e) irrational | e) -3 ; 7 ; 11 | e) $1,\overline{27}$ |
| | f) irrational | f) $\frac{14}{0}$ | f) $4,8\dot{3}$ |
| | g) rational, an integer, a whole number and a natural number | 8. | g) $2,\dot{1}$ |
| | h) irrational | a) (i) 555 (ii) rational | 12. |
| | i) irrational | b) (i) any three digits (ii) irrational | a) $\frac{5}{9}$ |
| 5. | a) rational | c) (i) any three digits (ii) irrational | b) $\frac{57}{90}$ |
| | | d) (i) any three digits (ii) irrational | c) $\frac{4}{9}$ |
| | | e) (i) 545 (ii) rational | d) $\frac{526}{99}$ |
| | | 9. | e) $\frac{163}{33}$ |
| | | a) $\frac{1}{10}$ | f) $\frac{130}{33}$ |

Exercise 1 – 2:

- | | | | | | | |
|----|-----------|----|---------------|----|---------------|---------------|
| 1. | a) 12,566 | 2. | a) 345,0440 | 3. | a) 9,87 | c) 648 768,22 |
| | b) 3,317 | | b) 1361,73 | | b) 4,93 | 5. 11 |
| | c) 0,267 | | c) 728,009052 | | c) 14,80 | 6. R 5,03 |
| | d) 1,913 | | d) 0,0370 | | d) 14,8044... | |
| | e) 6,325 | | e) 0,45455 | 4. | a) 0,01 | |
| | f) 0,056 | | f) 0,08 | | b) 519 854,59 | |

Exercise 1 – 3:

- | | | | | |
|----|------------|--------------|------------|---|
| 1. | a) 4 and 5 | e) 12 and 13 | i) 4 and 5 | c) 3,9 |
| | b) 5 and 6 | f) 7 and 8 | j) 4 and 5 | d) 3,5 |
| | c) 1 and 2 | g) 8 and 9 | 2. | a) 3,1 |
| | d) 4 and 5 | h) 4 and 5 | b) 9,1 | 3. $-\sqrt{8}$; $-\sqrt{\frac{9}{4}}$; $0,45$; $0,\overline{45}$; $\frac{27}{7}$; $\sqrt{19}$; 6 ; 2π ; $\sqrt{51}$ |

Exercise 1 – 4:

- | | | | |
|----|------------------------|---------------------------------|---|
| 1. | a) $2y^2 + 8y$ | t) $36d^2 - 49$ | m) $-20y^3 - 116y^2 - 13y + 6$ |
| | b) $y^2 + 7y + 10$ | u) $25z^2 - 1$ | n) $-24y^3 + 126y^2 - 3y - 9$ |
| | c) $2t^2 - 5t + 2$ | v) $1 - 9h^2$ | o) $-20y^2 - 80y - 30$ |
| | d) $x^2 - 16$ | w) $4p^2 + 10p + 6$ | p) $49y^3 + 42y^2 + 79y + 30$ |
| | e) $x^2 - 16$ | x) $8a^2 + 60a + 28$ | q) $a^3 + 4a^2b + 5ab^2 + 2b^3$ |
| | f) $a^2 - b^2$ | y) $10r^2 + 28r + 16$ | r) $x^3 + y^3$ |
| | g) $6p^2 + 29p + 9$ | z) $w^2 - 1$ | s) $52m^3 + 6m + 30m^2$ |
| | h) $3k^2 + 16k - 12$ | 2. | a) $g^2 - 121$ |
| | i) $s^2 + 12s + 36$ | b) $8b^2 - 20b + 8$ | t) $48x^5 + 16x^2 + 24x^3$ |
| | j) $x^2 - 49$ | c) $8b^2 - 10b + 3$ | u) $15k^5 + 9k^3 + 14k^2$ |
| | k) $9x^2 - 1$ | d) $18x^2 + 24x - 24$ | v) $81x^4 - 72x + 16$ |
| | l) $-14k^2 + 17k + 6$ | e) $6w^2 + 17w - 14$ | w) $-6y^6 + 107y^4 + 3y^3 - 176y^2 - 48y$ |
| | m) $16x^2 - 8x + 1$ | f) $4t^2 - 12t + 9$ | x) $x^4 + x^3 - 11x^2 - 9x + 18$ |
| | n) $y^2 - 2y - 15$ | g) $25p^2 - 80p + 64$ | y) $-3a^2 + 20a - 12$ |
| | o) $-x^2 + 64$ | h) $16y^2 + 40y + 25$ | 3. |
| | p) $x^2 + 18x + 81$ | i) $-10y^7 - 39y^6 - 36y^5$ | a) $3x^2 + 16x + 5$ |
| | q) $84y^2 - 153y + 33$ | j) $72y^2 - 18y + 27$ | b) $2a^4 + 5a^3 - 5a - 2$ |
| | r) $g^2 - 10g + 25$ | k) $-10y^3 + 4y^2 + 103y - 132$ | c) $-y^4 - 8y^3 - 14y^2 + 8y - 1$ |
| | s) $d^2 + 18d + 81$ | l) $-14y^3 + 26y^2 + 4y - 16$ | d) $2x^3 - 2x^2y - 2xy^2 - 2y^3$ |
| | | | e) $3a^3 - 30ab^2 + 9b^3$ |

- f) $8a^4 - 12a^3b + 2a^2b^2 + 3ab^3 - b^4$
 g) $9x^2 + 6xy - 3y^2$
 h) $5x^2 - 6xy - 2y^2$
 i) $\frac{x^2}{12} + \frac{7}{12} + \frac{3}{x^2}$
 j) $\frac{x^2}{3} + \frac{10}{3} - \frac{8}{x^2}$
 k) $10x - 12y$
 l) $2a^2 + 3ab + 2a + 3b$

4. 4

5. -4
 6. a) 5
 b) $k = -3$ or $k = -1$
 c) $k < 0$
 d) $k > 4$
 7. a) $x^2 + 8 + \frac{16}{x^2}$
 b) 6
 8. a) $a^2 + 2 + \frac{1}{a^2}$

- b) 9
 c) 13
 9. a) $9y^2 + 3 + \frac{1}{4y^2}$
 b) 16
 10. a) $a^2 + \frac{2}{3} + \frac{1}{9a^2}$
 b) $a^3 + \frac{1}{27a^3}$
 c) 6

Exercise 1 – 5:

1. $4(3x + 8y)$
 2. $-2ab(b + 2a)$
 3. $3b(6a - c)$
 4. $6k(2j + 3q)$
 5. $12a(-1 + 2a^2)$
 6. $-2a(b + 4)$
 7. $8kj(3 - 2k)$
 8. $-ab(a + b)$
 9. $18b^2q(4 - bq)$
 10. $5(5x^3 - y)(5x^3 + y)$
 11. $2x(3x + 1 + 5x^2)$
 12. $xy(2y + yz + 3)$
 13. $12k^2j(1 + 2j)$
 14. $3(a^2 + 2a - 6)$
 15. $7a + 4$

Exercise 1 – 6:

1. $(y - 3)(4 - k)$
 2. $(a - 1)(a - 5)(a + 5)$
 3. $(b + 4)(m)(b - 6)$
 4. $(a + 7)(a^2 + 9)$
 5. $(b - 4)(3b + 7)$
 6. $(z + 6)(3g + 2)$
 7. $(y + 2)(4b + 5)$
 8. $(r + 5)(3d + 14)$
 9. $(6x + y - 3)(6x + y + 3)$
 10. $3(2x - y)(3y - 2x)$
 11. $(4a + 3b + 4c)(4a - 3b - 4c)$
 12. $(-2b + 11)(4b - 19)$
 13. $(29 - 26a)(30a - 41)$
 14. $(4k - 2)(4k + 2)$
 15. $(abc - 1)(abc + 1)$
 16. $(\frac{1}{3}a - 2b)(\frac{1}{3}a + 2b)$
 17. $2(\frac{1}{2}x + 1)(\frac{1}{2}x - 1)$
 18. $(y - \sqrt{8})(y + \sqrt{8})$
 19. $(y - \sqrt{13})(y + \sqrt{13})$
 20. $(a - 3b)(a - 5b)(a + 3b)^2$

Exercise 1 – 7:

1. $(2d - 3r)(3 + t^5)$
 2. $(z - 2m)(9 + b^3)$
 3. $(7z - 2y)(5 + c^5)$
 4. $(3 + a)(2x + 1)$
 5. $(x + 5)(x - 6)$
 6. $(5 - a)(x + 2y)$
 7. $(a - x)(a - 2)$
 8. $(y + 2)(5x - 3)$
 9. $(-a + b)(a + 1)$
 10. $(7m - 2n)(2 + j)$
 11. $(7r - 5x)(4 + g)$
 12. $(5d - 3m)(5 + y)$
 13. $(5q - 2z)(9 + c)$
 14. $(2j - 5v)(3 + y)$
 15. $(2a - 5k)(8 + z)$
 16. $(a - b)(x + y + 2)$
 17. $(3a + b)(x - y - 3)$

Exercise 1 – 8:

1. $(x + 5)(x + 3)$
 2. $(x + 8)(x + 1)$
 3. $(x + 6)^2$
 4. $(h + 3)(2h - 1)$
 5. $(x + 1)(3x + 1)$
 6. $(s + 2)(3s - 5)$
 7. $(x + 3)(x - 5)$
 8. $(x + 3)(x - 1)$
 9. $(x + 5)(x - 4)$
 10. $(x - 5)(x + 4)$
 11. $2(x + 1)(x + 10)$
 12. $2(a + 1)(3a + 4)$
 13. $3(2v - 3)(v - 3)$
 14. $3(g - 3)(2g + 1)$
 15. $(3x + 1)(x + 6)$
 16. $(3x - 1)(x + 6)$
 17. $(7x + 1)(x - 1)$
 18. $3(2x + 1)(x - 3)$
 19. $(a - 4b)(a - 3b)$
 20. $(3a - 4b)(a + 3b)$
 21. $2((7x + 2)(7x - 1))$
 22. $(x - 6)(x - 5)$
 23. $(a - 7)(a - 1)$
 24. $(y - 3)(y + 6)$
 25. $3(b + 4)(b + 1)$
 26. $6(a + 7)(a - 4)$

Exercise 1 – 9:

1. $(w - 2)(w^2 + 2w + 4)$
 2. $(g + 4)(g^2 - 4g + 16)$
 3. $(h + 1)(h^2 - h + 1)$
 4. $(x + 2)(x^2 - 2x + 4)$
 5. $(3 - m)(9 + 3m + m^2)$
 6. $2(x - y)(x^2 + xy + y^2)$
 7. $3(k + 3q)(k^2 - 3kq + 9q^2)$
 8. $(4t - 1)(16t^2 + 4t + 1)$
 9. $(8x - 1)(8x + 1)$
 10. $(5x + 1)(25x^2 - 5x + 1)$
 11. $(\sqrt[3]{25}x + 1)((\sqrt[3]{25})^2x^2 - \sqrt[3]{25}x + 1)$
 12. $(z)(1 - 5z)(1 + 5z + 25z^2)$
 13. $(2m^2 + n^3)(4m^4 - 2m^2n^3 + n^6)$
 14. $(6n - k)(36n^2 + 6nk + k^2)$
 15. $(5s + d)(25s^2 - 5sd + d^2)$
 16. $(2k + r)(4k^2 - 2kr + r^2)$
 17. $(2jkl - b)(4j^2k^2l^2 + 2jklabc + b^2)$
 18. $(3xy + w)(9x^2y^2 - 3xyw + w^2)$
 19. $2(4m + f)(16m^2 - 4mf + f^2)$
 20. $(p^5 - \frac{1}{2}y^4)(p^{10} + \frac{1}{2}p^5y^4 + \frac{1}{4}y^8)$
 21. $(\frac{3}{t} - s)(\frac{9}{t^2} + \frac{3s}{t} + s^2)$
 22. $(\frac{1}{4q} - h)(\frac{1}{16q^2} + \frac{h}{4q} + h^2)$
 23. $\frac{1}{3}(6g + v)(36g^2 - 6gv + v^2)$
 24. $(1 - x + y)(1 - x + y + x^2 - 2xy + y^2)$
 25. $(h - 1)(h + 1)(h^2 + 1)(2g^2 + h)(4g^4 - 2g^2h + h^2)$
 26. $(x + y)(5w - h)(25w^2 + 5wh + h^2)$
 27. $(x - 6)(x + 1)(3p + w)(9p^2 - 3pw + w^2)$

Exercise 1 – 10:

1. a) $\frac{a}{5}$
b) $\frac{a+5}{2}$
c) 5
d) a
e) $\frac{3a}{2}$
f) $\frac{a+3}{a+2}$
g) $\frac{a(3b+1)}{b}$
h) $\frac{4xy}{3}$
i) $\frac{p(y-2)}{3y}$
j) $\frac{3x+4}{2}$
k) $-\frac{b+9}{2}$
l) $\frac{t+s}{t-s}$
m) $\frac{x+3}{5}$
n) $\frac{x-3}{x+3}$
o) $\frac{x+2}{x^2+3x+9}$
p) $\frac{a+8}{a^2+2a+4}$
2. a) $\frac{a-6b}{a+2b}$
r) $\frac{2a-3}{b}$
s) $\frac{2x+1}{x(x+1)}$
t) $q+16$
u) $p+5$
v) $h+13$
w) af
a) 5
b) 3
c) 4
d) $\frac{3(a+3)}{98}$
e) $\frac{4a^2(a-5)}{6(a+5)^2}$
f) $\frac{(3x+4)^2}{96p^2}$
g) $\frac{4(3a-1)}{3(a-1)}$
h) $2a$
i) $\frac{3q}{8}$
j) $\frac{30b^3}{4(a+b)}$
3. a) -1
l) $\frac{5(a^2-ab+b^2)}{a^3}$
m) $\frac{1}{a+4}$
n) $\frac{1}{(x-4)(x+2)}$
o) $\frac{2a^2-14a+15}{6}$
p) 1
q) $\frac{14-x}{6}$
r) $\frac{3(p^2-pq+q^2)}{p^2}$
a) $\frac{x-27}{12}$
b) $\frac{47-x}{36}$
c) $\frac{5x-8}{12}$
d) $\frac{19a}{(a+11)(a-8)}$
e) $\frac{6x}{(x-12)(x-6)}$
f) $\frac{20r}{(r+12)(r-8)}$
g) $\frac{2z+4y+3x}{xyz}$
h) $\frac{4t-12}{(t-2)(t-3)}$
- i) $\frac{2(k+2)}{(k^2+2)(k+2)}$
j) $\frac{5t+7}{6q}$
k) $\frac{5p-2}{(p+2)(p-2)^2}$
l) $\frac{2x^2-xy}{(x+y)(x-y)}$
m) $\frac{m+n}{m^2-mn+n^2}$
n) $\frac{f}{(h+f)(h^2+hf+f^2)}$
o) $\frac{2x-1}{6}$
p) 0
q) $\frac{-x^2+3x+1}{(x-1)^2(x^2+x+1)}$
r) $\frac{2t^2+5t-8}{t^2-9}$
s) $\frac{x^2-9x-1}{(x+3)(x+1)(x-2)}$
t) $\frac{a^2+4b-4b^2}{(a-2b)^2(a+2b)}$
4. a) $x \neq 2$
b) $x \neq -1$
c) $x \neq 0$ and $x \neq \pm 1$

Exercise 1 – 11:

1. a) between the rectangle and $\mathbb{Z}$
b) (i)
2. a) non-real
b) undefined
c) real
d) real
e) non-real
f) real
3. a) Irrational
b) Irrational
c) Rational
d) Rational
4. a) rational
b) rational
c) irrational
d) irrational
5. a) $\sqrt{-24}$
b) $-\sqrt{24}$; $\frac{3}{2}$; $\frac{\pi}{2}$; $\sqrt[3]{26}$; π ; $\sqrt{39}$; $7,ii$; $7,12$; $\sqrt{78}$; 9 ; 3π ; π^2
c) $-\sqrt{24}$; $\frac{\pi}{2}$; $\sqrt[3]{26}$; π ; $\sqrt{39}$; $\sqrt{78}$; 3π ; π^2
d) $\frac{3}{2}$; $7,ii$; $7,12$; 9
e) 9
f) $\frac{\sqrt{2}}{0}$
6. a) $\frac{3}{25}$
b) $\frac{3}{500}$
c) $\frac{410}{99}$
d) $1\frac{59}{100}$
e) $\frac{221}{18}$
f) $\frac{37}{45}$
g) $\frac{81}{11}$
8. a) 0,05
b) 1,5
9. $\frac{78}{99}$
10. a) (i) any three numbers (ii) irrational
b) (i) 111 (ii) rational
11. a) 0,50
b) 1,00
c) 0,11
- d) 1,00
e) 1,618
f) 1,414
g) 2,718
12. a) 3,142
b) 1,618
c) 1,414
d) 2,718
13. 1523,0020
14. 1982,940290
15. 101,5238
16. a) 1,414
b) 1,732
c) 2,236
d) 2,449
17. a) Irrational number.
b) Irrational number.
c) Rational number.
d) Rational number.
e) Rational number.
f) Rational number.
g) Irrational number.
h) Rational number.
i) Rational number.
j) Irrational number.
18. a) 0,71
b) 3,742
20. a) 2 and 3
b) 3 and 4
c) 4 and 5
d) 5 and 6
e) 1 and 2
f) 2 and 3
g) 2 and 3
h) 3 and 4
i) 9 and 10
j) 8 and 9
k) 3 and 4
l) 4 and 5
21. a) 3,7
b) 10,5
c) 6,9
d) 4,5
22. a) $a^2 + 10a + 25$
b) $n^2 + 24n + 144$
c) $d^2 - 8d + 16$
d) $49w^2 - 4$
e) $144q^2 - 1$
f) $x^2 + 4x + 4$
g) $25k^2 - 16$
h) $10f^2 + 18f + 8$
i) $18n^2 + 51n + 30$
j) $2g^2 + 18g + 36$
k) $16y^2 + 36y + 8$
l) $7d^2 - 19d - 6$
m) $6z^2 - 16z + 8$
n) $25w^2 - 110w + 121$
o) $25s^2 - 10s + 1$
p) $9d^2 - 48d + 64$
q) $36f^3 + 25f^2 + 49f$
r) $74d^4 + 16d + 24d^2$
s) $59x^3 + 10x^2 + 49x$
23. a) $y^6 - y^5 + y^4 - 2y^3 - 7y^2 - 2y$
b) $4x$
c) $x^4 - 2x^2 + 1$
d) $64a^3 - 27b^3$
e) $2x^3 + 4x^2y - 8xy^2 - 6y^3$
f) $9a^4 + 9a^3 - 34a^2b^2 + 25ab^3 - 25b^4$
g) $y^2 - \frac{1}{y^2}$
h) $\frac{a^2}{9} - \frac{3}{a^2}$
i) $6x$
j) $-4x - 8$
24. 4
25. a) -1 ; 0 ; 1 ; 6
b) 0 ; 1 ; 6
c) $k > 0$
d) $k > -2$
26. a) $9a^2 + 3 + \frac{1}{4a^2}$
b) $27a^3 - \frac{1}{8a^3}$
c) $374\frac{1}{2}$
27. a) 64

28. b) 25
c) 2400800
d) 100
a) 11×13
b) $2^3 \times 3 \times 7$
c) 29×31
d) $3^2 \times 11$
e) $3 \times 13 \times 41$
29. a) $(a-3)(a+3)$
b) $9(b-3)(b+3)$
c) $(m-\frac{1}{3})(m+\frac{1}{3})$
d) $5(1-ab^3)(1+ab^3)$
e) $b(2a-3)(2a+3)(4a^2+9)$
f) $(a-5)(a+5)$
g) $(4b+7)(4b+7)$
h) $-4b^2(6b^3-1)^2$
i) $(4+x^2)(2+x)(2-x)$
j) $7(x-2)(x+y)$
k) $(y-10)(y+3)$
l) $(1-x)^2(1+x)$
m) $(1+p)(-2+3p)$
n) $x(x-2)+(1+y)(1-y)(1+y^2)$
o) $(x-1)(x^2+x+1)(4b-x)$
p) $(v-7)(3m+19)$
q) $(3f+19)(z+3)$
r) $3(p-\frac{1}{3})(p^2+\frac{p}{3}+\frac{1}{9})$
s) $(2x^2-5y^3)(4x^4+10x^2y^3+25y^6)$
t) $(-p)(12+18p+7p^2)$
u) $\frac{(a+3b)^2(a-3b)}{3}$
v) $(2a-5)(3a-1)$
w) $(s-3)(s+5)$
x) $(2v+3h)(8+j^5)$
30. y) $(2h-5g)(9+m^3)$
z) $(7d-2s)(9+u^2)$
a) $2(a+1)(3a+4)$
b) $3(g-3)(2g+1)$
c) $(5g-r)(25g^2+5gr+r^2)$
d) $(2r+z)(4r^2-2rz+z^2)$
e) $(7m-2n)(2+j)$
f) $(5d-3m)(5+y)$
g) $(g-3)(g^2+3g+9)$
h) $(z+5)(z^2-5z+25)$
i) $3(b-a)(3a-b)$
j) $(4x+5y)(y-4x)$
k) $4(4x^3-3y^4)(4x^3+3y^4)$
l) $\frac{1}{6}(a-12b^2)(a+12b^2)$
m) $(a-3)(2-9x)(2+9x)$
n) $(b+3)(b-10)$
o) $(2x+5y)(x+y)$
p) $(x-5y)(x+3y)$
q) $(4x^2+3)(x^2+2)$
r) $2(3x^2-4)(x^2-5)$
s) $(3a+b)(3a-b)(x+y+3)$
t) $2(2y+3)(y-4)$
u) $\frac{(x-4)(x-3)(x+3)}{2}$
v) $(3rs-1)(9r^2s^2+3rs+1)$
w) $(\frac{1}{5h}+r)(\frac{1}{25h^2}-\frac{r}{5h}+r^2)$
x) $(j+k)(4n-b)(16n^2+4nb+b^2)$
31. a) $-8a+4$
b) $125a^3-64b^3$
c) $16m^4-81$
d) $a^2+4ab+4b^2-c^2$
e) 2
- f) 5
g) $-\frac{2+b}{3}$
h) $\frac{1}{x-2}$
i) $\frac{x-7}{3}$
j) 5
k) $\frac{a-5}{(a+2)^2}$
l) $\frac{1}{a^2+6a+36}$
m) $-2(2a+b)$
n) $s+31$
o) $n+8$
p) $p^2-2pq+q^2$
q) $\frac{12-x^2}{6x}$
r) $\frac{-14}{(a+7)(a-7)}$
s) $\frac{32x^3+x+2}{2x^3}$
t) $\frac{4a-1}{(2a+1)(2a-1)(a-1)}$
u) $\frac{5x+20}{6}$
v) $\frac{4x^3+11x^2+7x+3}{x(x+2)(x+3)}$
w) $\frac{2(b^2+9)}{(b-3)(b+3)}$
x) $\frac{x(x+1)}{x^2+x+6}$
y) $\frac{17z}{(z+12)(z-5)}$
z) $\frac{7w}{(w-11)(w-4)}$
32. $(3x-4)(x+2)$
33. $5x$
34. 8,85
35. $a^2+2ab+4b^2$
36. $9x^2-3x+1$
37. a) $x \neq \frac{1}{3}$ and $x \neq -1$
b) $a \neq b$ and $a \neq -3$

2 Exponents

Exercise 2 – 1:

1. 1
2. 16
3. 11^{11x}
4. 10^{8x}
5. $216c^3$
6. $125n^3$
7. $\frac{1}{36}$
8. 40
9. $\frac{27}{8}$
10. a^3
11. $\frac{1}{x^3y^4}$
12. x^{3t+3}
13. 3^{2a+3}
14. 1
15. 2
16. $6a^5b^2$
17. $56m^{10}n^9$
18. $-72a^{16}b^{16}$
19. $-\frac{1}{5}x^{14}y^{19}$
20. a^{2x}
21. $5ax$
22. $2c^4p^3$
23. $3a^5m^5$
24. 27
25. $7a^2$
26. $9a^5b^{27}$
27. $\frac{1}{9}$
28. $\frac{a^{30}}{b^{35}}$
29. $8t^{12}$
30. 3^{2n+6}
31. $\frac{1}{27}$
32. 85
33. 3^{9x+3}
34. 4^x+12^b
35. $5^{7y-8} \times 2^{9y-1}$
36. $\frac{2^{17}}{3^25^3}$
37. 12
38. $\frac{51}{49}$
39. $\frac{4^y-2^{5y}}{3}$

Exercise 2 – 2:

1. $3t^2$
2. $8x$
3. $\frac{1}{2}$
4. $\frac{1}{3}$
5. $3p^3$
6. $96a^3b^5$
7. $4a^3b$
8. $\frac{b^3}{a}$
9. $2 \cdot 2^{\frac{1}{3}}a^4b^2$

Exercise 2 – 3:

1. a) 0
b) $-\frac{5}{2}$
c) -7
d) $\frac{5}{9}$
- e) 6
f) -2
g) 4
h) $-\frac{1}{2}$
i) $\frac{4}{5}$
- j) 3
k) $x=2$ or $x=3$
l) $x=3$ or $x=1$
m) $x=0$ or $x=4$
n) 1
- o) $\frac{3}{2}$ or 0
p) 1
q) $-\frac{7}{2}$
r) -3
s) -3

- t) 3 or -1
u) 3

- v) -2
2. 7

3. $x \approx 2,81$
4. $x \approx 1,49$

Exercise 2 – 4:

1. a) $512x^3$
b) $2t^3$
c) $5^{5x+y+3z}$
d) 15^{15x}
e) 7
f) $21d^7$
g) $-\frac{18}{7}a^{15}b^{12}$
h) b^{k^2+k}
i) $4c^6m^2$
j) 2
k) $\frac{1}{7a^{10}b}$
l) $\frac{a^{14}}{b^8}$
m) $2^{5p} \cdot 3^{3p}$
n) $27m^t$
o) $\frac{1}{3x^5}$
p) $\frac{1}{625}$
q) $\frac{27}{4}$
r) $\frac{1}{2^7}$
s) $\frac{1}{2^7}$
t) $\frac{y^2}{x^5}$
u) -8
v) $8x^{6a}y^{3b}$
w) 4^{x+3}
x) 22
2. a) $-\frac{1}{3}$
b) $\frac{1}{8}$
c) 14
d) $2^{2p} - 2^p + 1$
e) a^5b^3
f) $3x^4y^2$
g) 34
h) 3^{20z+2}
i) $11^b - 4^p$
j) $11^{2c-2} \times 2^{14c-4}$
k) $\frac{3^4}{2^{13} \times 5}$
l) $\frac{9}{50}$
m) $108a^6b^9$
n) $22p^{10}q^6$
o) $-(ab)^{-1}$
p) x^6
q) $\left(\frac{2}{3}\right)^{2y}$
r) 4
s) -3
t) 3
3. a) 1
b) $\frac{2}{3}$
c) -9
d) -5
e) $\frac{1}{2}$
f) -1
g) 16
h) $\frac{1}{16}$
i) -1
j) 1 or 16
k) 2
l) $k = \frac{1}{81}$
m) $x = 81$
n) $\frac{1}{2}$
o) $x = 3$ or $x = 2$
p) $x = 2$
q) $x = 1$ or $x = -1$
r) $x = -8$
s) $x = 1$
t) $x = -1$
u) $x = -\frac{1}{3}$ or $x = -1$
v) no solution
w) $x \approx 2,73$
x) $x \approx 3,10$
4. a) The sum of two powers of the same degree is not the power of the sum of the bases
b) The sum of two powers of the same degree is not the power of the sum of the bases
c) A negative sign is missing, when a power is moved from the denominator to the numerator, the sign of the exponent changes.
d) We cannot multiply bases unless they are raised to the same power
e) The sign of a base is not changed when an exponent is moved from the denominator to the numerator in a fraction
f) The power of a product is the product of all the bases raised to the same power
5. 25×10^{2013} 2015 digits.

3 Number patterns

Exercise 3 – 1:

2. $T_{n-1} = -1$
3. $T_{n-4} = C$
4. a) no common difference
b) $d = 7$
c) no common difference
d) $d = -0,65$
5. a) 35, 45 and 55
b) 7, 12 and 17
c) 21, 18 and 15
d) $T_4 = -28,1; T_5 = -33,1; T_6 = -38,1$
e) $T_4 = -39x; T_5 = -49x; T_6 = -59x$
f) $T_4 = 44,2; T_5 = 64,2; T_6 = 84,2$
6. $T_6 = 28$ and $T_9 = 43$
7. $T_5 = G$ and $T_8 = J$
8. $T_5 = 23$ and $T_8 = 35$
9. a) 8
b) -1
c) -9 and -5
d) 28
e) 27
11. a) 13 triangles
b) $2n + 3$
c) 53
12. a) 47 ; 55 ; 63
b) $8n + 7$
13. a) 76 ; 106 ; 136
b) $30n - 74$
c) = 16
14. a) No common difference
b) $T_1 = -3$ and $T_2 = 3$
15. a) $d = -2n - 3$
b) $T_1 = 2$ and $T_3 = 0$
16. a) $-\frac{5}{3}$
b) $-\frac{14}{9}, \frac{43}{9}$ and $\frac{100}{9}$
17. a) $-\frac{1}{2}$
b) -2, -3 and -4
18. E

Exercise 3 – 2:

2. $d = -3$
3. no common difference
4. $T_{n-3} = 7$
5. a) $T_4 = -67,2$; $T_5 = -86,2$; $T_6 = -105,2$
b) $T_4 = 38r$; $T_5 = 34r$; $T_6 = 30r$
6. $T_6 = 31$ and $T_8 = 41$
7. $T_6 = F$ and $T_{10} = J$
8. a) $T_6 = 49$
b) $T_6 = -11$
c) $T_6 = 18,9$
9. a) $T_n = -4n - 14, T_{10} = -54, T_{15} = -74, T_{30} = -134$
b) $T_n = -7n + 8, T_{10} = -62, T_{15} = -97, T_{30} = -202$
10. a) 12 and 16
b) -10
c) 23 and 53
11. a) $T_n = 4n - 1$
b) $T_n = 3n - 5$
c) $T_n = 4n + 7$
d) $T_n = \frac{1}{3}n$
12. a) -49 ; -63 ; 77
b) $-14n + 7$
c) = 66
13. M
14. C
15. 111 seats
16. a) 17 boxes

- b) $T_n = 3n - 1$
 c) 89
 17. a) $T_1 = 4$
 b) 3
 c) $T_n = 3n + 1$
 d) 76 matchsticks
 18. 10
19. a) $d = 3y + 3$
 b) $T_1 = -7$ and $T_2 = -1$
 20. a) $-\frac{1}{3}$
 b) $-\frac{1}{6}, \frac{3}{2}$ and $\frac{19}{6}$
 21. $T_{85} = 7227$ blocks
 22. a) 29
- b) $T_n = (n + 1)^2 + n$
 c) $T_{14} = 239$ blocks
 23. 77 parts
 24. a) 7
 b) 0
 c) 4

4 Equations and inequalities

Exercise 4 – 1:

- | | | | | | |
|-------------------|----------------------|---------------------|----------------------|-----------------------|----------------------|
| 1. 5 | 7. 12 | 13. 1 | 19. 5 | 25. 10 | 31. $\frac{223}{45}$ |
| 2. -8 | 8. 11 | 14. 3 | 20. 4 | 26. $-\frac{22}{6}$ | 32. 12 |
| 3. -1 | 9. $\frac{217}{20}$ | 15. 6 | 21. $-\frac{36}{11}$ | 27. 6 | 33. $-\frac{17}{13}$ |
| 4. -3 | 10. 15 | 16. $\frac{17}{12}$ | 22. 8 | 28. -2 | 34. -19 |
| 5. 0 | 11. $\frac{9}{10}$ | 17. $\frac{20}{3}$ | 23. 5 | 29. -12,5 | 35. $-\frac{1}{6}$ |
| 6. $-\frac{7}{8}$ | 12. $-\frac{50}{13}$ | 18. 26 | 24. 24 | 30. $\frac{280}{451}$ | |

Exercise 4 – 2:

- | | | |
|--|--|---|
| 1. a) $5r^2 + 16r = 0$
b) $6r^2 - 25r + 39 = 0$
c) $2d^2 + 15d + 17 = 0$ | l) $x = -\frac{3}{2}$ or $x = 4$
m) $x = \frac{6}{5}$ or $x = \frac{8}{3}$
n) $x = 4$ or $x = 11$
o) 2
p) $x = \frac{3}{5}$ or $x = 3$
q) $t = 0$ or $t = 3$
r) 5
s) $x = \sqrt{18}$ or $x = -\sqrt{18}$
t) $p = 7$ or $p = -1$
u) $x = -\frac{11}{4}$ or $x = 7$
v) $x = -\frac{6}{7}$ or $x = \frac{1}{2}$
w) $x = 3$ or $x = -2$
x) $a = -\frac{1}{2}$ or $a = 3,5$
y) $x = 11$ or $x = 1$ | 3. a) $y = 3$ or $y = -3$
b) $z = -\frac{1}{5}$ or $z = \frac{1}{2}$
c) $x = 3$ or $x = -6$
d) $y = \frac{1}{4}$ or $y = 4$
e) $b = -\frac{1}{3}b$ or $b = 4$
f) $y = -\frac{4}{3}$ or $y = 1$
g) $x = 4$ or $x = -4$
h) $z = 1$ or $z = -1$
i) $b = \pm 2$ or $b = \pm 3$
j) $a = -\frac{43}{8}$ or $a = 1$
k) $x = 3$
l) $x = 4$
m) $a = -2$
n) $a = 0$ or $a = \frac{2}{3}$ |
| 2. a) $x = -5$ or $x = 3$
b) $p = -2$ or $p = 9$
c) $x = -\frac{2}{3}$ or $x = \frac{4}{3}$
d) $x = \frac{9}{5}$ or $x = -6$
e) $z = -2$ or $z = -1$
f) $b = 3$ or $b = 4$
g) $a = 3$ or $a = 6$
h) $y = \frac{3}{2}$ or $y = -\frac{3}{2}$
i) $x = \frac{1}{2}$ or $x = -\frac{9}{2}$
j) $\frac{3}{2}$
k) $m = 0$ or $m = -\frac{4}{5}$ | | |

Exercise 4 – 3:

- | | | |
|--|--|---|
| 1. $x = -2$ and $y = -3$
2. there is no solution to x and y
3. $x = 2$ and $y = -3$
4. a) $x = \frac{1}{10}$ and $y = -\frac{43}{50}$
b) $x = -7$ and $y = -\frac{3}{2}$
c) $x = 5$ and $y = 3$
d) $x = -1$ and $y = -1$
e) $x = 13$ and $y = -1$ | f) $x = 11$ and $y = 5$
g) $x = 9$ and $y = 2$
h) $a = \frac{16}{3}$ and $b = \frac{4}{3}$
i) $x = -\frac{1}{10}$ and $y = -2$
j) $x = -\frac{7}{5}$ and $y = \frac{8}{5}$
k) $x = \frac{1}{7}$ and $y = -\frac{1}{4}$
l) x can be any real number, $\frac{1}{2} \leq$ | $y < 2$
m) a and b can be any real number except for 0.
5. a) $x = -1$ and $y = 2$
b) $x = 9$ and $y = -4$
c) $x = -1$ and $y = -4$
d) $x = 2$ and $y = 1$
e) $x = 1,5$ and $y = 3,5$ |
|--|--|---|

Exercise 4 – 4:

- | | | |
|--|---|---|
| 1. 2 hours
2. 1 hour
3. Zwelibanzi achieved 80 marks and Jessica achieved 68 marks.
4. 18 large shirts and 2 small shirts
6. -34
7. a milkshake costs R 34 and a wrap costs R 27. | 8. 30° and 60°
9. $b = 8$ cm and $l = 2b = 16$ cm
10. 7 or -3
11. length: 6 cm, width: 4 cm
12. $\frac{19}{50}$ litres
13. 9 and 11
14. $\frac{1}{2}$ | 15. 8 years old
16. 7 and 35 years old.
17. $x = -\frac{1}{2}$
18. $x = 2$ or $x = -3$
19. 15
20. 34
21. 48 blue beads, 96 red beads and 36 purple beads, |
|--|---|---|

Exercise 4 – 5:

- | | | |
|---|---|---|
| 1. $x = 1 - 2y$
2. $\frac{2(s-ut)}{t^2} = a$
3. $\frac{pV}{RT} = n$ | 4. $x = \frac{-2b^2}{1-2b}$
5. $\pm \sqrt{\frac{V}{\pi h}} = r$
6. $\frac{E\lambda}{c} = h$ | 7. $\frac{A-2\pi r}{2\pi r} = h$
8. $\lambda = \frac{D}{tf}$
9. $\frac{E}{gh + \frac{1}{2}v^2} = m$ |
|---|---|---|

$$10. x = -a \text{ or } x = -b$$

$$11. b = \pm \sqrt{c^2 - a^2}$$

$$12. U = \frac{VW}{W-V}$$

$$13. r = \pm \sqrt{\frac{A - \pi R^2}{\pi}}$$

$$14. C = \frac{5}{9} (F - 32^\circ)$$

$$15. r = \sqrt[3]{\frac{3V}{\pi}}$$

$$16. x = a + 4 \text{ or } x = -1$$

$$17. x = 2 \text{ or } x = -2$$

$$18. x = 3,91$$

$$19. u = 2,6$$

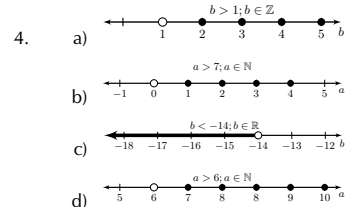
$$20. h = \pm 16$$

$$21. 6$$

Exercise 4 – 6:

1. a) $x < -1$ and $x \geq 6; x \in \mathbb{R}$
b) $3 < x < 6; x \in \mathbb{R}$
c) $x \neq 3; x \neq 6; x \in \mathbb{R}$
d) $x > -10; x \in \mathbb{R}$
- 2.
3. a) $(-34; \infty)$
b) $(-\infty; 5)$

- c) $x \in [\frac{29}{13}; \infty)$
d) $x \in (-\infty; \frac{6}{5}]$
e) $(-\infty; -\frac{55}{13})$
f) $(-\infty; -\frac{21}{11}]$
g) $(-\infty; -\frac{3}{2}) \cup (\frac{1}{2}; \infty)$
h) $[-\frac{5}{3}; -\frac{1}{2}]$



Exercise 4 – 7:

1. a) 1
b) -1
c) 1
d) -8
e) $\frac{1}{2}$
f) $\frac{11}{54}$
g) -2
h) -2
i) 10
j) $-\frac{8}{3}$
k) 42
l) 5
m) $\frac{1}{3}$
2. a) $b = -9$ or $b = 3$
b) $x = -1$ or $x = 6$
c) $b = -5$ or $b = 2$
d) $\frac{9}{7}$
e) $x = -\frac{1}{5}$ or $x = 3$
f) 1
g) -32
h) -23
i) $x = 2$ or $x = 1$
j) $y = -3$ or $y = 2$
k) $x = -\frac{9}{2}$ or $x = 2$
l) $d = 4$

$$3. x = -1 \text{ and } y = -1$$

4. there is no solution to x and y .

$$5. x = -1 \text{ and } y = 3$$

6. a) $x = 1$ and $y = 2$
b) $x = 4$ and $y = 2$
c) $x = 5$ and $y = -2$
d) $x = 6$ and $y = -5$
e) $x = -6$ and $y = 4$
f) $x = 3$ and $y = 6$
g) $x = -7$ and $y = 5$
h) $x = 14$ and $y = 2$
i) $x = -8$ and $y = -5$
j) $x = 1$ and $y = -1$
k) $x = \frac{5}{12}$ and $y = \frac{31}{12}$
l) $x = -\frac{31}{38}$ and $y = \frac{41}{38}$

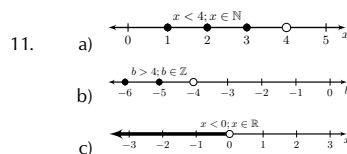
m) no solution

n) a and b can be any real number except for 0

o) x and y can be any real number except for 0

$$p) x = 48 \text{ and } y = 7$$

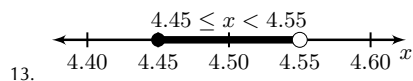
7. a) $\frac{120}{13}$
b) each ruler costs R 5 and each pen costs R 3
c) The price of the hotdog is R 31 while a milkshake costs R 25.
d) Lefu has 77 marks and Monique has 89 marks
e) 0,56 km
f) The trucks will meet after 1
g) 8 km
h) R 20
i) 33
j) 12
k) $x = 5$
8. a) $x = \frac{a-c}{b}, b \neq 0$
b) $\frac{P}{V} = I$
c) $\frac{E}{c^2} = m$
d) $\frac{v-u}{a} = t$
e) $f = \frac{uv}{v+u}$
f) $mx + c = y$
g) If $(a+b+c) \neq 0$ then $x = 4 - b$ If $a+b+c = 0, x \in \mathbb{R}$
h) $r = \frac{13}{15}$
i) $b = \pm 8$
9. a) $x < -1$ and $x \geq 4; x \in \mathbb{R}$
b) $x \geq -2; x \in \mathbb{R}$
c) $-1 < x \leq -2; x \in \mathbb{R}$
10. a) $(-\infty; -\frac{29}{2})$
b) $x \in (-\infty; -\frac{8}{5}]$
c) $[\frac{80}{31}; \infty)$
d) $(-\infty; \frac{1}{3}) \cup (\frac{17}{3}; \infty)$



12. a) $x = -\frac{317}{76}$
b) $x = -2$ or $x = 10$
c) $a = 5$ or $a = -11$
d) $x = 2$ or $x = -2$
e) $a = 4$ or $a = -3$
f) $a = 2$ or $a = -3$
g) $a = 2$ or $a = -9$
h) $b = \pm\sqrt{3}$ or $b = \pm 1$
i) $y = \pm\frac{2}{3}$ or $y = \pm 1$

- j) $b = 4$
 k) $a = 1$
 l) $x \geq 2$
 m) $x < -1$
 n) $x > \frac{588}{10}$
 o) $a \leq -7$
 p) $-3 \leq k < 2$
 q) $x = 7$ or $x = -6$
 r) $x = -1$ or $x = -2$
 s) $x = \frac{3}{8}$

- t) $x = \frac{a}{(a-b)}$ for $a, b \neq 0$ and $a \neq b$
 u) $x = -\frac{2}{3}y$ or $x = y$
 v) $x = \frac{1}{2}y$ or $x = -1$
 w) no solution since $x \neq 4$
 x) no solution since $\sqrt{-1}$
 y) $x > -13$



5 Trigonometry

Exercise 5 – 1:

3. a) $\cos \hat{O} = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{m}{o}$
 b) $\tan \hat{M} = \frac{\text{opposite}}{\text{adjacent}} = \frac{m}{o}$
 c) $\sin \hat{O} = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{o}{n}$
 d) $\cos \hat{M} = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{o}{n}$
 5. $\cos \hat{P} = \frac{r}{q}$
 6. $\sin \alpha = \frac{4}{5}$

Exercise 5 – 2:

1. a) 2,14 g) -0,82 m) 1,56 s) 0,67 x) 0,88 c) true
 b) 0,62 h) -1,48 n) -1,19 t) 5,01 y) non-real d) false
 c) 0,28 i) 2,37 o) -1,59 u) 0,91 z) 0,21 3. 71,57
 d) 0,21 j) 1,02 p) -1,39 v) undefined 2. a) true
 e) 0,90 k) 1,49 q) 0,23 w) 2,13 b) false
 f) 1,15 l) 1,60 r) 2,52

Exercise 5 – 3:

1. a) $\frac{1}{\sqrt{2}}$ d) $\frac{\sqrt{3}}{1}$ h) $\frac{1}{2}$ b) $\frac{3}{2}$ b) $\frac{-1}{2\sqrt{3}}$ f) $\frac{-\sqrt{3}}{2}$
 b) $\frac{1}{\sqrt{2}}$ e) $\frac{1}{\sqrt{2}}$ 2. $\frac{1}{\sqrt{2}}$ c) $\frac{\sqrt{3}-1}{2}$ c) -1 g) $\frac{-\sqrt{3}}{2}$
 c) $\frac{1}{\sqrt{3}}$ f) $\frac{1}{\sqrt{3}}$ 3. $\frac{1}{\sqrt{3}}$ 5. a) $\frac{2}{\sqrt{3}}$ d) $\frac{\sqrt{2}}{1}$ 6. 7.
 g) $\frac{1}{\sqrt{3}}$ 4. a) $\frac{1}{2}$

Exercise 5 – 4:

1. a) 36,11 e) 3,51 i) 2,87 c) $\frac{AC}{BC} = \frac{AD}{AB}$
 b) 8,91 f) 33,43 j) 9,06 3. $MN = 12,86$ and $NP = 15,32$
 c) 10,90 g) 29,46 2. a) $\frac{AC}{AB} = \frac{AD}{BD}$ 4. $x = 29,82$ and $y = 31,98$
 d) 21,65 h) 10 b) $\frac{AD}{BD} = \frac{CD}{AD}$

Exercise 5 – 5:

1. 23,96° 2. 35,23° 3. 39,40° 4. 26,31° 5. 36,87° 6. 45° 7. 30°

Exercise 5 – 6:

1. a) 59,5° i) 18,1° 2. a) -0,342 3. a) 54,49°
 b) 53,1° j) 40,5° b) 0,827 b) 90°
 c) 71,3° k) no solution c) -0,440 c) 36,12°
 d) 76,6° l) 18,4° d) 1,770 d) 63,07°
 e) 80,1° m) no solution e) 0,242 e) -18,97°
 f) 41,8° n) 109,9° f) -2,924 f) 43°
 g) no solution o) 26,6° g) 1,614
 h) 41,4° p) 17,7° h) 0,625

Exercise 5 – 7:

1. a) $\sqrt{10}$ d) -3 3. a) $\frac{\sqrt{t^2+4}}{2}$
 b) $\frac{1}{\sqrt{10}}$ 2. a) $\frac{-\sqrt{21}}{5}$ b) $\frac{2}{t}$
 c) $\frac{\sqrt{10}}{-3}$ b) -2 c) $\frac{4}{t^2+4}$

4. a) $-\frac{4}{5}$
b) $-\frac{82}{25}$
5. a) $-\frac{8}{17}$
b) $\frac{15}{8}$
c) 1
6. $\frac{7}{13}$

7. a) $\frac{8}{15}$
b) -45
8. a) $\sqrt{89}$
b) $\frac{8}{\sqrt{89}}$
c) $\frac{5}{\sqrt{89}}$
d) $\frac{8}{5}$
e) $\frac{\sqrt{89}}{8}$

- f) $\frac{\sqrt{89}}{5}$
g) $\frac{5}{8}$
h) 1
9. a) $(-24; -7)$ and $(-48; -14)$
b) $a = 4(-24) = -96$ and $b = 4(-7) = -28$
c) $-\frac{7}{25}$
10. -13

Exercise 5 – 8:

1. a) not been written correctly
b) been written correctly
c) not been written correctly
2. a) 5,67
b) 0,29
c) 0,29
d) $-1,07$
e) $-0,74$
f) 1,79
g) 3,08
h) 1,66
i) $-4,45$
j) 1,79
k) 2,33
l) 7,73
m) $-0,23$
n) 0,99
o) 1,31
p) 0,05
q) 10,46
r) 2,12
3. a) $\frac{\sqrt{3}}{2}$
b) $\frac{1}{2}$
c) $\sqrt{3}$
d) $\frac{1}{2}$
e) $\frac{\sqrt{3}}{2}$
f) $\frac{1}{\sqrt{3}}$

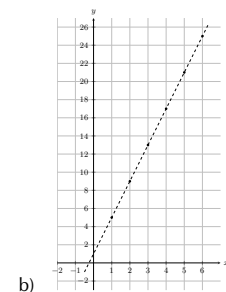
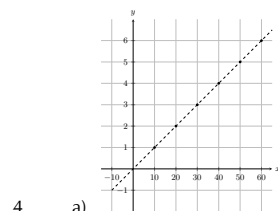
4. a) $\frac{1}{\sqrt{2}}$
b) $\frac{1}{\sqrt{2}}$
c) 1
5. a) $-\frac{\sqrt{3}}{2}$
b) $-\frac{1}{2\sqrt{3}}$
c) $-\frac{\sqrt{3}}{2}$
d) $\frac{1}{8}$
e) $\frac{\sqrt{3}}{\sqrt{2}}$
f) $\frac{\sqrt{3}}{2\sqrt{2}}$
g) $\frac{\sqrt{3}}{2}$
h) $\frac{3}{8}$
6. $\frac{3}{2}$
7. $\frac{3}{2\sqrt{3}}$
8. $\sqrt{3}$
9. 9,96 mm and 8,35 mm
10. a) 35°
b) 4
c) 7
d) 6
e) 3
f) 45°
g) 55°
h) 65°
i) 5
12. a) 17,32 cm

- b) 10 cm
c) $25,08^\circ$
13. $44,44^\circ$
14. 11,88
15. a) $\sqrt{55}$
b) $\frac{3}{\sqrt{55}}$
c) $\theta = 67,976^\circ$
16. 27,46 cm
17. a) $42,07^\circ$
b) $63,43^\circ$
c) 25°
d) $56,25^\circ$
e) $14,32^\circ$
f) 30°
g) 59,26
h) 45°
18. a) 4,01
b) 6,39
c) 0,91
d) 1,85
19. a) $\frac{3}{\sqrt{34}}$
b) -1
20. $33,69^\circ$
21. $23,96^\circ$
22. $97,12^\circ$
23. 8,7 cm, 5,65 cm and 5,65 cm
24. Area of $\triangle ABC = 16944$ units²

6 Functions

Exercise 6 – 1:

1. a) $\{x : x \in \mathbb{R}, x \leq 7\}$
b) $\{x : x \in \mathbb{R}, -13 \leq x < 4\}$
c) $\{x : x \in \mathbb{R}, x > 35\}$
d) $\{x : x \in \mathbb{R}, \frac{3}{4} \leq x < 21\}$
e) $\{x : x \in \mathbb{R}, -\frac{1}{2} \leq x \leq \frac{1}{2}\}$
f) $\{x : x \in \mathbb{R}, x > -\sqrt{3}\}$
2. a) $(-\infty; 6]$
b) $(-5; 5)$
c) $(\frac{1}{5}; \infty)$
d) $[21; 41)$
3. a) $y = 5x$
b) $y = 5$
c) $y = \frac{1}{2}x$



6. a) 2

- b) -11
c) -2
d) 3
e) 8
f) 1
g) 10
h) 3
7. a) R 108,88
b) R 199,36
- c) 22,043 L
d) 22,071 L
e) 1,15 V
8. a) 7 m
b) $0 \text{ m} \leq s(t) \leq 10 \text{ m}$
c) The domain is $0 \text{ s} \leq t \leq 20 \text{ s}$. It represents the total time taken to reach the bottom of the slope.
9. a) 5 m
b) 10 m
c) 5 m

Exercise 6 – 2:

1. a) x -intercept = 1 and y -intercept = -1
b) x -intercept = -2 and y -intercept = 2
c) x -intercept = 3 and y -intercept = -3
2. $m = 3$ and $c = -1$.
3. $m = 1$ and $c = -1$.
4. a) (0; 1) and (-1; 0). The graph is increasing
b) (0; -1) and (1; 0). The graph is increasing
c) (0; -1) and ($\frac{1}{2}$; 0). The graph is increasing
d) (0; 1) and ($\frac{1}{3}$; 0). The graph is decreasing
e) (0; 2) and (-3; 0). The graph is increasing
f) (0; 3). The graph is horizontal.
g) (0; 0). The graph is increasing
h) (0; -3) and (2; 0). The graph is increasing
5. a) False
b) True
c) True
- d) False
e) False
6. a) $y = -\frac{3}{2}x + \frac{1}{2}$
b) $y = -3x + 5$
c) $y = \frac{1}{3}x + \frac{4}{3}$
d) $y = -2x + 4$
7. a) E
b) A
c) B
d) F
e) D
f) C
8. a) $a(x) = -\frac{3}{4}x + 3$
b) $b(x) = \frac{3}{2}x - 6$
c) $c(x) = 3$
d) $d(x) = -\frac{3}{4}x$

Exercise 6 – 3:

1. $a = -\frac{1}{3}$; $q = 4$
2. $a = 2$; $q = -3$
3. a) -2
b) (-0,63; 0) and (0,63; 0)
4. a) 1
b) (-0,71; 0) and (0,71; 0)
5. a) $h(x)$
b) $g(x)$
c) $f(x)$
d) $k(x)$
6. a) $h(x)$
b) $f(x)$
- c) $g(x)$
7. a) $a = 1$; $p = -9$
b) $b = -1$; $q = 23$
c) $x \leq -4$ or $x \geq 4$
d) $x \geq 0$

Exercise 6 – 4:

1. $a = -1$ and $q = 2$
2. $a = -2$ and $q = 3$
3. a) There is no y -intercept.
b) $-\frac{3}{2}$
4. a) There is no y -intercept.
5. a) $h(x)$
b) $g(x)$
c) $k(x)$
d) $f(x)$
6. b) the point does lie on the graph
c) -24
e) $y = 0$ and $x = 0$
f) (-3; 2)

Exercise 6 – 5:

1. a) (0; 0,33)
b) (0,4; 0)
2. $a = -1$ and $q = 4$
3. $a = -1$ and $q = 5$
4. a) (0; -0,75)
b) (1; 0)
5. a) $g(x)$
- b) $h(x) = -3^x$
c) $(-\infty; 0)$
d) $g(x) = 3^{-x}$
e) $j(x) = 2 \cdot 3^x$
f) $k(x) = 3^x - 3$
6. b) asymptote
c) $y = (\frac{1}{2})^x$
d) (0; 1)
7. a) $f(x) = 3^x$

Exercise 6 – 6:

1. $a = \frac{3}{2}$ and $q = \frac{3}{2}$
2. $a = 4$ and $q = -2$
3. $a = \frac{1}{2}$, and $q = -1$
4. $a = \frac{1}{2}$, and $q = 0$
5. $a = 3$ and $q = \frac{1}{3}$
6. $a = 1$ and $q = 0$
7. a) $h(x)$
b) $k(x)$
c) $f(x)$
d) $g(x)$
8. $g(x) = 4 \sin \theta$
9. a) $E(180^\circ; -2)$ and $-2 \leq y \leq 0$
b) $E(360^\circ; 2)$ and $0 \leq y \leq 4$
c) $E(90^\circ; -0,5)$ and $-2,5 \leq y \leq -0,5$
d) $E(180^\circ; 0,5)$ and $0,5 \leq y \leq$

- 4,5
11. $E(180^\circ; 1)$, range $y \in \mathbb{R}$ and domain $0 \leq \theta \leq 360, x \neq 90, x \neq 270$
13. a) $y = -2 \cos \theta$
b) $y = \sin \theta + 1$
14. $90^\circ < \theta < 270^\circ$
15. $0^\circ < \theta < 210^\circ$ and $330^\circ < \theta < 360^\circ$
16. $60^\circ < \theta < 300^\circ$
17. a) $A = (90^\circ; 4)$, $B = (90^\circ; -2)$, $C = (180^\circ; 4)$ and $D = (180^\circ; -2)$
b) 2
c) 2
d) 2
18. a) $A = (90^\circ; 1)$, $B = (180^\circ; -3)$, $C = (270^\circ; -1)$ and $D = (360^\circ; 3)$
19. a) $A = (90^\circ; 3)$, $B = (90^\circ; 2)$, $C = (180^\circ; -4)$ and $D = (270^\circ; 2)$
b) 3
c) 2
d) -1

Exercise 6 – 7:

2. a) $y = x + 2$ and $y = -\frac{1}{2}x^2 + 2$
b) $y = 8x$ and $y = \frac{8}{x}$
3. a) (iii)
b) (iv)
- c) (iii)
d) (i)

Exercise 6 – 8:

1. a) $y = 3x$
b) $y = x - 4$
4. a) x -intercept $= -\frac{5}{3}$ and y -intercept $= -5$
b) x -intercept $= -2$ and y -intercept $= 4$
5. $m = 1$, and $y = x - 4$
6. a) E
b) B
c) A
d) C
e) F
f) D
7. a) True
b) False
c) False
8. a) $y = \frac{5}{2}x + 3$
b) $y = \frac{4}{3}x + \frac{1}{6}$
10. a) 120 mL
b) 600 mL
c) 33,33 s
d) 240 mL
11. a) 50 km
b) The domain is $0 \leq t \leq 120$ min.
c) The range is $0 \leq s \leq 100$ km, it represents the total distance travelled.
12. $a = -\frac{1}{3}$; $q = 6$
13. a) 3
b) $(-0,77; 0)$ and $(0,77; 0)$
14. a) $k(x)$
b) $g(x)$
c) $h(x)$
d) $f(x)$
15. a) $g(x)$
b) $f(x)$
c) $h(x)$
17. a) 8 m
b) 6 m
c) 3 m
d) 1 m
18. $a = 2$ and $q = -3$
19. a) no solution
b) $\frac{3}{4}$
20. a) $k(x)$
b) $f(x)$
c) $g(x)$
d) $h(x)$
22. a) Translation by 2 in the positive y -direction.
- b) Reduction by 4
c) Translation by 3 units in the positive y -direction.
d) Reflection on x -axis
23. a) 2,50
b) 1,3
25. $a = 3$ and $q = -4$
26. a) $h(x)$
b) $f(x)$
c) $k(x)$
d) $g(x)$
27. a) -4
b) -2
c) $-\frac{323}{81}$
d) $-\frac{7}{10}$
e) $\frac{1}{27}$
f) 77
g) $-\frac{3}{2}$
28. a) False
b) False
c) True
d) False
e) False
f) True
29. b) $(-3; 12)$ and $(2; 2)$
d) $y = -2x^2 + 6$
30. 1,6 units
31. a) $x + y = 15$; $y = x + 3$
b)
c) 6 R 5 coins and 9 R 2 coins
32. $a = \frac{3}{2}$ and $q = 3$
33. $a = 3$ and $q = 0$
34. $a = 1$ and $q = -3$
35. a) $g(x)$
b) $f(x)$
c) $k(x)$
d) $h(x)$
36. $f(x) = -3,5 \cos \theta$
40. a) $E = (270^\circ; -2)$ and $-2 \leq y \leq 2$
b) $E = (360^\circ; 2)$ and $-2 \leq y \leq 2$
41. $E = (45^\circ; 0)$, range $y \in \mathbb{R}$ and domain $0 \leq x \leq 360, x \neq 90, \theta \neq 270$
42. $0^\circ < \theta < 180^\circ$
43. $0^\circ < \theta < 90^\circ$ and $270^\circ < \theta < 360^\circ$
44. $0^\circ < \theta < 360^\circ$
45. a) $y = 3x$
b) $y = -2x^2 + 3$

- c) $y = \frac{-3}{x}$
d) $y = x + 2$
e) $y = 5 \sin \theta + 1$
f) $y = 2 \times 2^x + 1$
g) $y = -\tan \theta - 2$
46. a) $A(90^\circ; 1)$, $B(90^\circ; -1)$, $C(180^\circ; 2)$ and $D(360^\circ; 1)$
b) 0
c) 1
d) 5
47. a) $A(90^\circ; 1)$, $B(270^\circ; 3)$, $C(270^\circ; 1)$ and $D(360^\circ; 2)$
b) 2
c) 3
d) 2
48. a) $M(0; 1)$ and $N(0; -1)$
b) 2
c) 1
d) $y = 2^{-x}$
e) Range $y = 2^x$: $(0; \infty)$
Range $y = -2^x$: $(-\infty; 0)$
50. a) $q = 1$
- b) $BC = 3 + 4 = 7$ units
c) $y = -4^x$
d) $y = 4^x + 1$
e) $y = 0$
f) Range $f(x)$: $(0; \infty)$, Range $g(x)$: $(-\infty; 1]$
51. c) $y = x^2 + 4$
d) Domain h : $(-\infty; \infty)$. Range h : $[-4; \infty)$
53. a) $A(\sqrt{8}; \sqrt{8})$ and $B(-\sqrt{8}; -\sqrt{8})$
b) $2\sqrt{8}$
c) 8
d) 2
54. a) $A(-1; 0)$, $B(1; 0)$, $C(0; 3)$
c) $D(2; -9)$
d) $y = -6x + 3$
55. a) $\{\theta : 0^\circ \leq \theta \leq 360^\circ, \theta \neq 90^\circ; 270^\circ\}$
b) 3
56. a) $y = -2x$ en $y = x^2 - 3$
b) $y = 4x + 1$ and $y = 3^x$
57. a) (iv)
b) (ii)

7 Euclidean geometry

Exercise 7 – 1:

3. 70°
4. a) $\hat{x} = 55^\circ$
b) 35°
c) $\hat{r} = 135^\circ$
d) 45°
e) $\hat{p} = 45^\circ$
- f) EF is not parallel to CG
5. a) $\hat{a} = 50^\circ$
b) 40°
c) $\hat{c} = 140^\circ$
d) 40°
e) $\hat{d} = 40^\circ$
- f) $PQ \parallel NR$
6. a) no parallel lines
b) not parallel
c) $\therefore TY \parallel MN$

Exercise 7 – 2:

1. a) 72°
b) 98°
c) $y = 112^\circ$ and $x = 44^\circ$
d) 29
e) 25
- f) $x = 18$ and $y = 4$
g) $x = 12$ and $y = 13$
2. diagram A
3. diagram B
5. $\triangle PNM \equiv \triangle QSR$, reason: SAS
6. a) $\triangle ABC \equiv \triangle EDC$
b) not congruent
c) not enough information
d) not enough information
e) $\triangle ABC \equiv \triangle ADC$

Exercise 7 – 3:

1. a) $S\hat{R}O$ and $O\hat{R}Q$ b) $2x$ c) 28°

Exercise 7 – 4:

Exercise 7 – 5:

Exercise 7 – 6:

1. rectangle
2. rhombus
3. quadrilateral
4. kite and quadrilateral
5. square, rectangle, rhombus, parallelogram, kite, trapezium and quadrilateral
6. 315

Exercise 7 – 7:

1. ED
2. TS
3. ED and AC
4. no parallel lines
5. a) 91°
b) $\triangle DEC \parallel \triangle DAB$
6. a) 60°
- b) $\triangle HKG \parallel \triangle JKL$
7. 6
8. 3
9. 11
10. 4
11. a) 14
b) 3,5
- c) 5
d) $x = 28$ units and $y = 100^\circ$
e) $x = 24^\circ$ and $y = 13,93$
13. a) $MN = -4a + 8$
b) $-\frac{5}{2}$
14. a) $\frac{a}{3} + 2$
b) 18

15. a) $\hat{A} = -4q + 121^\circ$
b) 18°

16. a) $\hat{K} = -9b + 122^\circ$
b) $\frac{16}{3}$

Exercise 7 – 8:

1. a) straight angle
b) obtuse angle
c) acute angle
d) right angle
e) Reflex angle
f) obtuse angle
g) straight angle
h) reflex angle
2. a) False
b) True
c) True
d) False
e) True
f) True
g) True
h) True
i) False
j) False
3. a) $AB \parallel CD$
b) $MN \parallel OP$
c) $GH \parallel KL$ and $GK \parallel HL$
4. a) $a = 107^\circ, b = 73^\circ, c = 107^\circ, d = 73^\circ$
b) $a = b = c = d = 80^\circ$
c) $a = 50^\circ, b = 45^\circ, c = 95^\circ, d = 85^\circ$
5. a) $\hat{x} = 45^\circ$
b) 45°
c) $\hat{r} = 135^\circ$
d) 45°
e) $\hat{p} = 45^\circ$
f) parallel
6. diagram B
7. diagram A
8. not enough information
9. $(\triangle VWU \equiv \triangle YZX)$, reason: SSS
10. a) $\triangle ABC \equiv \triangle DEF$
b) $\triangle GHI \equiv \triangle JKL$
c) $\triangle MNO \equiv \triangle RPQ$
d) $\triangle QRS$ not congruent $\triangle TUV$
11. a) 4,24 cm
b) 12 cm
c) 7,28 cm
d) 40 mm
12. a) 25°
b) 145°
c) $x = 10$ and $y = 12,5$
d) 60°
e) 36°
f) $x = 6$ and $y = 10$
14. $x = 2,75$ and $y = 30^\circ$
15. kite
16. rhombus, parallelogram, kite, trapezium and quadrilateral
17. $x = 15^\circ$
18. a) $ACDF, ABEF$ and $BCDE$
b) $ACDF, ABEF, BCDE$ and $BCEF$
c) $ACEF$ and $BCDF$
d) $ABEF$ and $BCDE$
19. TS
20. ZY and VX
21. a) 39°
b) $\triangle DEC \parallel \triangle DAB$
22. a) 98°
b) $\triangle QMN \parallel \triangle QRP$
23. 5
24. 9
25. no solution
26. a) $PR = \frac{z}{4} - 1$
b) $z = 12$
27. a) $-8b + 119^\circ$
b) $\frac{19}{2}$
28. a) $-24d + 180^\circ$
b) $\frac{20}{3}$
29. $a = 5$ and $b = 12$
33. b) rhombus
c) 44°

8 Analytical geometry

Exercise 8 – 1:

1. (3; 3)
2. $A(3; -4), B(3; -3), C(-3; -4), D(5; -3)$ and $E(5; -4)$.
3. E
4. B
5. shape Z
6. shape Z

Exercise 8 – 2:

1. 6,71
2. 2,24
3. $x = 1$
4. $y = 3,5$
5. a) $\sqrt{29}$
b) $\sqrt{52}$
c) $\sqrt{17}$
6. a) $x = 3$ or $x = 9$
b) $y = 3$ or $y = -5$
7. $p = 3$ or $p = -9$

Exercise 8 – 3:

1. a) $\frac{9}{11}$
b) $\frac{11}{8}$
- c) $\frac{4}{3}$
2. -0,5
3. 1,5
4. a) $\frac{-10}{3}$
b) 5
5. 4
6. 3

Exercise 8 – 4:

1. a) parallel
b) neither
c) perpendicular
2. a) collinear
b) collinear
c) not collinear
3. $y = -2x + 0,5$
4. $y = 0,5x - 0,5$
5. $\frac{1}{2}$
6. 6
7. $y = x - 2$
8. $y = -x + 1,5$
9. 4
10. -2
11. $y = -x + 2$
12. $y = 2x - 1,5$
13. 3
14. -2
15. 0,5
16. -0,5
17. 6
18. $\frac{-2}{3}$
19. b) 42,5 units
20. b) (3; 13)
c) $y = 2x + 12$

Exercise 8 – 5:

1. $M(1; 0)$
2. $M(-0,5; -1,25)$
3. a) $(-1; 6)$
b) $(14; 32)$
- c) $\left(\frac{2x-3}{2}; \frac{2y-5}{2}\right)$
4. $(8; 13)$
5. $S(4; -5)$

Exercise 8 – 6:

1. $A(7; -4)$, $B(3; -3)$, $C(7; 7)$, $D(-2; -8)$ and $E(-5; 4)$
2. A
3. Shape Z
5. 9,81
6. 3
7. 2
8. 4
9. 4
10. a) $y = \frac{1}{3}x + 3$
b) $\sqrt{40}$
11. $y = -1,5x - 1$
12. 2
13. 0
14. $y = -x - 1,5$
15. -4
16. $(-0,5; 0,5)$
18. a) $y = -x + 10$
b) $y = 3x + 12$
c) $y = \frac{1}{6}x + \frac{2}{3}$
19. b) opposite sides are not parallel
c) do not bisect
d) ordinary quadrilateral
21. b) isosceles trapezium
22. b) i. $\sqrt{10}$
ii. 3
c) Trapezium
23. $H(3; 3)$
24. a) $\sqrt{34}$
b) $\frac{1}{3}$
c) $\left(\frac{3}{2}; \frac{-1}{2}\right)$
25. c) $\left(\frac{7}{2}; \frac{7}{2}\right)$
d) $\frac{-2}{3}$
29. a) $\frac{1}{2}$
c) $C(11; 8)$
30. a) 9,8
b) $\frac{-2}{3}$
d) $y = \frac{3}{2}x + 6$
31. a) $a = 1$ and $b = \frac{7}{2}$
b) $y = \frac{4}{3}x$
c) $\left(\frac{7}{2}; 1\right)$
e) $C(7; 0)$
32. a) $-\frac{1}{3}$
c) 9,49
d) $y = -x + 6$
e) 45
33. a) no value of k
b) $k = 1$
34. a) $5\sqrt{5}$
b) $-\frac{1}{2}$
c) $y = -\frac{1}{2}x + \frac{5}{2}$
d) 1
35. a) $M(1; 3)$
b) 2
c) $-\frac{1}{2}$
d) $y = -\frac{1}{2}x + \frac{7}{2}$
e) $y = 2x + 6$
36. a) $N(3; 5)$
b) rhombus
37. a) $\frac{1}{4}$
c) 4,1
d) $\frac{1}{4}$
e) $y = -\frac{1}{2}x - 4$
38. a) $\sqrt{41}$
b) $M(1; 2)$
- c) $T(-1; \frac{1}{2})$
e) $S(1; 6)$
39. a) isosceles
c) $\frac{3}{4}$
40. a) $Q(0; 2)$
c) $m_{PR} = 1$
d) 6
e) $y = 2x - 3$
41. a) -1
42. a) $d_{AC} = d_{BD} = \sqrt{26}$
b) $(-0,5; -4,5)$
e) square
43. a) $(1; -5)$
b) 3
c) $y = 3x - 8$
d) $y = -\frac{1}{3}x - \frac{4}{3}$
44. a) $E = (\frac{1}{2}; -\frac{3}{2})$
b) $ABCD$ is a parallelogram
c) $y = \frac{1}{3}x - \frac{5}{3}$
d) $y = -3x$
e) $m_{AC} = -\frac{15}{7}$
f) not a rhombus
g) 10
45. a) 5
b) $7\frac{1}{2}$
c) $8\frac{1}{3}$
46. a) $m_{MN} = -\frac{1}{3}$
b) $y = -\frac{1}{3}x + 3\frac{2}{3}$
c) $AB \parallel MN$
d) 2
e) $P(x; y) = (8; 12)$
47. a) -13
b) $(8; 12)$

9 Finance and growth

Exercise 9 – 1:

1. R 4025
2. R 5398,80
3. R 1700,00
4. R 7030,80
5. a) R 324
- b) R 3937,50
6. R 11 538,46
7. 19
8. 16,25%
9. 5,7% per annum
10. 3,8% per annum
11. 4,3% per annum
12. 20 years
13. 32 years

Exercise 9 – 2:

1. R 4044,69
2. R 5930,94
3. R 9327,76
4. R 59 345,13
5. R 24 002,00
6. R 17 942,00
7. 8,45% p.a
8. 4,3% per annum
9. 1,8% per annum

Exercise 9 – 3:

1. a) R 3960,00
b) R 4316,40
c) R 359,70
d) R 4756,40
2. a) R 4760,00
b) R 6092,80
3. a) R 3825
b) R 4743
c) R 197,63
d) R 5418
4. a) R 12 962,50
b) R 4462,50
c) R 360,07
5. a) R 10 240
b) R 3840
c) R 213,33

6. a) R 6324
b) R 1224
c) R 263,50

7. a) R 5400
b) R 4251,97
8. store A

9. R 156,84
10. R 210,22

Exercise 9 – 4:

1. R 33,28
2. R 29,61
3. R 22,77
4. R 14,72
5. R 14,24
6. R 2174,77
7. R 38,64

Exercise 9 – 5:

1. 4 142 255
2. 4 217 645
3. 553

Exercise 9 – 6:

1. a) R 1400
b) R 200
c) R 100
2. a) R 1430
b) R 260
c) R 1040
3. a) R 1680
b) R 600
c) R 480
4. a) USA
b) Sollie
5. New York publisher
6. R 36 780
7. a) Brazilian tourists
b) Japanese tourists
8. 530 523 TZS

Exercise 9 – 7:

1. R 11 204,10
2. R 2470,80
3. R 35 087,72
4. 3,6% per annum
5. 3,6% per annum
6. 25 years
7. 22 years
8. R 938
9. R 7319,78
10. R 4158,88
11. R 44 872
12. R 18 731,00
13. a) R 205
b) R 286,52
c) R 128
14. 3,3% per annum
15. 5,5% per annum
16. a) R 534,25
b) R 520
17. Bank B
18. a) R 200
b) R 200
21. a) R 4800,00
b) R 5232,00
c) R 436,00
d) R 6432,00
22. a) R 4320
b) R 4838,40
c) R 403,20
d) R 5318,40
23. a) R 10 880
b) R 2880
c) R 302,22
24. a) R 13 860
b) R 6160
c) R 231,00
25. 8,5%
26. R 1106,04
27. R 24,53
28. R 27,49
29. R 12,60
30. R 8,06
31. a) 62,3 million people
b) 1,7
32. 4 065 346
33. 4 083 001
34. a) R 2100
b) R 840
c) R 700
35. a) R 1960
b) R 560
c) R 840
36. R 1840
37. UK publisher
38. 3521,37 BRL
40. R 13 343,92

10 Statistics

Exercise 10 – 1:

1. qualitative anecdotal
2. quantitative discrete
3. quantitative discrete

Exercise 10 – 2:

1. 9,7
2. 13
3. 6
4. a) mean: 13,2; median: 11; mode: 8
b) mean: 26; median: 25; mode: 24
c) mean: 11,2; median: 11; mode: 11
d) mean: 34,29; median: 31; mode: none
5. mean: 38,3; median 38; mode 33 and 42.
6. 23
7. 26
8. 20
9. a) 5
b) 7
10.
11. 36

Exercise 10 – 3:

1. 1
2. 1
3. 3
4. 5
5. E
6. B
7. Data Set C
8. Data Set C

Exercise 10 – 4:

- Mean: 52; Modal group: $50 < m \leq 55$; Median group: $50 < m \leq 55$
 - Mean: 70,66; Modal group: $65 < t \leq 75$; Median group: $65 < t \leq 75$
 - a) $700 < x \leq 800$
- 33 600
 - 700
 - 750
 - R 588 000

Exercise 10 – 5:

- 10
- 9
- 9
- $Q_1 = 6,5; Q_2 = 18; Q_3 = 29$

Exercise 10 – 6:

Exercise 10 – 7:

- quantitative continuous
- qualitative categorical
- 10
- 10
- 44
- 6
- 10
- 15
- $x = 19$
- $\bar{x} = p + 4$
- 1
- 4
- 3
- C
- E
- data set C
- data set B
- 18
- 14
- a) Mean and mode
- a) 19,9
- a) R 182 222,22
- b) R 100 000
- c) R 100 000
- a) The mean and range are 89,2 and 27 respectively.
- a) $x = 17$
- b) x is any integer with $x \neq \{1; 3; 4; 5; 9\}$
- c) $x = 7$
- d) $x = 10$
- e) $x = 4$
- $\{2; 5; 7; 10; 10\}$
- a) median = 6,5; mode = 8; mean = 5,45
- b) $\phi_{21} = 4$
- $\{3; 3; 5; 5; 8; 8\}$ or $\{3; 4; 4; 5; 8; 8\}$
- a) 22; 36,5; 50; 55; 64
- b) no men can be retrenched
- $\bar{x} = 42,643$
- a) B
- b) median
- 3
- brand C

11 Trigonometry

Exercise 11 – 1:

- 5
- 9
- $53,13^\circ$
- $35,30^\circ$
- 26 m
- 15 m

Exercise 11 – 2:

- $19,47^\circ$
- a) 18°
- b) 23°
- 8
- $44,99^\circ$
- 8°
- 10 m
- b) $48,59^\circ$
- b) 54,69 m
- 7,71 m
- the ship is not safe
- 473,52 m
- $106,62^\circ$ and $67,38^\circ$
- 11,47 cm and 8,03 cm
- 4,24 cm
- a) 5 cm
- a) $36,53^\circ$
- b) 34,82 m
- b) 379,73 m
- 793,77 m
- a) 11,21 cm
- b) $56,4^\circ$
- 167 m

12 Euclidean geometry

Exercise 12 – 1:

- a) $AECF$ is a parm
- $AD = EF$
- 105°
- 42°
- $ABCD$ is a parm
- b) 34°
- b) 36°

Exercise 12 – 2:

- b) 24 cm^2
- a) (i) In $\triangle QRT$ and $\triangle RST$ side $RT = RT$, (ii) common side, (iii) $\therefore QR = TS$ and $QT = RS$ and (iv) opp. sides are equal
- b) 63°
- c) 79°

13 Measurements

Exercise 13 – 1:

- a) 25 cm^2
- 60 cm^2
- $9\pi z^2 + 12\pi z + 4\pi$
- 50 cm^2
- 12 cm^2
- $-2hz - h$
- $78,54 \text{ cm}^2$
- $43,30 \text{ cm}^2$
- $\pi x^2 + 8\pi x + 16\pi$
- 40 cm^2
- 276 cm^2
- $-hx + 2h$

Exercise 13 – 2:

- | | | |
|--------------------------|--------------------------|------------|
| 1. a) 344 cm^2 | d) $471,24 \text{ cm}^2$ | 2. a) 24 L |
| b) $277,82 \text{ cm}^2$ | e) 270 | b) 22 L |
| c) $87,96 \text{ cm}^2$ | f) $532,84 \text{ cm}^2$ | |

Exercise 13 – 3:

- | | | |
|-----------------------|-------------------------|-----------|
| 1. 420 cm^3 | 3. $785,4 \text{ cm}^3$ | 5. 1056 |
| 2. 500 cm^3 | 4. 32,06 | 6. 552,92 |

Exercise 13 – 4:

- | | | |
|----------------------------|--------------------------|---------------------|
| 1. a) $282,7 \text{ cm}^2$ | d) $1256,6 \text{ cm}^2$ | 4. 189 square units |
| b) $45,6 \text{ cm}^2$ | 2. 175,93 | |
| c) 180 cm^2 | 3. $804,25$ square units | |

Exercise 13 – 5:

- | | | |
|------------------------------|--|--|
| 1. $2144,66 \text{ units}^3$ | b) $52,0 \text{ cm}^3$ | 6. a) $91,39 \text{ cm}^2$ |
| 2. $29,32 \text{ units}^3$ | c) 144 cm^3 | b) $29,39 \text{ cm}^3$ |
| 3. $170,67 \text{ units}^3$ | d) $4188,8 \text{ cm}^3$ | 7. surface area is 190 cm^2 and the volume is 175 cm^3 |
| 4. a) $314,16 \text{ cm}^3$ | 5. surface area is 393 cm^2 and the volume is 507 cm^3 | |

Exercise 13 – 6:

- | | |
|----------------------------------|--|
| 1. $\frac{1}{9}$ | b) increases by a factor of 27 |
| 2. 4 | 5. 64 times |
| 3. doubles | 6. volume is $31\,552 \text{ cm}^3$ and the surface area is $96\,112 \text{ cm}^2$. |
| 4. a) increases by a factor of 9 | |

Exercise 13 – 7:

- | | |
|---|--|
| 1. a) 75 cm^2 | b) $V_{\text{cylinder}} = 502,7 \text{ cm}^3$ $V_{\text{triangular prism}} = 240 \text{ cm}^3$ $V_{\text{rectangular prism}} = 40 \text{ cm}^3$ |
| b) $153,94 \text{ mm}^2$ | c) $A_{\text{cylinder}} = 3166,7 \text{ cm}^2$ $A_{\text{triangular prism}} = 3456 \text{ cm}^2$ $A_{\text{rectangular prism}} = 684 \text{ cm}^2$ |
| c) $252,76 \text{ cm}^2$ | d) $V_{\text{cylinder}} = 13\,571,9 \text{ cm}^3$ $V_{\text{triangular prism}} = 6480 \text{ cm}^3$ $V_{\text{rectangular prism}} = 1080 \text{ cm}^3$ |
| 2. a) $15y^2 - 10y$ | 20. a) $460,64 \text{ cm}^2$ |
| b) $\frac{15y^2}{2} - 5y$ | b) 588 cm^3 |
| 3. 420 | 21. $30\,159,52 \text{ cm}^2$ and $301\,592,88 \text{ cm}^3$ |
| 4. 210 | 22. a) $3665,19 \text{ cm}^3$ and $1194,67 \text{ cm}^2$ |
| 5. $640,88$ | b) $2148,85 \text{ m}^3$ and $867,08 \text{ m}^2$ |
| 6. 360 | c) 600 ft^3 and $600,37 \text{ ft}^2$ |
| 7. 300 | 23. Yes |
| 8. $1847,26$ | 24. a) 90 cm |
| 9. $615,75$ | b) $261,5 \text{ l}$ |
| 10. 88 square units | 25. a) $47\,123,89 \text{ cm}^3$ and $1570,80 \text{ cm}^2$ |
| 11. $103,67$ square units | b) 92,38 and 180,04 |
| 12. $113,1 \text{ units}^3$ | c) 1200 and 886 |
| 13. $183,26 \text{ units}^3$ | 26. a) 2412,743 |
| 14. $130,67$ units cubed | b) $645,07 \text{ cm}^3$ |
| 15. a) $A_{\text{cone}} = 126,67 \text{ cm}^2$ $A_{\text{square pyramid}} = 437,26 \text{ cm}^2$ $A_{\text{half sphere}} = 150,80 \text{ cm}^2$ | 27. b) $28,57 \text{ units}^2$ |
| b) $V_{\text{cone}} = 94,25 \text{ cm}^3$ $V_{\text{square pyramid}} = 900 \text{ cm}^3$ $V_{\text{half sphere}} = 134,04 \text{ cm}^3$ | c) $285,664 \text{ units}^3$ |
| 16. $\frac{1}{16}$ | 28. b) $74,626 \text{ cm}^3$ |
| 17. $\frac{1}{9}$ | 29. a) $4a^2$ |
| 18. $\frac{1}{8}$ | b) $612,00538 \text{ feet}$ |
| 19. a) $A_{\text{cylinder}} = 351,9 \text{ cm}^2$ $A_{\text{triangular prism}} = 384 \text{ cm}^2$ $A_{\text{rectangular prism}} = 76 \text{ cm}^2$ | c) 1,620 |
| | d) $91\,661\,532,5 \text{ feet}^3$ and $1\,551\,425,432 \text{ feet}^2$ |

14 Probability

Exercise 14 – 1:

1. event set = $\{(2; 6); (3; 5); (4; 4); (5; 3); (6; 2)\}$ 6. a) $\frac{1}{2}$ b) $\frac{1}{2}$
2. $\{1; 2; 3; 4; 5; 6\}$ b) $\frac{1}{3}$ c) $\frac{3}{13}$
3. $= \frac{1}{3}$ c) $\frac{5}{6}$ d) $\frac{1}{13}$
4. 0,50 d) $\frac{1}{3}$ e) $\frac{3}{13}$
5. $\frac{1}{5}$ 7. a) $\frac{1}{52}$

Exercise 14 – 2:

1. 0,11
2. 0,57
3. 0,22

Exercise 14 – 3:

1. $\{1; 2; 3; 4; 5; 7; 8; 9; 10; 11; 12; 13; 14; 15\}$ b) $\{1; 2; 3; 4; 6; 12\}$ c) 29
2. $\{1; 2; 4; 5; 6; 7; 9; 10; 11; 13; 14; 15\}$ c) $\{2; 3; 5; 7; 11\}$ d) 2
3. a) $\{1; 2; \dots; 12\}$ 5. b) 6

Exercise 14 – 4:

1. $\{7; 10\}$
2. $\{1; 2; 4; 5; 6; 7; 8; 9; 10; 11; 12; 13; 14; 15\}$

Exercise 14 – 5:

1. 0,83
2. 0,17

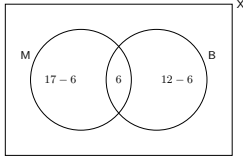
Exercise 14 – 6:

1. mutually exclusive
2. mutually exclusive
3. not mutually exclusive
4. mutually exclusive

Exercise 14 – 7:

1. $\{2; 3; 4; 6; 10; 11; 12\}$
2. $\{2; 4; 9; 11; 13; 15\}$
3. yes
4. no

Exercise 14 – 8:

1. the die lands on the number 5 c) 0,80
2. $\{1; 2; 3; 4; 5; 7; 8; 9; 10; 11; 12; 14; 15\}$ f) 0,55
3. $\{1; 2; 3; 4; 5; 7; 8; 9; 10; 11; 12; 13; 14; 15\}$ g) 0,16
4. yes h) 0,01
5. $\{1; 3; 6; 12; 14; 15\}$ 14. a) $\frac{1}{6}$
6. 0,08 b) $\frac{1}{2}$
7. 0,1 c) $\frac{1}{3}$
8. 0,50 15. $\frac{5}{12}$
9. 0,18 16. a) $\frac{19}{30}$
10. a) $\frac{1}{6}$ b) $\frac{11}{30}$
11. a) 0,21 b) $\frac{4}{9}$
- b) 0,56 c) $\frac{5}{9}$
- c) 0 d) $\frac{1}{9}$
- d) 0,72 e) $\frac{8}{9}$
12. a) 0,5 e) $\frac{5}{9}$
- b) 0,23 f) $\frac{4}{9}$
- c) 0,67 18. a) $\frac{11}{21}$
- d) 0 b) $\frac{11}{14}$
- e) 0,67 19. a) $\frac{3}{56}$
- f) 0,56 b) $\frac{53}{56}$
13. a) 0,11 20. a) 103
- b) 0,33 c) $\frac{14}{19}$
- c) 0,16 21. b) 10
- d) 0,01
- c) 4
22. a) $P = \frac{1}{3}$
- b) $P = \frac{2}{3}$
- c) $P = \frac{1}{2}$
- d) $P = \frac{1}{6}$
23. a) (i) $P = \frac{1}{52}$; (ii) $P = \frac{7}{26}$ (iii) $P = \frac{4}{13}$
- b) $\frac{13}{17}$
24. 0,17
27. e) Mutually exclusive and complementary.
29. a) 50%
- b) 31,25%
- c) 6,25%
30. 31
31. a) 
- b) 13
- c) (i) $P = \frac{1}{6}$; (ii) $P = \frac{1}{6}$; (iii) $P = \frac{2}{3}$
32. a) $w = 4, x = 2, y = 6$ and $z = 1$
- b) (i) $P = \frac{4}{15}$; (ii) $P = \frac{1}{10}$

1 Mathematics, Paper 1, Exemplar 2012

MARKS: 100
TIME: 2 hours

Instructions and information

Read the following instructions carefully before answering the questions.

1. This question paper consists of 7 questions.
2. Answer ALL the questions.
3. Clearly show ALL calculations, diagrams, et cetera that you have used in determining your answers.
4. Answers only will not necessarily be awarded full marks.
5. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.
6. If necessary, round off answers to TWO decimal places, unless stated otherwise.
7. Diagrams are NOT necessarily drawn to scale.
8. Number the answers correctly according to the numbering system used in this question paper.
9. Write neatly and legibly.

Exercise 1 – 1:

1. a) Simplify the following expressions fully:
 - i. $(m - 2n)(m^2 - 6mn - n^2)$ (3 marks)
 - ii. $\frac{x^3 + 1}{x^2 - x + 1} - \frac{4x^2 - 3x - 1}{4x + 1}$ (5 marks)
- b) Factorise the following expressions fully:
 - i. $6x^2 - 7x - 20$ (2 marks)
 - ii. $a^2 + a - 2ab - 2b$ (3 marks)
- c) Determine, **without the use of a calculator**, between which two consecutive integers $\sqrt{51}$ lies. (2 marks)
- d) Prove that $0,2\dot{4}\dot{5}$ is rational. (4 marks)

[TOTAL: 19 marks]

2. a) Determine, **without the use of a calculator**, the value of x in each of the following:
 - i. $x^2 - 4x = 21$ (3 marks)
 - ii. $96 = 3x^{\frac{5}{4}}$ (3 marks)
 - iii. $R = \frac{2\sqrt{x}}{3S}$ (2 marks)
- b) Solve for p and q simultaneously if:

$$\begin{aligned} 6q + 7p &= 3 \\ 2q + p &= 5 \end{aligned}$$

(5 marks)

[TOTAL: 13 marks]

3. a) $3x + 1$; $2x$; $3x - 7$; ... are the first three terms of a linear number pattern.
 - i. If the value of x is three, write down the FIRST THREE terms. (3 marks)
 - ii. Determine the formula for T_n , the general term of the sequence. (2 marks)
 - iii. Which term in the sequence is the first to be less than -31 ? (3 marks)
- b) The multiples of three form the number pattern: 3 ; 6 ; 9 ; 12 ; ...
Determine the 13th number in this pattern that is even. (3 marks)

[TOTAL: 11 marks]

4. a) Thando has R 4500 in his savings account. The bank pays him a compound interest rate of 4,25% p.a. Calculate the amount Thando will receive if he decides to withdraw the money after 30 months. (3 marks)
- b) The following advertisement appeared with regard to buying a bicycle on a hire-purchase agreement loan:

Purchase price R 5999
Required deposit R 600
Loan term only 18 months, at 8% p.a. simple interest

- i. Calculate the monthly amount that a person has to budget for in order to pay for the bicycle. (6 marks)
 - ii. How much interest does one have to pay over the full term of the loan? (1 marks)
- c) The following information is given:

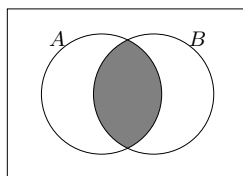
$$1 \text{ ounce} = 28,35 \text{ g}$$

$$\text{\$}1 = \text{R } 8,79$$

Calculate the rand value of a 1 kg gold bar, if 1 ounce of gold is worth \\$978,34

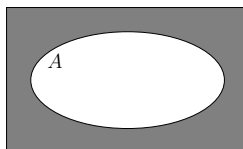
[TOTAL: 14 marks]

5. a) What expression BEST represents the shaded area of the following Venn diagrams?



i.

(1 mark)



ii.

(1 mark)

- b) State which of the following sets of events is mutually exclusive:

A Event 2: The learners in Grade 10 in the debating team	Event 1: The learners in Grade 10 in the swimming team
B Event 2: The learners in Grade 12	Event 1: The learners in Grade 8
C Event 2: The learners who take Physical Sciences in Grade 10	Event 1: The learners who take Mathematics in Grade 10

(1 mark)

- c) In a class of 40 learners the following information is TRUE:

- 7 learners are left-handed
- 18 learners play soccer
- 4 learners play soccer and are left-handed
- All 40 learners are either right-handed or left-handed

Let L be the set of all left-handed people and S be the set of all learners who play soccer.

- How many learners in the class are right-handed and do NOT play soccer? (1 mark)
- Draw a Venn diagram to represent the above information. (4 marks)
- Determine the probability that a learner is:
 - Left-handed or plays soccer (3 marks)
 - Right-handed and plays soccer (2 marks)

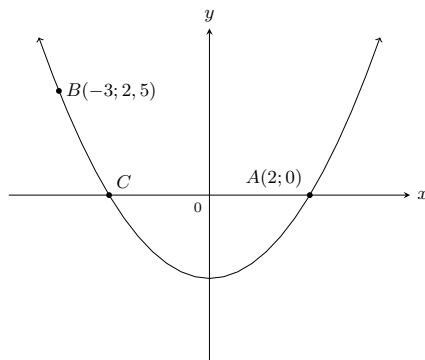
[TOTAL: 13 marks]

6. Given: $f(x) = \frac{3}{x} + 1$ and $g(x) = -2x - 4$

- Sketch the graphs of f and g on the same set of axes. (4 marks)
- Write down the equations of the asymptotes of f . (2 marks)
- Write down the domain of f . (2 marks)
- Solve for x if $f(x) = g(x)$. (5 marks)
- Determine the values of x for which $-1 \leq g(x) \leq 3$. (3 marks)
- Determine the y -intercept of k if $k(x) = 2g(x)$. (2 marks)
- Write down the coordinates of the x - and y -intercepts of h if h is the graph of g reflected about the y -axis. (2 marks)

[TOTAL: 20 marks]

7. The graph of $f(x) = ax^2 + q$ is sketched below.
 Points $A(2; 0)$ and $B(-3; 2,5)$ lie on the graph of f .
 Points A and C are x -intercepts of f .



- Write down the coordinates of C . (1 mark)
- Determine the equation of f . (3 marks)
- Write down the range of f . (1 mark)
- Write down the range of h , where $h(x) = -f(x) - 2$. (2 marks)
- Determine the equation of an exponential function, $g(x) = b^x + q$, with range $y > -4$ and which passes through the point A . (3 marks)

[TOTAL: 10 marks]

2 Mathematics, Paper 2, Exemplar 2012

MARKS: 100
TIME: 2 hours

Instructions

- This question paper consists of 9 questions.
- Answer ALL the questions.
- Clearly show ALL calculations, diagrams, graphs, et cetera which you have used in determining the answers.
- Answers only will NOT necessarily be awarded full marks.
- You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.
- If necessary, round off answers to TWO decimal places, unless stated otherwise.
- Diagrams are NOT necessarily drawn to scale.
- ONE diagram sheet for QUESTION 6.1.1 and QUESTION 9 is attached at the end of this question paper. Write your centre number and examination number on this sheet in the spaces provided and insert the sheet inside the back cover of your ANSWER BOOK.
- Number the answers correctly according to the numbering system used in this question paper.
- Write neatly and legibly.

Exercise 2 – 1:

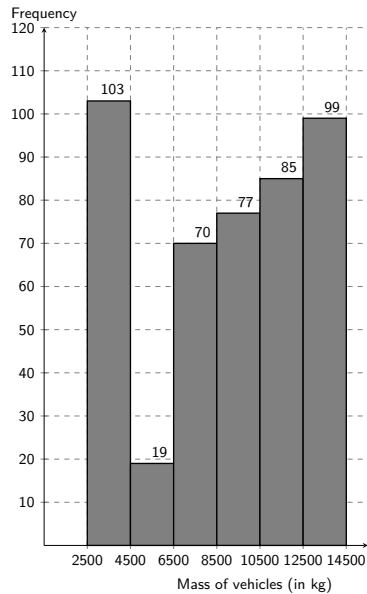
- A baker keeps a record of the number of scones that he sells each day. The data for 19 days is shown below.

31	36	62	74	65	63	60	34	46	56
37	46	40	52	48	39	43	31	66	

- Determine the mean of the given data.
(2 marks)
- Rearrange the data in ascending order and then determine the median.
(2 marks)
- Determine the lower and upper quartiles for the data.
(2 marks)
- Draw a box and whisker diagram to represent the data.
(2 marks)

[TOTAL: 8 marks]

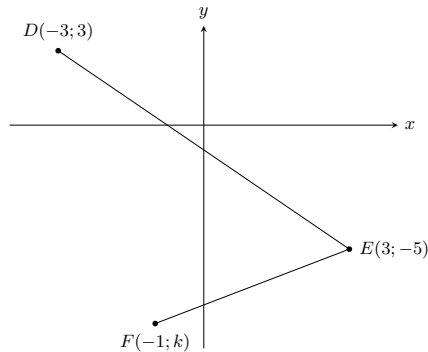
- Traffic authorities are concerned that heavy vehicles (trucks) are often overloaded. In order to deal with this problem, a number of weighbridges have been set up along the major routes in South Africa. The gross (total) vehicle mass is measured at these weighbridges. The histogram below shows the data collected at a weighbridge over a month.



- Write down the modal class of the data.
(1 mark)
- Estimate the mean gross vehicle mass for the month.
(5 marks)
- Which of the measures of central tendency, the modal class or the estimated mean, will be most appropriate to describe the data set? Explain your choice.
(1 mark)

[TOTAL: 7 marks]

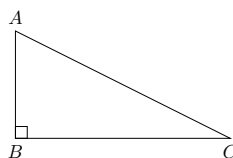
- In the diagram below, $D(-3; 3)$, $E(3; -5)$ and $F(-1; k)$ are three points in the Cartesian plane.



- Calculate the length of DE .
(2 marks)
 - Calculate the gradient of DE .
(2 marks)
 - Determine the value of k if $\hat{DEF} = 90^\circ$.
(4 marks)
 - If $k = -8$, determine the coordinates of M , the midpoint of DF .
(2 marks)
 - Determine the coordinates of a point G such that the quadrilateral $DEFG$ is a rectangle.
(4 marks)
- C is the point $(1; -2)$. The point D lies in the second quadrant and has coordinates $(x; 5)$. If the length of CD is $\sqrt{53}$ units, calculate the value of x .
(4 marks)

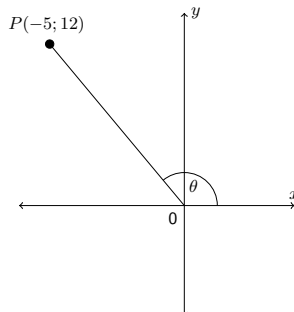
[TOTAL: 18 marks]

- In the diagram below, $\triangle ABC$ is right-angled at B .



Complete the following statements:

- i. $\sin C = \frac{AB}{?}$
(1 mark)
 - ii. $?A = \frac{AB}{BC}$
(1 mark)
- b) **Without using a calculator**, determine the value of $\frac{\sin 60^\circ \cdot \tan 30^\circ}{\sec 45^\circ}$
(4 marks)
- c) In the diagram, $P(-5; 12)$ is a point in the Cartesian plane and $\hat{R}OP = \theta$.



Determine the value of:

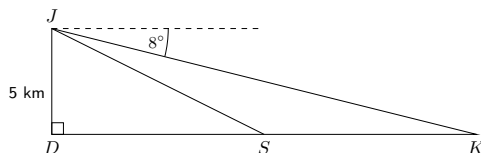
- i. $\cos \theta$
(3 marks)
- ii. $\operatorname{cosec}^2 \theta + 1$
(3 marks)

[TOTAL: 12 marks]

5. a) Solve for x , correct to ONE decimal place, in each of the following equations where $0^\circ \leq x \leq 90^\circ$.

- i. $5 \cos x = 3$
(2 marks)
- ii. $\tan 2x = 1,19$
(3 marks)
- iii. $4 \sec x - 3 = 5$
(4 marks)

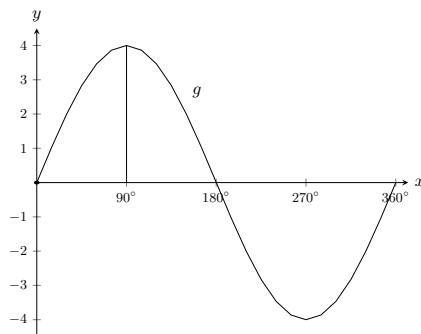
- b) An aeroplane at J is flying directly over a point D on the ground at a height of 5 kilometres. It is heading to land at point K . The angle of depression from J to K is 8° . S is a point along the route from D to K .



- i. Write down the size of $\hat{JKD}$.
(1 mark)
- ii. Calculate the distance DK , correct to the nearest metre.
(3 marks)
- iii. If the distance SK is 8 kilometres, calculate the distance DS .
(1 mark)
- iv. Calculate the angle of elevation from point S to J , correct to ONE decimal place.
(2 marks)

[TOTAL: 16 marks]

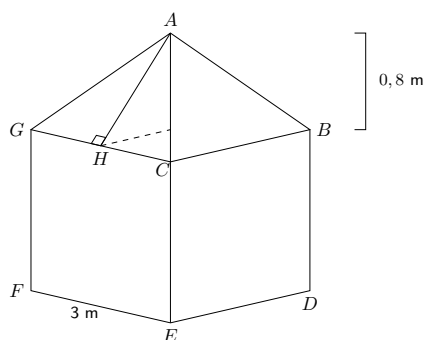
6. a) Consider the function $y = 2 \tan x$.
- i. Make a neat sketch of $y = 2 \tan x$ for $0^\circ \leq x \leq 360^\circ$ on the axes provided on DIAGRAM SHEET 1. Clearly indicate on your sketch the intercepts with the axes and the asymptotes.
(4 marks)
 - ii. If the graph of $y = 2 \tan x$ is reflected about the x -axis, write down the equation of the new graph obtained by this reflection.
(1 mark)
- b) The diagram below shows the graph of $g(x) = a \sin x$ for $0^\circ \leq x \leq 360^\circ$.



- Determine the value of a .
(1 mark)
- If the graph of g is translated 2 units upwards to obtain a new graph h , write down the range of h .
(2 marks)

[TOTAL: 8 marks]

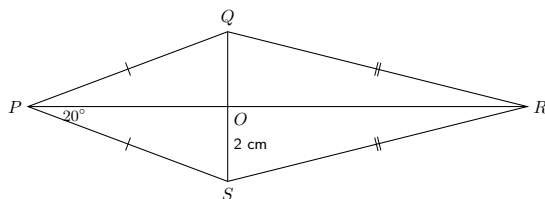
- The roof of a canvas tent is in the shape of a right pyramid having a perpendicular height of 0,8 metres on a square base. The length of one side of the base is 3 metres.



- Calculate the length of AH .
(2 marks)
 - Calculate the surface area of the roof.
(2 marks)
 - If the height of the walls of the tent is 2,1 metres, calculate the total amount of canvas required to make the tent if the floor is excluded.
(2 marks)
- A metal ball has a radius 8 millimetres.
 - Calculate the volume of metal used to make this ball, correct to TWO decimal places.
(2 marks)
 - If the radius of the ball is doubled, write down the ratio of the new volume : the original volume.
(2 marks)
 - You would like this ball to be silver plated to a thickness of 1 millimetre. What is the volume of silver required? Give your answer correct to TWO decimal places.
(2 marks)

[TOTAL: 12 marks]

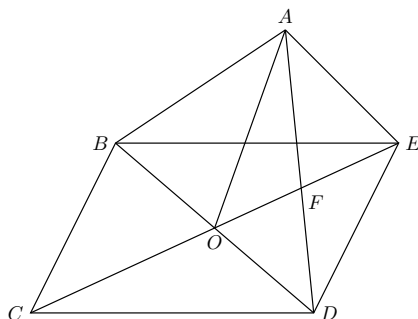
- $PQRS$ is a kite such that the diagonals intersect in O .
 $OS = 2$ cm and $\angle OPS = 20^\circ$.



- Write down the length of OQ .
(2 marks)
- Write down the size of $\angle POQ$.
(2 marks)
- Write down the size of $\angle QPS$.
(2 marks)

[TOTAL: 6 marks]

- In the diagram, $BCDE$ and $AODE$ are parallelograms.



- Prove that $OF \parallel AB$
(4 marks)
- Prove that $ABOE$ is a parallelogram.
(4 marks)
- Prove that $\triangle ABO \equiv \triangle EOD$.
(5 marks)

[TOTAL: 13 marks]

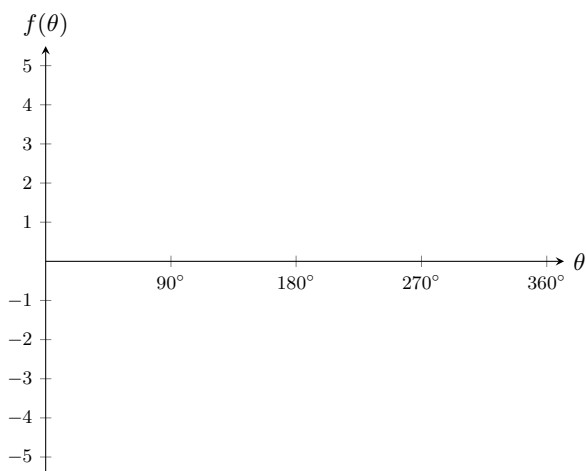
2.0 Diagram sheet 1

EMA85

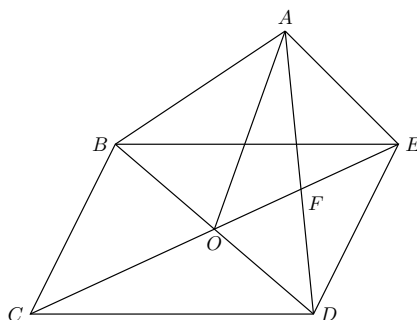
Centre number:

Examination number:

Question 6.1.1



Question 9



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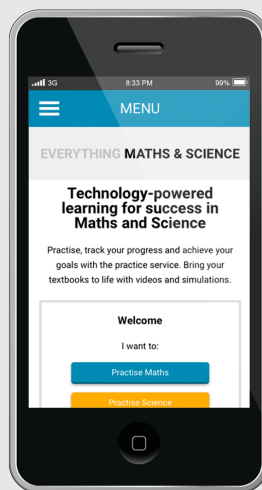
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